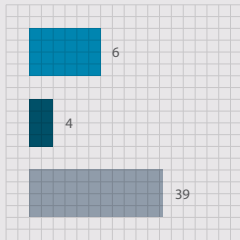
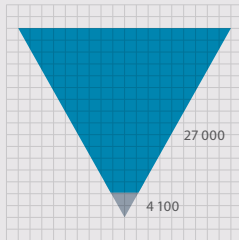


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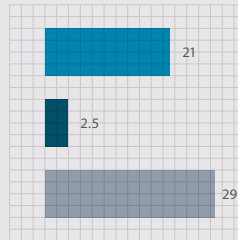
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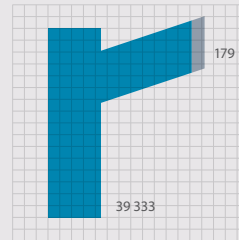
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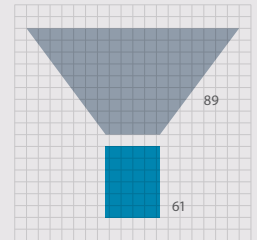
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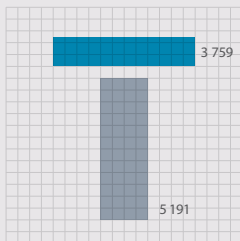
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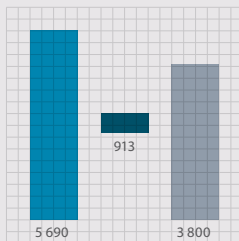
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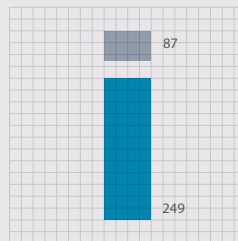
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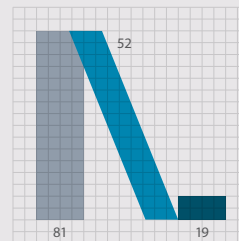
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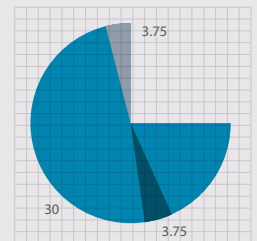
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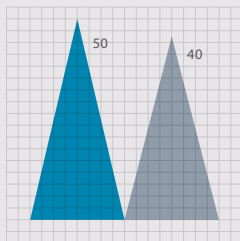
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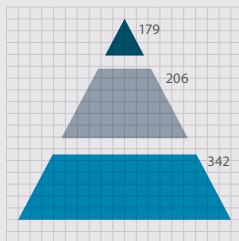
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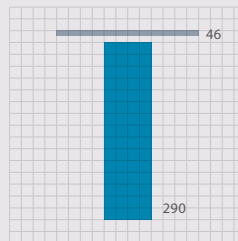
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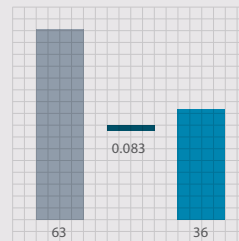
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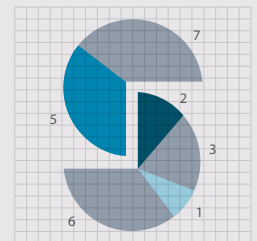
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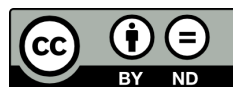
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# EVERYTHING MATHS

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Mathematics is commonly thought of as being about numbers but mathematics is actually a language! Mathematics is the language that nature speaks to us in. As we learn to understand and speak this language, we can discover many of nature's secrets. Just as understanding someone's language is necessary to learn more about them, mathematics is required to learn about all aspects of the world – whether it is physical sciences, life sciences or even finance and economics.

The great writers and poets of the world have the ability to draw on words and put them together in ways that can tell beautiful or inspiring stories. In a similar way, one can draw on mathematics to explain and create new things. Many of the modern technologies that have enriched our lives are greatly dependent on mathematics. DVDs, Google searches, bank cards with PIN numbers are just some examples. And just as words were not created specifically to tell a story but their existence enabled stories to be told, so the mathematics used to create these technologies was not developed for its own sake, but was available to be drawn on when the time for its application was right.

There is in fact not an area of life that is not affected by mathematics. Many of the most sought after careers depend on the use of mathematics. Civil engineers use mathematics to determine how to best design new structures; economists use mathematics to describe and predict how the economy will react to certain changes; investors use mathematics to price certain types of shares or calculate how risky particular investments are; software developers use mathematics for many of the algorithms (such as Google searches and data security) that make programmes useful.

But, even in our daily lives mathematics is everywhere – in our use of distance, time and money. Mathematics is even present in art, design and music as it informs proportions and musical tones. The greater our ability to understand mathematics, the greater our ability to appreciate beauty and everything in nature. Far from being just a cold and abstract discipline, mathematics embodies logic, symmetry, harmony and technological progress. More than any other language, mathematics is everywhere and universal in its application.



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The merger between Metropolitan and Momentum was lauded for the complementary fit between two companies. This complementary fit is also evident in the focus areas of CSI programmes where Metropolitan and Momentum together cover and support the most important sectors and where the greatest need is in terms of social participation.

HIV/AIDS is becoming a manageable disease in many developed countries but in a country such as ours, it remains a disease where people are still dying of this scourge unnecessarily. Metropolitan continues to make a difference in making sure that HIV AIDS moves away from being a death sentence to a manageable disease. Metropolitan's other focus area is education which remains the key to economic prosperity for our country.

Momentum's focus on persons with disabilities ensures that this community is included and allowed to make their contribution to society. Orphaned and vulnerable children are another focus area for Momentum and projects supported ensure that children are allowed to grow up safely, to assume their role along with other children in inheriting a prosperous future.

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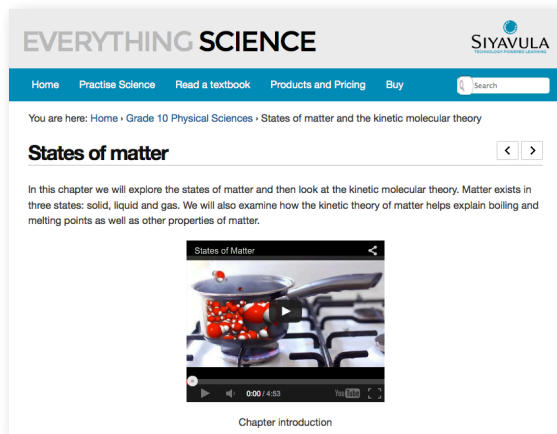
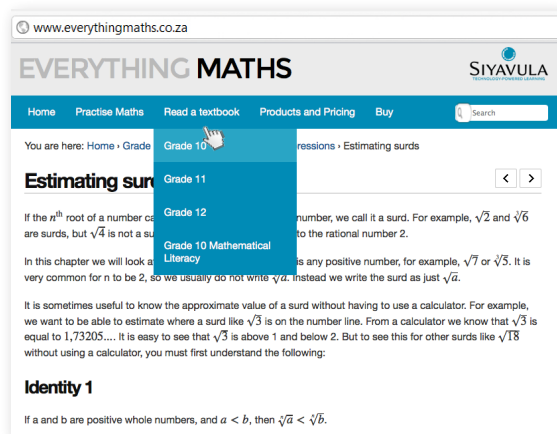


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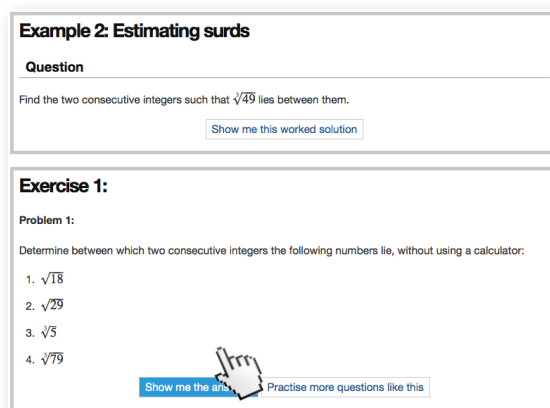
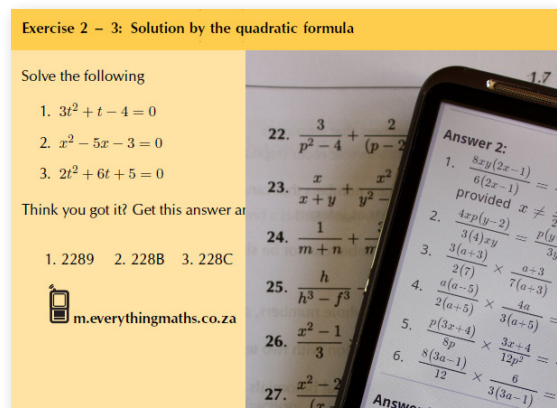
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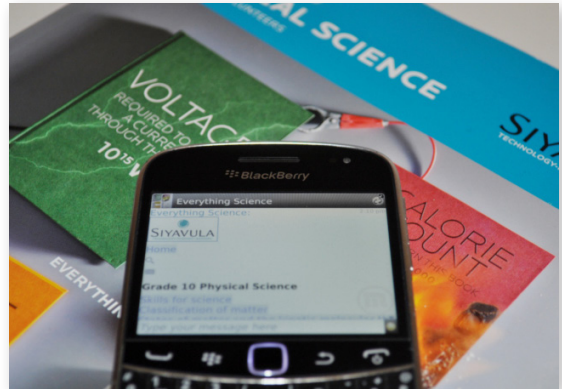


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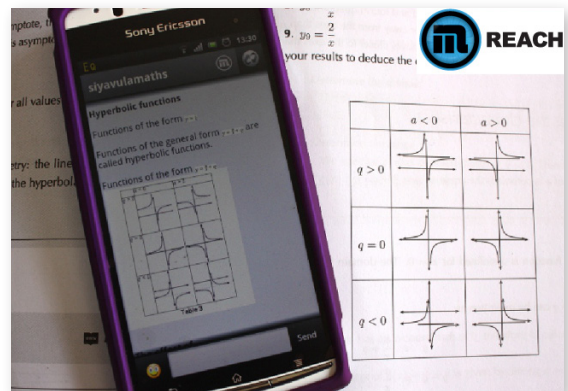
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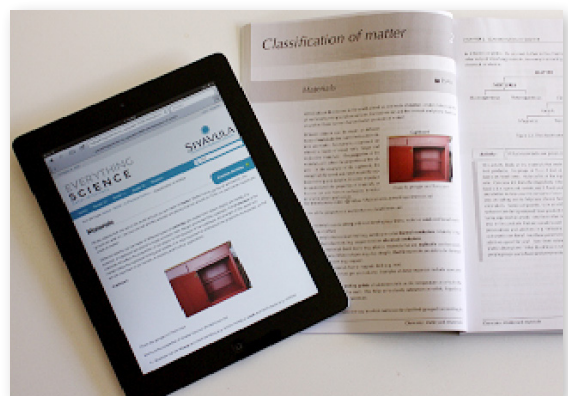
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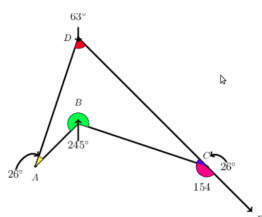
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### Angles in quadrilaterals

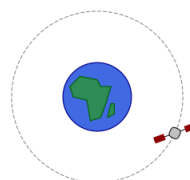
The diagram below represents quadrilateral ABCD with extended line  $\overline{CE}$ . Quadrilateral ABCD is a polygon with four sides and four angles. The sum of the interior angles in a quadrilateral =  $360^\circ$ . Angles on a straight line like  $\overline{CE} = 180^\circ$ .



### Effect of mass on gravitational force

The International Space Station (ISS) has a mass  $M$ , as it orbits the Earth, it experiences a gravitational force of  $F$ . A space shuttle docks onto the ISS. The gravitational force the ISS experiences once the mass of the shuttle is added increases by a factor of 3.

By what factor does the mass of the ISS increase for it to experience this increase of gravitational force? Write your answer as a fraction of the original mass  $M_{ISS}$  of the ISS.

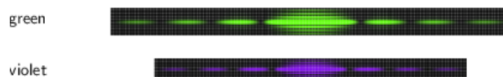


Answer:   $M_{ISS}$  [2 points] [Check answer](#)

[Help! How should I type my answer?](#)

### Wavelength and diffraction

Two diffraction patterns are presented, determine which one has the longer wavelength based on the features of the diffraction pattern. The first pattern is for green light and the second pattern is for violet light:



The same diffraction grating is used to generate both diffraction patterns.

Answer: [Select an answer](#) [2 points] [Check answer](#)

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| The atom                                          | 395 / 526 | ☆☆☆☆    |
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## *Sequences and series*

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# 1 Sequences and series

In earlier grades we learnt about number patterns, which included linear sequences with a common difference and quadratic sequences with a common second difference. We also looked at completing a sequence and how to determine the general term of a sequence.

In this chapter we also look at geometric sequences, which have a constant ratio between consecutive terms. We will learn about arithmetic and geometric series, which are the summing of the terms in sequences.

## 1.1 Arithmetic sequences

EMCDP

An arithmetic sequence is a sequence where consecutive terms are calculated by adding a constant value (positive or negative) to the previous term. We call this constant value the common difference ( $d$ ).

For example,

$$3; 0; -3; -6; -9; \dots$$

This is an arithmetic sequence because we add  $-3$  to each term to get the next term:

|             |          |               |          |
|-------------|----------|---------------|----------|
| First term  | $T_1$    |               | 3        |
| Second term | $T_2$    | $3 + (-3) =$  | 0        |
| Third term  | $T_3$    | $0 + (-3) =$  | -3       |
| Fourth term | $T_4$    | $-3 + (-3) =$ | -6       |
| Fifth term  | $T_5$    | $-6 + (-3) =$ | -9       |
| $\vdots$    | $\vdots$ | $\vdots$      | $\vdots$ |

▶ See video: 284G at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

### Exercise 1 – 1: Arithmetic sequences

Find the common difference and write down the next 3 terms of the sequence.

- 2; 6; 10; 14; 18; 22; ...
- 1; -4; -7; -10; -13; -16; ...
- 5; -3; -1; 1; 3; ...
- 1; 10; 21; 32; 43; 54; ...
- $a - 3b; a - b; a + b; a + 3b; \dots$
- $-2; -\frac{3}{2}; -1; -\frac{1}{2}; 0; \frac{1}{2}; 1; \dots$
- More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 284H   2. 284J   3. 284K   4. 284M   5. 284N   6. 284P



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For a general arithmetic sequence with first term  $a$  and a common difference  $d$ , we can generate the following terms:

$$\begin{aligned}
 T_1 &= a \\
 T_2 &= T_1 + d = a + d \\
 T_3 &= T_2 + d = (a + d) + d = a + 2d \\
 T_4 &= T_3 + d = (a + 2d) + d = a + 3d \\
 &\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 T_n &= T_{n-1} + d = (a + (n-2)d) + d = a + (n-1)d
 \end{aligned}$$

Therefore, the general formula for the  $n^{\text{th}}$  term of an arithmetic sequence is:

$$T_n = a + (n-1)d$$

**DEFINITION:** *Arithmetic sequence*

An arithmetic (or linear) sequence is an ordered set of numbers (called terms) in which each new term is calculated by adding a constant value to the previous term:

$$T_n = a + (n-1)d$$

where

- $T_n$  is the  $n^{\text{th}}$  term;
- $n$  is the position of the term in the sequence;
- $a$  is the first term;
- $d$  is the common difference.

**Test for an arithmetic sequence**

To test whether a sequence is an arithmetic sequence or not, check if the difference between any two consecutive terms is constant:

$$d = T_2 - T_1 = T_3 - T_2 = \dots = T_n - T_{n-1}$$

If this is not true, then the sequence is not an arithmetic sequence.

**Worked example 1: Arithmetic sequence**

**QUESTION**

Given the sequence  $-15; -11; -7; \dots 173$ .

1. Is this an arithmetic sequence?
2. Find the formula of the general term.
3. Determine the number of terms in the sequence.



## SOLUTION

### Step 1: Check if there is a common difference between successive terms

$$T_2 - T_1 = -11 - (-15) = 4$$

$$T_3 - T_2 = -7 - (-11) = 4$$

$\therefore$  This is an arithmetic sequence with  $d = 4$

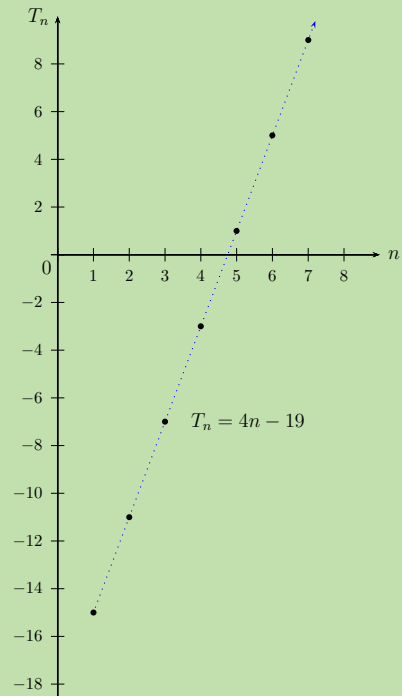
### Step 2: Determine the formula for the general term

Write down the formula and the known values:

$$T_n = a + (n - 1)d$$

$$a = -15; \quad d = 4$$

$$\begin{aligned} T_n &= a + (n - 1)d \\ &= -15 + (n - 1)(4) \\ &= -15 + 4n - 4 \\ &= 4n - 19 \end{aligned}$$



A graph was not required for this question but it has been included to show that the points of the arithmetic sequence lie in a straight line.

Note: The numbers of the sequence are natural numbers ( $n \in \{1; 2; 3; \dots\}$ ) and therefore we should not connect the plotted points. In the diagram above, a dotted line has been used to show that the graph of the sequence lies on a straight line.

### Step 3: Determine the number of terms in the sequence

$$T_n = a + (n - 1)d$$

$$173 = 4n - 19$$

$$192 = 4n$$

$$\therefore n = \frac{192}{4}$$

$$= 48$$

$$\therefore T_{48} = 173$$

### Step 4: Write the final answer

Therefore, there are 48 terms in the sequence.

## Arithmetic mean

The arithmetic mean between two numbers is the number half-way between the two numbers. In other words, it is the average of the two numbers. The arithmetic mean and the two terms form an arithmetic sequence.

For example, the arithmetic mean between 7 and 17 is calculated:

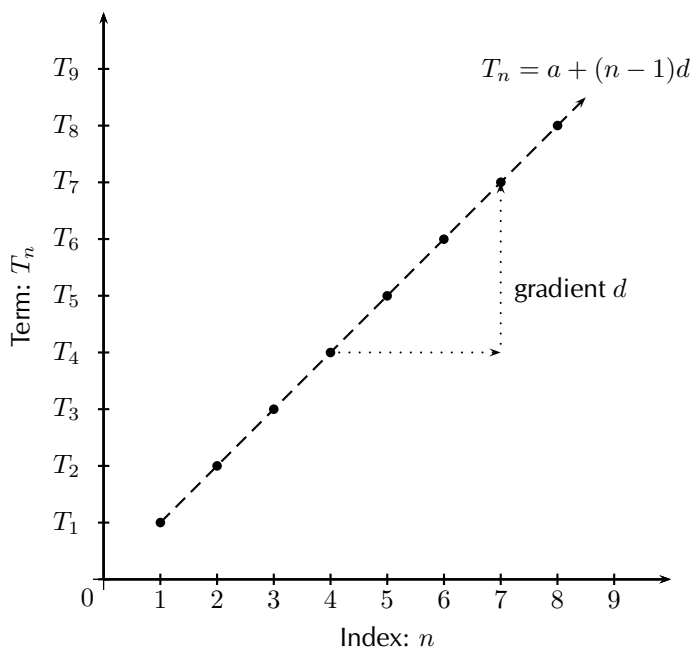
$$\begin{aligned}\text{Arithmetic mean} &= \frac{7 + 17}{2} \\ &= 12\end{aligned}$$

$\therefore 7; 12; 17$  is an arithmetic sequence

$$T_2 - T_1 = 12 - 7 = 5$$

$$T_3 - T_2 = 17 - 12 = 5$$

Plotting a graph of the terms of a sequence sometimes helps in determining the type of sequence involved. For an arithmetic sequence, plotting  $T_n$  vs.  $n$  results in the following graph:

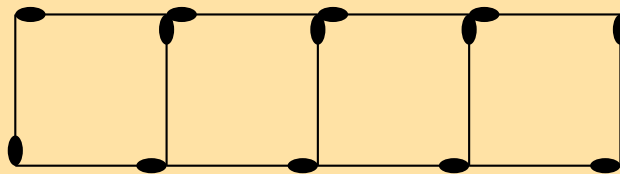


- If the sequence is arithmetic, the plotted points will lie in a straight line.
- Arithmetic sequences are also called linear sequences, where the common difference ( $d$ ) is the gradient of the straight line.

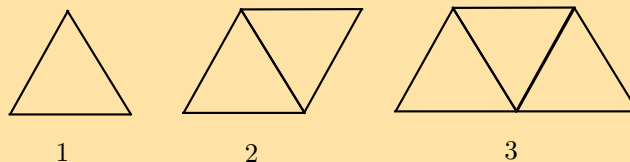
$$\begin{aligned}T_n &= a + (n - 1)d \\ \text{can be written as } T_n &= d(n - 1) + a \\ \text{which is of the same form as } y &= mx + c\end{aligned}$$

## Exercise 1 – 2: Arithmetic Sequences

- Given the sequence  $7; 5,5; 4; 2,5; \dots$ 
  - Find the next term in the sequence.
  - Determine the general term of the sequence.
  - Which term has a value of  $-23$ ?
- Given the sequence  $2; 6; 10; 14; \dots$ 
  - Is this an arithmetic sequence? Justify your answer by calculation.
  - Calculate  $T_{55}$ .
  - Which term has a value of  $322$ ?
  - Determine by calculation whether or not  $1204$  is a term in the sequence?
- An arithmetic sequence has the general term  $T_n = -2n + 7$ .
  - Calculate the second, third and tenth terms of the sequence.
  - Draw a diagram of the sequence for  $0 < n \leq 10$ .
- The first term of an arithmetic sequence is  $-\frac{1}{2}$  and  $T_{22} = 10$ . Find  $T_n$ .
- What are the important characteristics of an arithmetic sequence?
- You are given the first four terms of an arithmetic sequence. Describe the method you would use to find the formula for the  $n^{\text{th}}$  term of the sequence.
- A single square is made from 4 matchsticks. To make two squares in a row takes 7 matchsticks, while three squares in a row takes 10 matchsticks.



- Write down the first four terms of the sequence.
  - What is the common difference?
  - Determine the formula for the general term.
  - How many matchsticks are in a row of 25 squares?
  - If there are 109 matchsticks, calculate the number of squares in the row.
- A pattern of equilateral triangles decorates the border of a girl's skirt. Each triangle is made by three stitches, each having a length of 1 cm.



- Complete the table:

| Figure no.      | 1 | 2 | 3   | $q$ | $r$ | $n$ |
|-----------------|---|---|-----|-----|-----|-----|
| No. of stitches | 3 | 5 | $p$ | 15  | 71  | $s$ |

- The border of the skirt is 2 m in length. If the entire length of the border is decorated with the triangular pattern, how many stitches will there be?

9. The terms  $p$ ;  $(2p + 2)$ ;  $(5p + 3)$  form an arithmetic sequence. Find  $p$  and the 15<sup>th</sup> term of the sequence.  
[IEB, Nov 2011]
10. The arithmetic mean of  $3a - 2$  and  $x$  is  $4a - 4$ . Determine the value of  $x$  in terms of  $a$ .
11. Insert seven arithmetic means between the terms  $(3s - t)$  and  $(-13s + 7t)$ .
12. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [284Q](#) 2. [284R](#) 3. [284S](#) 4. [284T](#) 5. [284V](#) 6. [284W](#)  
7. [284X](#) 8. [284Y](#) 9. [284Z](#) 10. [2852](#) 11. [2853](#)



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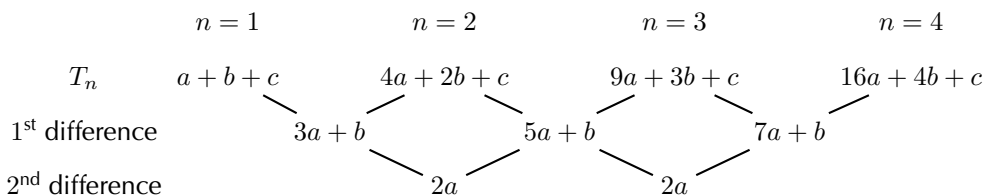
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### DEFINITION: Quadratic sequence

A quadratic sequence is a sequence of numbers in which the second difference between any two consecutive terms is constant.

The general formula for the  $n^{\text{th}}$  term of a quadratic sequence is:

$$T_n = an^2 + bn + c$$



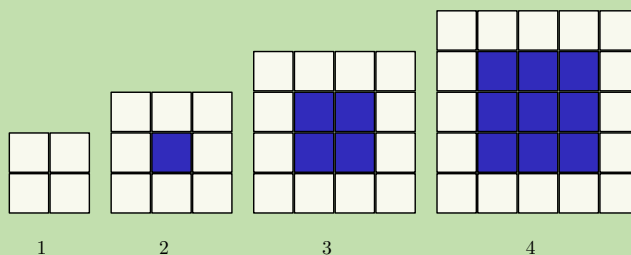
It is important to note that the first differences of a quadratic sequence form an arithmetic sequence. This sequence has a common difference of  $2a$  between consecutive terms. In other words, a linear sequence results from taking the first differences of a quadratic sequence.

### Worked example 2: Quadratic sequence

#### QUESTION

Consider the pattern of white and blue blocks in the diagram below.

- Determine the sequence formed by the white blocks ( $w$ ).
- Find the sequence formed by the blue blocks ( $b$ ).



| Pattern number ( $n$ )      | 1 | 2 | 3 | 4 | 5 | 6 | $n$ |
|-----------------------------|---|---|---|---|---|---|-----|
| No. of white blocks ( $w$ ) |   |   |   |   |   |   |     |
| Common difference ( $d$ )   |   |   |   |   |   |   |     |

| Pattern number ( $n$ )     | 1 | 2 | 3 | 4 | 5 | 6 | $n$ |
|----------------------------|---|---|---|---|---|---|-----|
| No. of blue blocks ( $b$ ) |   |   |   |   |   |   |     |
| Common difference ( $d$ )  |   |   |   |   |   |   |     |

### SOLUTION

**Step 1: Use the diagram to complete the table for the white blocks**

| Pattern number ( $n$ )      | 1 | 2 | 3  | 4  | 5  | 6  | $n$  |
|-----------------------------|---|---|----|----|----|----|------|
| No. of white blocks ( $w$ ) | 4 | 8 | 12 | 16 | 20 | 24 | $4n$ |
| Common difference ( $d$ )   |   | 4 | 4  | 4  | 4  | 4  |      |

We see that the next term in the sequence is obtained by adding 4 to the previous term, therefore the sequence is linear and the common difference ( $d$ ) is 4.

The general term is:

$$\begin{aligned}
 T_n &= a + (n - 1)d \\
 &= 4 + (n - 1)(4) \\
 &= 4 + 4n - 4 \\
 &= 4n
 \end{aligned}$$

**Step 2: Use the diagram to complete the table for the blue blocks**

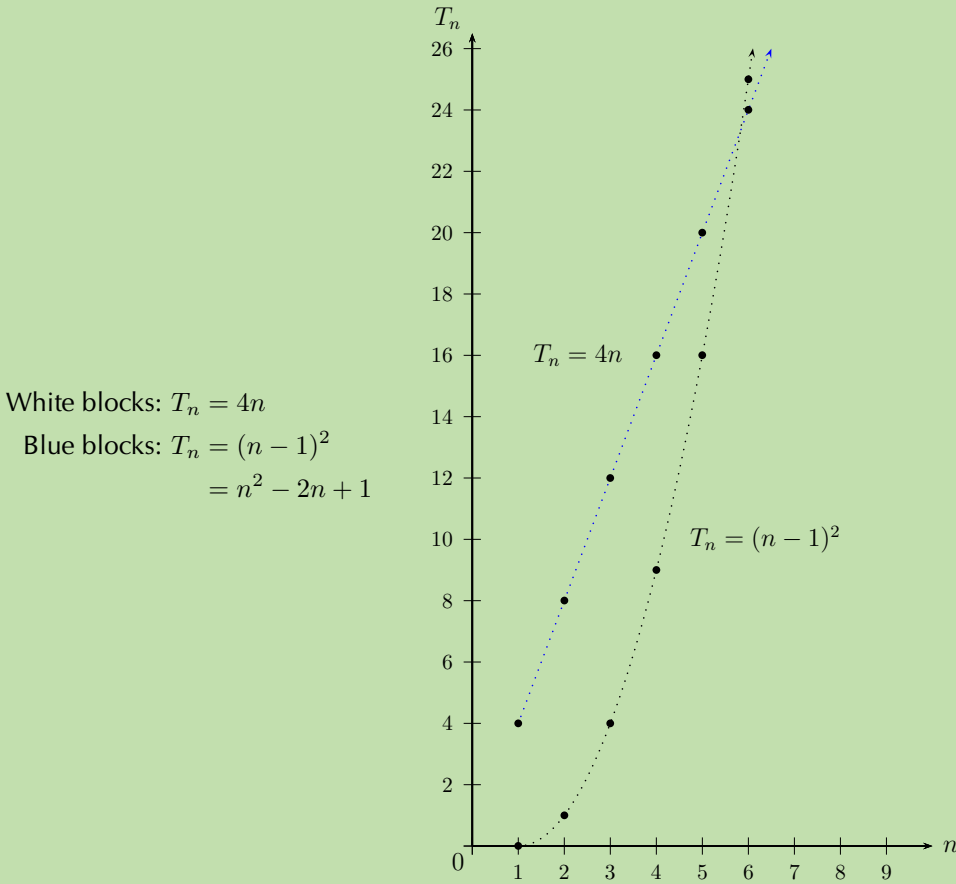
| Pattern number ( $n$ )     | 1 | 2 | 3 | 4 | 5  | 6  |
|----------------------------|---|---|---|---|----|----|
| No. of blue blocks ( $b$ ) | 0 | 1 | 4 | 9 | 16 | 25 |
| Difference                 |   | 1 | 3 | 5 | 7  | 9  |

We notice that there is no common difference between successive terms. However, there is a pattern and on further investigation we see that this is in fact a quadratic sequence:

| Pattern number ( $n$ )     | 1         | 2         | 3         | 4         | 5         | 6         | $n$       |
|----------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| No. of blue blocks ( $b$ ) | 0         | 1         | 4         | 9         | 16        | 25        | $(n-1)^2$ |
| First difference           | —         | 1         | 3         | 5         | 7         | 9         | —         |
| Second difference          | —         | —         | 2         | 2         | 2         | 2         | —         |
| Pattern                    | $(1-1)^2$ | $(2-1)^2$ | $(3-1)^2$ | $(4-1)^2$ | $(5-1)^2$ | $(6-1)^2$ | $(n-1)^2$ |

$$T_n = (n - 1)^2$$

**Step 3: Draw a graph of  $T_n$  vs.  $n$  for each sequence**



Since the numbers of the sequences are natural numbers ( $n \in \{1; 2; 3; \dots\}$ ), we should not connect the plotted points. In the diagram above, a dotted line has been used to show that the graph of the sequence formed by the white blocks ( $w$ ) is a straight line and the graph of the sequence formed by the blue blocks ( $b$ ) is a parabola.

### Exercise 1 – 3: Quadratic sequences

1. Determine whether each of the following sequences is:

- a linear sequence;
- a quadratic sequence;
- or neither.

a)  $8; 17; 32; 53; 80; \dots$

e)  $5; 19; 41; 71; 109; \dots$

b)  $3p^2; 6p^2; 9p^2; 12p^2; 15p^2; \dots$

f)  $3; 9; 16; 21; 27; \dots$

c)  $1; 2,5; 5; 8,5; 13; \dots$

g)  $2k; 8k; 18k; 32k; 50k; \dots$

d)  $2; 6; 10; 14; 18; \dots$

h)  $2\frac{1}{2}; 6; 10\frac{1}{2}; 16; 22\frac{1}{2}; \dots$

2. A quadratic pattern is given by  $T_n = n^2 + bn + c$ . Find the values of  $b$  and  $c$  if the sequence starts with the following terms:

$$-1; 2; 7; 14; \dots$$

3.  $a^2; -a^2; -3a^2; -5a^2; \dots$  are the first 4 terms of a sequence.

- Is the sequence linear or quadratic? Motivate your answer.
- What is the next term in the sequence?
- Calculate  $T_{100}$ .

4. Given  $T_n = n^2 + bn + c$ , determine the values of  $b$  and  $c$  if the sequence starts with the terms:

$$2; 7; 14; 23; \dots$$

5. The first term of a quadratic sequence is 4, the third term is 34 and the common second difference is 10. Determine the first six terms in the sequence.

6. A quadratic sequence has a second term equal to 1, a third term equal to  $-6$  and a fourth term equal to  $-14$ .

- Determine the second difference for this sequence.
- Hence, or otherwise, calculate the first term of the pattern.

7. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 2854   1b. 2855   1c. 2856   1d. 2857   1e. 2858   1f. 2859

1g. 285B   1h. 285C   2. 285D   3. 285F   4. 285G   5. 285H

6. 285J



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**DEFINITION:** *Geometric sequence*

A geometric sequence is a sequence of numbers in which each new term (except for the first term) is calculated by multiplying the previous term by a constant value called the constant ratio ( $r$ ).

► See video: 285K at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

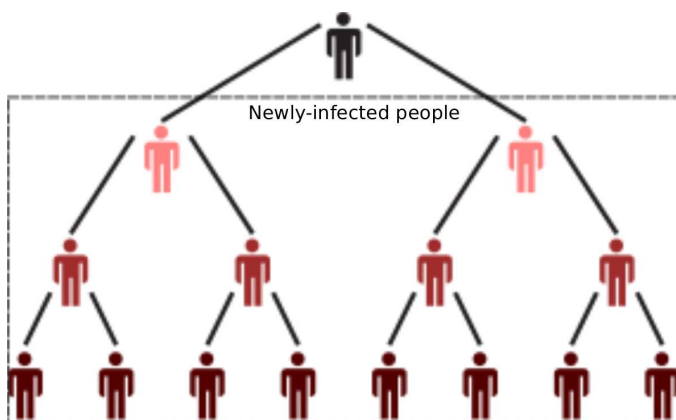
This means that the ratio between consecutive numbers in a geometric sequence is a constant (positive or negative). We will explain what we mean by ratio after looking at the following example.

**Example:** A flu epidemic

EMCDS

Influenza (commonly called “flu”) is caused by the influenza virus, which infects the respiratory tract (nose, throat, lungs). It can cause mild to severe illness that most of us get during winter time. The influenza virus is spread from person to person in respiratory droplets of coughs and sneezes. This is called “droplet spread”. This can happen when droplets from a cough or sneeze of an infected person are propelled through the air and deposited on the mouth or nose of people nearby. It is good practice to cover your mouth when you cough or sneeze so as not to infect others around you when you have the flu. Regular hand washing is an effective way to prevent the spread of infection and illness.

Assume that you have the flu virus, and you forgot to cover your mouth when two friends came to visit while you were sick in bed. They leave, and the next day they also have the flu. Let’s assume that each friend in turn spreads the virus to two of their friends by the same droplet spread the following day. Assuming this pattern continues and each sick person infects 2 other friends, we can represent these events in the following manner:



Each person infects two more people with the flu virus.



We can tabulate the events and formulate an equation for the general case:

| Day (n)  | No. of newly-infected people                                              |
|----------|---------------------------------------------------------------------------|
| 1        | $2 = 2$                                                                   |
| 2        | $4 = 2 \times 2 = 2 \times 2^1$                                           |
| 3        | $8 = 2 \times 4 = 2 \times 2 \times 2 = 2 \times 2^2$                     |
| 4        | $16 = 2 \times 8 = 2 \times 2 \times 2 \times 2 = 2 \times 2^3$           |
| 5        | $32 = 2 \times 16 = 2 \times 2 \times 2 \times 2 \times 2 = 2 \times 2^4$ |
| $\vdots$ | $\vdots$                                                                  |
| $n$      | $2 \times 2 \times 2 \times 2 \times \dots \times 2 = 2 \times 2^{n-1}$   |

The above table represents the number of **newly-infected** people after  $n$  days since you first infected your 2 friends.

You sneeze and the virus is carried over to 2 people who start the chain ( $a = 2$ ). The next day, each one then infects 2 of their friends. Now 4 people are newly-infected. Each of them infects 2 people the third day, and 8 new people are infected, and so on. These events can be written as a geometric sequence:

$$2; 4; 8; 16; 32; \dots$$

Note the constant ratio ( $r = 2$ ) between the events. Recall from the linear arithmetic sequence how the common difference between terms was established. In the geometric sequence we can determine the constant ratio ( $r$ ) from:

$$\frac{T_2}{T_1} = \frac{T_3}{T_2} = r$$

More generally,

$$\frac{T_n}{T_{n-1}} = r$$

#### Exercise 1 – 4: Constant ratio of a geometric sequence

Determine the constant ratios for the following geometric sequences and write down the next three terms in each sequence:

1.  $5; 10; 20; \dots$
2.  $\frac{1}{2}; \frac{1}{4}; \frac{1}{8}; \dots$
3.  $7; 0,7; 0,07; \dots$
4.  $p; 3p^2; 9p^3; \dots$
5.  $-3; 30; -300; \dots$
6. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [285M](#) 2. [285N](#) 3. [285P](#) 4. [285Q](#) 5. [285R](#)



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From the flu example above we know that  $T_1 = 2$  and  $r = 2$ , and we have seen from the table that the  $n^{\text{th}}$  term is given by  $T_n = 2 \times 2^{n-1}$ .

The general geometric sequence can be expressed as:

$$\begin{array}{ll} T_1 = a & = ar^0 \\ T_2 = a \times r & = ar^1 \\ T_3 = a \times r \times r & = ar^2 \\ T_4 = a \times r \times r \times r & = ar^3 \\ T_n = a \times [r \times r \dots (n-1) \text{ times}] & = ar^{n-1} \end{array}$$

Therefore the general formula for a geometric sequence is:

$$T_n = ar^{n-1}$$

where

- $a$  is the first term in the sequence;
- $r$  is the constant ratio.

### Test for a geometric sequence

To test whether a sequence is a geometric sequence or not, check if the ratio between any two consecutive terms is constant:

$$\frac{T_2}{T_1} = \frac{T_3}{T_2} = \frac{T_n}{T_{n-1}} = r$$

If this condition does not hold, then the sequence is not a geometric sequence.

### Exercise 1 – 5: General term of a geometric sequence

Determine the general formula for the  $n^{\text{th}}$  term of each of the following geometric sequences:

1. 5; 10; 20; ...
2.  $\frac{1}{2}$ ;  $\frac{1}{4}$ ;  $\frac{1}{8}$ ; ...
3. 7; 0,7; 0,07; ...
4.  $p$ ;  $3p^2$ ;  $9p^3$ ; ...
5. -3; 30; -300; ...
6. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 285S   2. 285T   3. 285V   4. 285W   5. 285X



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### Worked example 3: Flu epidemic

#### QUESTION

We continue with the previous flu example, where  $T_n$  is the number of newly-infected people after  $n$  days:

$$T_n = 2 \times 2^{n-1}$$

1. Calculate how many newly-infected people there are on the tenth day.
2. On which day will 16 384 people be newly-infected?

#### SOLUTION

**Step 1: Write down the known values and the general formula**

$$a = 2$$

$$r = 2$$

$$T_n = 2 \times 2^{n-1}$$

**Step 2: Use the general formula to calculate  $T_{10}$**

Substitute  $n = 10$  into the general formula:

$$\begin{aligned} T_n &= a \times r^{n-1} \\ \therefore T_{10} &= 2 \times 2^{10-1} \\ &= 2 \times 2^9 \\ &= 2 \times 512 \\ &= 1024 \end{aligned}$$

On the tenth day, there are 1024 newly-infected people.

**Step 3: Use the general formula to calculate  $n$**

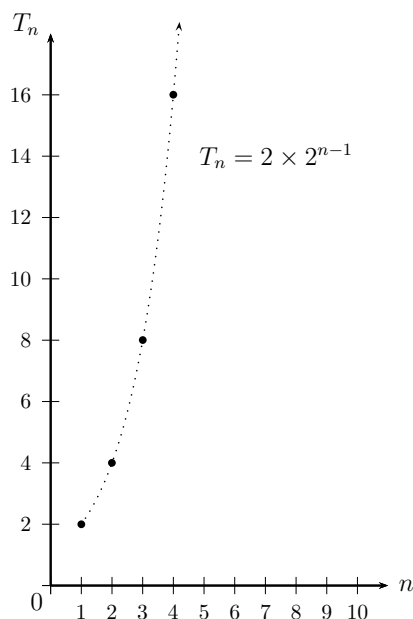
We know that  $T_n = 16\,384$  and can use the general formula to calculate the corresponding value of  $n$ :

$$\begin{aligned} T_n &= ar^{n-1} \\ 16\,384 &= 2 \times 2^{n-1} \\ \frac{16\,384}{2} &= 2^{n-1} \\ 8192 &= 2^{n-1} \\ \text{We can write } 8192 &\text{ as } 2^{13} \\ \text{So } 2^{13} &= 2^{n-1} \\ \therefore 13 &= n - 1 \quad (\text{same bases}) \\ \therefore n &= 14 \end{aligned}$$

There are 16 384 newly-infected people on the 14<sup>th</sup> day.

For this geometric sequence, plotting the number of newly-infected people ( $T_n$ ) vs. the number of days ( $n$ ) results in the following graph:

| Day ( $n$ ) | No. of newly-infected people |
|-------------|------------------------------|
| 1           | 2                            |
| 2           | 4                            |
| 3           | 8                            |
| 4           | 16                           |
| 5           | 32                           |
| 6           | 64                           |
| $n$         | $2 \times 2^{n-1}$           |



In this example we are only dealing with positive integers ( $n \in \{1; 2; 3; \dots\}$ ,  $T_n \in \{1; 2; 3; \dots\}$ ), therefore the graph is not continuous and we do not join the points with a curve (the dotted line has been drawn to indicate the shape of an exponential graph).

### Geometric mean

The geometric mean between two numbers is the value that forms a geometric sequence together with the two numbers.

For example, the geometric mean between 5 and 20 is the number that has to be inserted between 5 and 20 to form the geometric sequence: 5;  $x$ ; 20

$$\begin{aligned}
 \text{Determine the constant ratio: } \frac{x}{5} &= \frac{20}{x} \\
 \therefore x^2 &= 20 \times 5 \\
 x^2 &= 100 \\
 x &= \pm 10
 \end{aligned}$$

**Important:** remember to include both the positive and negative square root. The geometric mean generates two possible geometric sequences:

$$5; 10; 20; \dots$$

$$5; -10; 20; \dots$$

In general, the geometric mean ( $x$ ) between two numbers  $a$  and  $b$  forms a geometric sequence with  $a$  and  $b$ :

For a geometric sequence:  $a; x; b$

$$\begin{aligned}
 \text{Determine the constant ratio: } \frac{x}{a} &= \frac{b}{x} \\
 x^2 &= ab \\
 \therefore x &= \pm \sqrt{ab}
 \end{aligned}$$

## Exercise 1 – 6: Mixed exercises

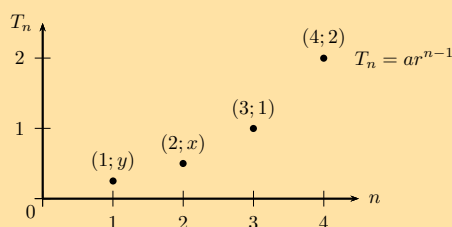
- The  $n^{\text{th}}$  term of a sequence is given by the formula  $T_n = 6 \left(\frac{1}{3}\right)^{n-1}$ .
  - Write down the first three terms of the sequence.
  - What type of sequence is this?
- Consider the following terms:

$$(k-4); (k+1); m; 5k$$

The first three terms form an arithmetic sequence and the last three terms form a geometric sequence. Determine the values of  $k$  and  $m$  if both are positive integers.

[IEB, Nov 2006]

- Given a geometric sequence with second term  $\frac{1}{2}$  and ninth term 64.
  - Determine the value of  $r$ .
  - Find the value of  $a$ .
  - Determine the general formula of the sequence.
- The diagram shows four sets of values of consecutive terms of a geometric sequence with the general formula  $T_n = ar^{n-1}$ .



- Determine  $a$  and  $r$ .
  - Find  $x$  and  $y$ .
  - Find the fifth term of the sequence.
- Write down the next two terms for the following sequence:
$$1; \sin \theta; 1 - \cos^2 \theta; \dots$$
  - $5; x; y$  is an arithmetic sequence and  $x; y; 81$  is a geometric sequence. All terms in the sequences are integers. Calculate the values of  $x$  and  $y$ .
  - The two numbers  $2x^2y^2$  and  $8x^4$  are given.
    - Write down the geometric mean between the two numbers in terms of  $x$  and  $y$ .
    - Determine the constant ratio of the resulting sequence.
  - Insert three geometric means between  $-1$  and  $-\frac{1}{81}$ . Give all possible answers.
  - More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- 285Y
- 285Z
- 2862
- 2863
- 2864
- 2865
- 2866
- 2867



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It is often important and valuable to determine the sum of the terms of an arithmetic or geometric sequence. The sum of any sequence of numbers is called a series.

### Finite series

We use the symbol  $S_n$  for the sum of the first  $n$  terms of a sequence  $\{T_1; T_2; T_3; \dots; T_n\}$ :

$$S_n = T_1 + T_2 + T_3 + \dots + T_n$$

If we sum only a finite number of terms, we get a finite series.

For example, consider the following sequence of numbers

$$1; 4; 9; 16; 25; 36; 49; \dots$$

We can calculate the sum of the first four terms:

$$S_4 = 1 + 4 + 9 + 16 = 30$$

This is an example of a finite series since we are only summing four terms.

### Infinite series

If we sum infinitely many terms of a sequence, we get an infinite series:

$$S_\infty = T_1 + T_2 + T_3 + \dots$$

## Sigma notation

Sigma notation is a very useful and compact notation for writing the sum of a given number of terms of a sequence.

A sum may be written out using the summation symbol  $\sum$  (Sigma), which is the capital letter “S” in the Greek alphabet. It indicates that you must sum the expression to the right of the summation symbol:

For example,

$$\sum_{n=1}^5 2n = 2 + 4 + 6 + 8 + 10 = 30$$

In general,

$$\sum_{i=m}^n T_i = T_m + T_{m+1} + \dots + T_{n-1} + T_n$$

where

- $i$  is the index of the sum;
- $m$  is the lower bound (or start index), shown below the summation symbol;
- $n$  is the upper bound (or end index), shown above the summation symbol;
- $T_i$  is a term of a sequence;
- the number of terms in the series = end index – start index + 1.

The index  $i$  increases from  $m$  to  $n$  by steps of 1.

Note that this is also sometimes written as:

$$\sum_{i=m}^n a_i = a_m + a_{m+1} + \cdots + a_{n-1} + a_n$$

When we write out all the terms in a sum, it is referred to as the expanded form.

If we are summing from  $i = 1$  (which implies summing from the first term in a sequence), then we can use either  $S_n$  or  $\sum$  notation:

$$S_n = \sum_{i=1}^n a_i = a_1 + a_2 + \cdots + a_n \quad (n \text{ terms})$$

#### Worked example 4: Sigma notation

##### QUESTION

Expand the sequence and find the value of the series:

$$\sum_{n=1}^6 2^n$$

##### SOLUTION

**Step 1: Expand the formula and write down the first six terms of the sequence**

$$\begin{aligned} \sum_{n=1}^6 2^n &= 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 \quad (6 \text{ terms}) \\ &= 2 + 4 + 8 + 16 + 32 + 64 \end{aligned}$$

This is a geometric sequence 2; 4; 8; 16; 32; 64 with a constant ratio of 2 between consecutive terms.

**Step 2: Determine the sum of the first six terms of the sequence**

$$\begin{aligned} S_6 &= 2 + 4 + 8 + 16 + 32 + 64 \\ &= 126 \end{aligned}$$

### Worked example 5: Sigma notation

#### QUESTION

Find the value of the series:

$$\sum_{n=3}^7 2an$$

#### SOLUTION

**Step 1: Expand the sequence and write down the five terms**

$$\begin{aligned}\sum_{n=3}^7 2an &= 2a(3) + 2a(4) + 2a(5) + 2a(6) + 2a(7) \quad (5 \text{ terms}) \\ &= 6a + 8a + 10a + 12a + 14a\end{aligned}$$

**Step 2: Determine the sum of the five terms of the sequence**

$$\begin{aligned}S_5 &= 6a + 8a + 10a + 12a + 14a \\ &= 50a\end{aligned}$$

### Worked example 6: Sigma notation

#### QUESTION

Write the following series in sigma notation:

$$31 + 24 + 17 + 10 + 3$$

#### SOLUTION

**Step 1: Consider the series and determine if it is an arithmetic or geometric series**

First test for an arithmetic series: is there a common difference?

We let:

$$\begin{aligned}T_1 &= 31; & T_4 &= 10; \\ T_2 &= 24; & T_5 &= 3; \\ T_3 &= 17;\end{aligned}$$

We calculate:

$$\begin{aligned}d &= T_2 - T_1 \\ &= 24 - 31 \\ &= -7 \\ d &= T_3 - T_2 \\ &= 17 - 24 \\ &= -7\end{aligned}$$

There is a common difference of  $-7$ , therefore this is an arithmetic series.



### Step 2: Determine the general formula of the series

$$\begin{aligned}T_n &= a + (n - 1)d \\&= 31 + (n - 1)(-7) \\&= 31 - 7n + 7 \\&= -7n + 38\end{aligned}$$

**Be careful:** brackets must be used when substituting  $d = -7$  into the general term. Otherwise the equation would be  $T_n = 31 + (n - 1) - 7$ , which would be incorrect.

### Step 3: Determine the sum of the series and write in sigma notation

$$\begin{aligned}31 + 24 + 17 + 10 + 3 &= 85 \\ \therefore \sum_{n=1}^5 (-7n + 38) &= 85\end{aligned}$$

### Rules for sigma notation

1. Given two sequences,  $a_i$  and  $b_i$ :

$$\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$$

2. For any constant  $c$  that is not dependent on the index  $i$ :

$$\begin{aligned}\sum_{i=1}^n (c \cdot a_i) &= c \cdot a_1 + c \cdot a_2 + c \cdot a_3 + \cdots + c \cdot a_n \\&= c(a_1 + a_2 + a_3 + \cdots + a_n) \\&= c \sum_{i=1}^n a_i\end{aligned}$$

3. Be accurate with the use of brackets:

Example 1:

$$\begin{aligned}\sum_{n=1}^3 (2n + 1) &= 3 + 5 + 7 \\&= 15\end{aligned}$$

Example 2:

$$\begin{aligned}\sum_{n=1}^3 (2n) + 1 &= (2 + 4 + 6) + 1 \\&= 13\end{aligned}$$

Note: the series in the second example has the general term  $T_n = 2n$  and the  $+1$  is added to the sum of the three terms. It is very important in sigma notation to use brackets correctly.

4.

$$\sum_{i=m}^n a_i$$

The values of  $i$ :

- start at  $m$  ( $m$  is not always 1);
- increase in steps of 1;
- and end at  $n$ .

### Exercise 1 – 7: Sigma notation

1. Determine the value of the following:

a)  $\sum_{k=1}^4 2$

b)  $\sum_{i=-1}^3 i$

c)  $\sum_{n=2}^5 (3n - 2)$

2. Expand the series:

a)  $\sum_{k=1}^6 0^k$

b)  $\sum_{n=-3}^0 8$

c)  $\sum_{k=1}^5 (ak)$

3. Calculate the value of  $a$ :

a)  $\sum_{k=1}^3 (a \cdot 2^{k-1}) = 28$

b)  $\sum_{j=1}^4 (2^{-j}) = a$

4. Write the following in sigma notation:

$$\frac{1}{9} + \frac{1}{3} + 1 + 3$$

5. Write the sum of the first 25 terms of the series below in sigma notation:

$$11 + 4 - 3 - 10 \dots$$

6. Write the sum of the first 1000 natural, odd numbers in sigma notation.

7. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 286B   1b. 286G   1c. 286B   2a. 286C   2b. 286D   2c. 286F  
3a. 286G   3b. 286H   4. 286J   5. 286K   6. 286M



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An arithmetic sequence is a sequence of numbers, such that the difference between any term and the previous term is a constant number called the common difference ( $d$ ):

$$T_n = a + (n - 1) d$$

where

- $T_n$  is the  $n^{\text{th}}$  term of the sequence;
- $a$  is the first term;
- $d$  is the common difference.

When we sum a finite number of terms in an arithmetic sequence, we get a finite arithmetic series.

### The sum of the first one hundred integers

A simple arithmetic sequence is when  $a = 1$  and  $d = 1$ , which is the sequence of positive integers:

$$\begin{aligned} T_n &= a + (n - 1) d \\ &= 1 + (n - 1) (1) \\ &= n \\ \therefore \{T_n\} &= 1; 2; 3; 4; 5; \dots \end{aligned}$$

If we wish to sum this sequence from  $n = 1$  to any positive integer, for example 100, we would write

$$\sum_{n=1}^{100} n = 1 + 2 + 3 + \dots + 100$$

This gives the answer to the sum of the first 100 positive integers.

The mathematician, Karl Friedrich Gauss, discovered the following proof when he was only 8 years old. His teacher had decided to give his class a problem which would distract them for the entire day by asking them to add all the numbers from 1 to 100. Young Karl quickly realised how to do this and shocked the teacher with the correct answer, 5050. This is the method that he used:

- Write the numbers in ascending order.
- Write the numbers in descending order.
- Add the corresponding pairs of terms together.
- Simplify the equation by making  $S_n$  the subject of the equation.

$$\begin{aligned}
S_{100} &= 1 + 2 + 3 + \cdots + 98 + 99 + 100 \\
+ \quad S_{100} &= \underline{100 + 99 + 98 + \cdots + 3 + 2 + 1} \\
\therefore 2S_{100} &= 101 + 101 + 101 + \cdots + 101 + 101 + 101 \\
\therefore 2S_{100} &= 101 \times 100 \\
&= 10\,100 \\
\therefore S_{100} &= \frac{10\,100}{2} \\
&= 5050
\end{aligned}$$

## General formula for a finite arithmetic series

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If we sum an arithmetic sequence, it takes a long time to work it out term-by-term. We therefore derive the general formula for evaluating a finite arithmetic series. We start with the general formula for an arithmetic sequence of  $n$  terms and sum it from the first term ( $a$ ) to the last term in the sequence ( $l$ ):

$$\begin{aligned}
\sum_{n=1}^l T_n &= S_n \\
S_n &= a + (a + d) + (a + 2d) + \cdots + (l - 2d) + (l - d) + l \\
+ \quad S_n &= \underline{l + (l - d) + (l - 2d) + \cdots + (a + 2d) + (a + d) + a} \\
\therefore 2S_n &= (a + l) + (a + l) + (a + l) + \cdots + (a + l) + (a + l) + (a + l) \\
\therefore 2S_n &= n \times (a + l) \\
\therefore S_n &= \frac{n}{2}(a + l)
\end{aligned}$$

This general formula is useful if the last term in the series is known.

We substitute  $l = a + (n - 1)d$  into the above formula and simplify:

$$\begin{aligned}
S_n &= \frac{n}{2}(a + [a + (n - 1)d]) \\
\therefore S_n &= \frac{n}{2}[2a + (n - 1)d]
\end{aligned}$$

The general formula for determining the sum of an arithmetic series is given by:

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

or

$$S_n = \frac{n}{2}(a + l)$$

For example, we can calculate the sum  $S_{20}$  for the arithmetic sequence  $T_n = 3 + 7(n - 1)$  by summing all the individual terms:

$$\begin{aligned}
S_{20} &= \sum_{n=1}^{20} [3 + 7(n - 1)] \\
&= 3 + 10 + 17 + 24 + 31 + 38 + 45 + 52 \\
&\quad + 59 + 66 + 73 + 80 + 87 + 94 + 101 \\
&\quad + 108 + 115 + 122 + 129 + 136 \\
&= 1390
\end{aligned}$$

or, more sensibly, we could use the general formula for determining an arithmetic series by substituting  $a = 3$ ,  $d = 7$  and  $n = 20$ :

$$\begin{aligned}S_n &= \frac{n}{2}(2a + (n-1)d) \\S_{20} &= \frac{20}{2}[2(3) + 7(20-1)] \\&= 1390\end{aligned}$$

This example demonstrates how useful the general formula for determining an arithmetic series is, especially when the series has a large number of terms.

▶ See video: 286N at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

### Worked example 7: General formula for the sum of an arithmetic sequence

#### QUESTION

Find the sum of the first 30 terms of an arithmetic series with  $T_n = 7n - 5$  by using the formula.

#### SOLUTION

**Step 1: Use the general formula to generate terms of the sequence and write down the known variables**

$$\begin{aligned}T_n &= 7n - 5 \\ \therefore T_1 &= 7(1) - 5 \\ &= 2 \\ T_2 &= 7(2) - 5 \\ &= 9 \\ T_3 &= 7(3) - 5 \\ &= 16\end{aligned}$$

This gives the sequence: 2; 9; 16 ...

$$a = 2; \quad d = 7; \quad n = 30$$

**Step 2: Write down the general formula and substitute the known values**

$$\begin{aligned}S_n &= \frac{n}{2}[2a + (n-1)d] \\ S_{30} &= \frac{30}{2}[2(2) + (30-1)(7)] \\ &= 15(4 + 203) \\ &= 15(207) \\ &= 3105\end{aligned}$$

**Step 3: Write the final answer**

$$S_{30} = 3105$$

### Worked example 8: Sum of an arithmetic sequence if first and last terms are known

#### QUESTION

Find the sum of the series  $-5 - 3 - 1 + \dots + 123$

#### SOLUTION

**Step 1: Identify the type of series and write down the known variables**

$$\begin{aligned}d &= T_2 - T_1 \\&= -3 - (-5) \\&= 2 \\d &= T_3 - T_2 \\&= -1 - (-3) \\&= 2\end{aligned}$$

$$a = -5; \quad d = 2; \quad l = 123$$

**Step 2: Determine the value of  $n$**

$$\begin{aligned}T_n &= a + (n - 1)d \\ \therefore 123 &= -5 + (n - 1)(2) \\ &= -5 + 2n - 2 \\ \therefore 130 &= 2n \\ \therefore n &= 65\end{aligned}$$

**Step 3: Use the general formula to find the sum of the series**

$$\begin{aligned}S_n &= \frac{n}{2}(a + l) \\ S_{65} &= \frac{65}{2}(-5 + 123) \\ &= \frac{65}{2}(118) \\ &= 3835\end{aligned}$$

**Step 4: Write the final answer**

$$S_{65} = 3835$$

### Worked example 9: Finding $n$ given the sum of an arithmetic sequence

#### QUESTION

Given an arithmetic sequence with  $T_2 = 7$  and  $d = 3$ , determine how many terms must be added together to give a sum of 2146.

#### SOLUTION

**Step 1: Write down the known variables**

$$\begin{aligned}d &= T_2 - T_1 \\ \therefore 3 &= 7 - a \\ \therefore a &= 4 \\ a &= 4; \quad d = 3; \quad S_n = 2146\end{aligned}$$

**Step 2: Use the general formula to determine the value of  $n$**

$$\begin{aligned}S_n &= \frac{n}{2}(2a + (n-1)d) \\2146 &= \frac{n}{2}(2(4) + (n-1)(3)) \\4292 &= n(8 + 3n - 3) \\\therefore 0 &= 3n^2 + 5n - 4292 \\&= (3n + 116)(n - 37) \\\therefore n &= -\frac{116}{3} \text{ or } n = 37\end{aligned}$$

but  $n$  must be a positive integer, therefore  $n = 37$ .

We could have solved for  $n$  using the quadratic formula but factorising by inspection is usually the quickest method.

**Step 3: Write the final answer**

$$S_{37} = 2146$$

### Worked example 10: Finding $n$ given the sum of an arithmetic sequence

#### QUESTION

The sum of the second and third terms of an arithmetic sequence is equal to zero and the sum of the first 36 terms of the series is equal to 1152. Find the first three terms in the series.

#### SOLUTION

**Step 1: Write down the given information**

$$\begin{aligned}T_2 + T_3 &= 0 & S_n &= \frac{n}{2}(2a + (n-1)d) \\ \text{So } (a + d) + (a + 2d) &= 0 & S_{36} &= \frac{36}{2}(2a + (36-1)d) \\ \therefore 2a + 3d &= 0 \dots\dots (1) & 1152 &= 18(2a + 35d) \\ & & \therefore 64 &= 2a + 35d \dots\dots (2)\end{aligned}$$

**Step 2: Solve the two equations simultaneously**

$$\begin{aligned}2a + 3d &= 0 \dots\dots (1) \\ 2a + 35d &= 64 \dots\dots (2) \\ \text{Eqn (2)} - (1) : & 32d = 64 \\ \therefore d &= 2 \\ \text{And } 2a + 3(2) &= 0 \\ 2a &= -6 \\ \therefore a &= -3\end{aligned}$$

### Step 3: Write the final answer

The first three terms of the series are:

$$T_1 = a = -3$$

$$T_2 = a + d = -3 + 2 = -1$$

$$T_3 = a + 2d = -3 + 2(2) = 1$$

$$-3 - 1 + 1$$

### Calculating the value of a term given the sum of $n$ terms:

If the first term in a series is  $T_1$ , then  $S_1 = T_1$ .

We also know the sum of the first two terms  $S_2 = T_1 + T_2$ , which we rearrange to make  $T_2$  the subject of the equation:

$$T_2 = S_2 - T_1$$

$$\text{Substitute } S_1 = T_1$$

$$\therefore T_2 = S_2 - S_1$$

Similarly, we could determine the third and fourth term in a series:

$$T_3 = S_3 - S_2$$

$$\text{And } T_4 = S_4 - S_3$$

$$T_n = S_n - S_{n-1}, \text{ for } n \in \{2; 3; 4; \dots\} \text{ and } T_1 = S_1$$

### Exercise 1 – 8: Sum of an arithmetic series

1. Determine the value of  $k$ :

$$\sum_{n=1}^k (-2n) = -20$$

2. The sum to  $n$  terms of an arithmetic series is  $S_n = \frac{n}{2} (7n + 15)$ .

- a) How many terms of the series must be added to give a sum of 425?
- b) Determine the sixth term of the series.

3. a) The common difference of an arithmetic series is 3. Calculate the values of  $n$  for which the  $n^{\text{th}}$  term of the series is 93, and the sum of the first  $n$  terms is 975.

- b) Explain why there are two possible answers.

4. The third term of an arithmetic sequence is  $-7$  and the seventh term is 9. Determine the sum of the first 51 terms of the sequence.

5. Calculate the sum of the arithmetic series  $4 + 7 + 10 + \dots + 901$ .

6. Evaluate without using a calculator:  $\frac{4 + 8 + 12 + \dots + 100}{3 + 10 + 17 + \dots + 101}$



7. The second term of an arithmetic sequence is  $-4$  and the sum of the first six terms of the series is 21.
- Find the first term and the common difference.
  - Hence determine  $T_{100}$ .
- [IEB, Nov 2004]

8. Determine the value of the following:

a) 
$$\sum_{w=0}^8 (7w + 8)$$

b) 
$$\sum_{j=1}^8 7j + 8$$

9. Determine the value of  $n$ .

$$\sum_{c=1}^n (2 - 3c) = -330$$

10. The sum of  $n$  terms of an arithmetic series is  $5n^2 - 11n$  for all values of  $n$ . Determine the common difference.
11. The sum of an arithmetic series is 100 times its first term, while the last term is 9 times the first term. Calculate the number of terms in the series if the first term is not equal to zero.
12. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [286P](#)   2a. [286Q](#)   2b. [286R](#)   3. [286S](#)   4. [286T](#)   5. [286V](#)  
 6. [286W](#)   7. [286X](#)   8a. [286Y](#)   8b. [286Z](#)   9. [2872](#)   10. [2873](#)  
 11. [2874](#)



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## 1.5 Finite geometric series

EMCDZ

When we sum a known number of terms in a geometric sequence, we get a finite geometric series. We generate a geometric sequence using the general form:

$$T_n = a \cdot r^{n-1}$$

where

- $n$  is the position of the sequence;
- $T_n$  is the  $n^{\text{th}}$  term of the sequence;
- $a$  is the first term;
- $r$  is the constant ratio.

$$S_n = a + ar + ar^2 + \cdots + ar^{n-2} + ar^{n-1} \dots (1)$$

$$r \times S_n = ar + ar^2 + \cdots + ar^{n-2} + ar^{n-1} + ar^n \dots (2)$$

Subtract eqn. (2) from eqn. (1)

$$\therefore S_n - rS_n = a + 0 + 0 + \cdots - ar^n$$

$$S_n - rS_n = a - ar^n$$

$$S_n(1 - r) = a(1 - r^n)$$

$$\therefore S_n = \frac{a(1 - r^n)}{1 - r} \quad (\text{where } r \neq 1)$$

The general formula for determining the sum of a geometric series is given by:

$$S_n = \frac{a(1 - r^n)}{1 - r} \quad \text{where } r \neq 1$$

This formula is easier to use when  $r < 1$ .

#### Alternative formula:

$$S_n = a + ar + ar^2 + \cdots + ar^{n-2} + ar^{n-1} \dots (1)$$

$$r \times S_n = ar + ar^2 + \cdots + ar^{n-2} + ar^{n-1} + ar^n \dots (2)$$

Subtract eqn. (1) from eqn. (2)

$$\therefore rS_n - S_n = ar^n - a$$

$$S_n(r - 1) = a(r^n - 1)$$

$$\therefore S_n = \frac{a(r^n - 1)}{r - 1} \quad (\text{where } r \neq 1)$$

The general formula for determining the sum of a geometric series is given by:

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{where } r \neq 1$$

This formula is easier to use when  $r > 1$ .

### Worked example 11: Sum of a geometric series

#### QUESTION

Calculate:

$$\sum_{k=1}^6 32 \left(\frac{1}{2}\right)^{k-1}$$

#### SOLUTION

**Step 1: Write down the first three terms of the series**

$$k = 1; \quad T_1 = 32 \left(\frac{1}{2}\right)^0 = 32$$

$$k = 2; \quad T_2 = 32 \left(\frac{1}{2}\right)^{2-1} = 16$$

$$k = 3; \quad T_3 = 32 \left(\frac{1}{2}\right)^{3-1} = 8$$

We have generated the series  $32 + 16 + 8 + \dots$

**Step 2: Determine the values of  $a$  and  $r$**

$$a = T_1 = 32$$

$$r = \frac{T_2}{T_1} = \frac{T_3}{T_2} = \frac{1}{2}$$

**Step 3: Use the general formula to find the sum of the series**

$$\begin{aligned} S_n &= \frac{a(1-r^n)}{1-r} \\ S_6 &= \frac{32(1-(\frac{1}{2})^6)}{1-\frac{1}{2}} \\ &= \frac{32(1-\frac{1}{64})}{\frac{1}{2}} \\ &= 2 \times 32 \left(\frac{63}{64}\right) \\ &= 64 \left(\frac{63}{64}\right) \\ &= 63 \end{aligned}$$

**Step 4: Write the final answer**

$$\sum_{k=1}^6 32 \left(\frac{1}{2}\right)^{k-1} = 63$$

### Worked example 12: Sum of a geometric series

#### QUESTION

Given a geometric series with  $T_1 = -4$  and  $T_4 = 32$ . Determine the values of  $r$  and  $n$  if  $S_n = 84$ .

#### SOLUTION

**Step 1: Determine the values of  $a$  and  $r$**

$$\begin{aligned}a &= T_1 = -4 \\T_4 &= ar^3 = 32 \\\therefore -4r^3 &= 32 \\r^3 &= -8 \\\therefore r &= -2\end{aligned}$$

Therefore the geometric series is  $-4 + 8 - 16 + 32 \dots$ . Notice that the signs of the terms alternate because  $r < 0$ .

We write the general term for this series as  $T_n = -4(-2)^{n-1}$ .

**Step 2: Use the general formula for the sum of a geometric series to determine the value of  $n$**

$$\begin{aligned}S_n &= \frac{a(1 - r^n)}{1 - r} \\\therefore 84 &= \frac{-4(1 - (-2)^n)}{1 - (-2)} \\84 &= \frac{-4(1 - (-2)^n)}{3} \\-\frac{3}{4} \times 84 &= 1 - (-2)^n \\-63 &= 1 - (-2)^n \\(-2)^n &= 64 \\(-2)^n &= (-2)^6 \\\therefore n &= 6\end{aligned}$$

**Step 3: Write the final answer**

$$r = -2 \text{ and } n = 6$$

### Worked example 13: Sum of a geometric series

#### QUESTION

Use the general formula for the sum of a geometric series to determine  $k$  if

$$\sum_{n=1}^8 k \left(\frac{1}{2}\right)^n = \frac{255}{64}$$

#### SOLUTION

**Step 1: Write down the first three terms of the series**

$$n = 1; \quad T_1 = k \left(\frac{1}{2}\right)^1 = \frac{1}{2}k$$

$$n = 2; \quad T_2 = k \left(\frac{1}{2}\right)^2 = \frac{1}{4}k$$

$$n = 3; \quad T_3 = k \left(\frac{1}{2}\right)^3 = \frac{1}{8}k$$

We have generated the series  $\frac{1}{2}k + \frac{1}{4}k + \frac{1}{8}k + \dots$

We can take out the common factor  $k$  and write the series as:  $k \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots\right)$

$$\therefore k \sum_{n=1}^8 \left(\frac{1}{2}\right)^n = \frac{255}{64}$$

**Step 2: Determine the values of  $a$  and  $r$**

$$a = T_1 = \frac{1}{2}$$

$$r = \frac{T_2}{T_1} = \frac{T_3}{T_2} = \frac{1}{2}$$

**Step 3: Calculate the sum of the first eight terms of the geometric series**

$$\begin{aligned} \therefore S_n &= \frac{a(1-r^n)}{1-r} \\ S_8 &= \frac{\frac{1}{2}(1-(\frac{1}{2})^8)}{1-\frac{1}{2}} \\ &= \frac{\frac{1}{2}(1-(\frac{1}{2})^8)}{\frac{1}{2}} \\ &= 1 - \frac{1}{256} \\ &= \frac{255}{256} \end{aligned}$$

$$\therefore \sum_{n=1}^8 \left(\frac{1}{2}\right)^n = \frac{255}{256}$$

So then we can write:

$$\begin{aligned}
 k \sum_{n=1}^8 \left(\frac{1}{2}\right)^n &= \frac{255}{64} \\
 k \left(\frac{255}{256}\right) &= \frac{255}{64} \\
 \therefore k &= \frac{255}{64} \times \frac{256}{255} \\
 &= \frac{256}{64} \\
 &= 4
 \end{aligned}$$

**Step 4: Write the final answer**

$$k = 4$$

### Exercise 1 – 9: Sum of a geometric series

1. Prove that  $a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(r^n - 1)}{r - 1}$  and state any restrictions.

2. Given the geometric sequence 1; -3; 9; ... determine:

- The eighth term of the sequence.
- The sum of the first eight terms of the sequence.

3. Determine:

$$\sum_{n=1}^4 3 \cdot 2^{n-1}$$

4. Find the sum of the first 11 terms of the geometric series  $6 + 3 + \frac{3}{2} + \frac{3}{4} + \dots$

5. Show that the sum of the first  $n$  terms of the geometric series  $54 + 18 + 6 + \dots + 5\left(\frac{1}{3}\right)^{n-1}$  is given by  $(81 - 3^{4-n})$ .

6. The eighth term of a geometric sequence is 640. The third term is 20. Find the sum of the first 7 terms.

7. Given:

$$\sum_{t=1}^n 8\left(\frac{1}{2}\right)^t$$

- Find the first three terms in the series.
- Calculate the number of terms in the series if  $S_n = 7\frac{63}{64}$ .

8. The ratio between the sum of the first three terms of a geometric series and the sum of the 4<sup>th</sup>, 5<sup>th</sup> and 6<sup>th</sup> terms of the same series is 8 : 27. Determine the constant ratio and the first 2 terms if the third term is 8.

9. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [2876](#)   2a. [2877](#)   2b. [2878](#)   3. [2879](#)   4. [287B](#)   5. [287C](#)  
 6. [287D](#)   7a. [287F](#)   7b. [287G](#)   8. [287H](#)



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So far we have been working only with finite sums, meaning that whenever we determined the sum of a series, we only considered the sum of the first  $n$  terms. We now consider what happens when we add an infinite number of terms together. Surely if we sum infinitely many numbers, no matter how small they are, the answer goes to infinity? In some cases the answer does indeed go to infinity (like when we sum all the positive integers), but surprisingly there are some cases where the answer is a finite real number.

### Investigation: Sum of an infinite series

1. Cut a piece of string 1 m in length.
2. Now cut the piece of string in half and place one half on the desk.
3. Cut the other half in half again and put one of the pieces on the desk.
4. Repeat this process until the piece of string is too short to cut easily.
5. Draw a diagram to illustrate the sequence of lengths of the pieces of string.
6. Can this sequence be expressed mathematically? Hint: express the shorter lengths of string as a fraction of the original length of string.
7. What is the sum of the lengths of all the pieces of string?
8. Predict what would happen if these steps could be repeated infinitely many times.
9. Will the sum of the lengths of string ever be greater than 1?
10. What can you conclude?

### Worked example 14: Sum to infinity

#### QUESTION

Complete the table below for the geometric series  $T_n = \left(\frac{1}{2}\right)^n$  and answer the questions that follow:

|                         | Terms         | $S_n$         | $1 - S_n$     |
|-------------------------|---------------|---------------|---------------|
| $T_1$                   | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ |
| $T_1 + T_2$             |               |               |               |
| $T_1 + T_2 + T_3$       |               |               |               |
| $T_1 + T_2 + T_3 + T_4$ |               |               |               |

1. As more and more terms are added, what happens to the value of  $S_n$ ?
2. As more and more terms are added, what happens to the value of  $1 - S_n$ ?
3. Predict the maximum value of  $S_n$  for the sum of infinitely many terms in the series.

## SOLUTION

### Step 1: Complete the table

|                         | Terms                                                    | $S_n$           | $1 - S_n$      |
|-------------------------|----------------------------------------------------------|-----------------|----------------|
| $T_1$                   | $\frac{1}{2}$                                            | $\frac{1}{2}$   | $\frac{1}{2}$  |
| $T_1 + T_2$             | $\frac{1}{2} + \frac{1}{4}$                              | $\frac{3}{4}$   | $\frac{1}{4}$  |
| $T_1 + T_2 + T_3$       | $\frac{1}{2} + \frac{1}{4} + \frac{1}{8}$                | $\frac{7}{8}$   | $\frac{1}{8}$  |
| $T_1 + T_2 + T_3 + T_4$ | $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$ | $\frac{15}{16}$ | $\frac{1}{16}$ |

### Step 2: Consider the value of $S_n$ and $1 - S_n$

As more terms in the series are added together, the value of  $S_n$  increases:

$$\frac{1}{2} < \frac{3}{4} < \frac{7}{8} < \dots$$

However, by considering  $1 - S_n$ , we notice that the amount by which  $S_n$  increases gets smaller and smaller as more terms are added:

$$\frac{1}{2} > \frac{1}{4} > \frac{1}{8} > \dots$$

We can therefore conclude that the value of  $S_n$  is approaching a maximum value of 1; it is converging to 1.

### Step 3: Write conclusion mathematically

We can conclude that the sum of the series

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$$

gets closer to 1 ( $S_n \rightarrow 1$ ) as the number of terms approaches infinity ( $n \rightarrow \infty$ ), therefore the series converges.

$$\sum_{i=1}^{\infty} \left(\frac{1}{2}\right)^i = 1$$

We express the sum of an infinite number of terms of a series as

$$S_{\infty} = \sum_{i=1}^{\infty} T_i$$

### Convergence and divergence

If the sum of a series gets closer and closer to a certain value as we increase the number of terms in the sum, we say that the series converges. In other words, there is a limit to the sum of a converging series. If a series does not converge, we say that it diverges. The sum of an infinite series usually tends to infinity, but there are some special cases where it does not.



## Exercise 1 – 10: Convergent and divergent series

For each of the general terms below:

- Determine if it forms an arithmetic or geometric series.
- Calculate  $S_1, S_2, S_{10}$  and  $S_{100}$ .
- Determine if the series is convergent or divergent.

1.  $T_n = 2n$

2.  $T_n = (-n)$

3.  $T_n = \left(\frac{2}{3}\right)^n$

4.  $T_n = 2^n$

5. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 287J   2. 287K   3. 287M   4. 287N



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Note the following:

- An arithmetic series never converges: as  $n$  tends to infinity, the series will always tend to positive or negative infinity.
- Some geometric series converge (have a limit) and some diverge (as  $n$  tends to infinity, the series does not tend to any limit or it tends to infinity).

## Infinite geometric series

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There is a simple test for determining whether a geometric series converges or diverges; if  $-1 < r < 1$ , then the infinite series will converge. If  $r$  lies outside this interval, then the infinite series will diverge.

### Test for convergence:

- If  $-1 < r < 1$ , then the infinite geometric series converges.
- If  $r < -1$  or  $r > 1$ , then the infinite geometric series diverges.

We derive the formula for calculating the value to which a geometric series converges as follows:

$$S_n = \sum_{i=1}^n ar^{i-1} = \frac{a(1-r^n)}{1-r}$$

Now consider the behaviour of  $r^n$  for  $-1 < r < 1$  as  $n$  becomes larger.

Let  $r = \frac{1}{2}$ :

$$n = 1 : r^n = r^1 = \left(\frac{1}{2}\right)^1 = \frac{1}{2}$$

$$n = 2 : r^n = r^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} < \frac{1}{2}$$

$$n = 3 : r^n = r^3 = \left(\frac{1}{2}\right)^3 = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} < \frac{1}{4}$$

Since  $r$  is in the range  $-1 < r < 1$ , we see that  $r^n$  gets closer to 0 as  $n$  gets larger. Therefore  $(1 - r^n)$  gets closer to 1.

Therefore,

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

If  $-1 < r < 1$ , then  $r^n \rightarrow 0$  as  $n \rightarrow \infty$

$$\begin{aligned}\therefore S_\infty &= \frac{a(1 - 0)}{1 - r} \\ &= \frac{a}{1 - r}\end{aligned}$$

The sum of an infinite geometric series is given by the formula

$$\therefore S_\infty = \sum_{i=1}^{\infty} ar^{i-1} = \frac{a}{1 - r} \quad (-1 < r < 1)$$

where

- $a$  is the first term of the series;
- $r$  is the constant ratio.

Alternative notation:

$$\underbrace{S_n}_{n \rightarrow \infty} \rightarrow \frac{a}{1 - r} \quad \text{if } -1 < r < 1$$

In words: as the number of terms ( $n$ ) tends to infinity, the sum of a converging geometric series ( $S_n$ ) tends to the value  $\frac{a}{1-r}$ .

► See video: [287P](#) at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

### Worked example 15: Sum to infinity of a geometric series

#### QUESTION

Given the series  $18 + 6 + 2 + \dots$ . Find the sum to infinity if it exists.

#### SOLUTION

##### Step 1: Determine the value of $r$

We need to know the value of  $r$  to determine whether the series converges or diverges.

$$\begin{aligned}\frac{T_2}{T_1} &= \frac{6}{18} \\ &= \frac{1}{3} \\ \frac{T_3}{T_2} &= \frac{2}{6} \\ &= \frac{1}{3} \\ \therefore r &= \frac{1}{3}\end{aligned}$$

Since  $-1 < r < 1$ , we can conclude that this is a convergent geometric series.

##### Step 2: Determine the sum to infinity

Write down the formula for the sum to infinity and substitute the known values:

$$\begin{aligned}a &= 18; \quad r = \frac{1}{3} \\ S_{\infty} &= \frac{a}{1-r} \\ &= \frac{18}{1-\frac{1}{3}} \\ &= \frac{18}{\frac{2}{3}} \\ &= 18 \times \frac{3}{2} \\ &= 27\end{aligned}$$

As  $n$  tends to infinity, the sum of this series tends to 27; no matter how many terms are added together, the value of the sum will never be greater than 27.

### Worked example 16: Using the sum to infinity to convert recurring decimals to fractions

#### QUESTION

Use two different methods to convert the recurring decimal  $0,\dot{5}$  to a proper fraction.

#### SOLUTION

##### Step 1: Convert the recurring decimal to a fraction using equations

$$\begin{aligned}\text{Let } x &= 0,\dot{5} \\ \therefore x &= 0,555\dots\dots (1) \\ 10x &= 5,55\dots\dots (2) \\ (2) - (1) : \quad 9x &= 5 \\ \therefore x &= \frac{5}{9}\end{aligned}$$

## Step 2: Convert the recurring decimal to a fraction using the sum to infinity

$$0,\dot{5} = 0,5 + 0,05 + 0,005 + \dots$$
$$\text{or } 0,\dot{5} = \frac{5}{10} + \frac{5}{100} + \frac{5}{1000} + \dots$$

This is a geometric series with  $r = 0,1 = \frac{1}{10}$ . And since  $-1 < r < 1$ , we can conclude that the series is convergent.

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} \\ &= \frac{\frac{5}{10}}{1 - \frac{1}{10}} \\ &= \frac{\frac{5}{10}}{\frac{9}{10}} \\ &= \frac{5}{9} \end{aligned}$$

## Worked example 17: Sum to infinity

### QUESTION

Determine the possible values of  $a$  and  $r$  if

$$\sum_{n=1}^{\infty} ar^{n-1} = 5$$

### SOLUTION

**Step 1: Write down the sum to infinity formula and substitute known values**

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} \\ \therefore 5 &= \frac{a}{1-r} \\ a &= 5(1-r) \\ \therefore a &= 5 - 5r \\ \text{And } 5r &= 5 - a \\ \therefore r &= \frac{5-a}{5} \end{aligned}$$

**Step 2: Apply the condition for convergence to determine possible values of  $a$**

For a series to converge:  $-1 < r < 1$

$$\begin{aligned} -1 &< r < 1 \\ -1 &< \frac{5-a}{5} < 1 \\ -5 &< 5-a < 5 \\ -10 &< -a < 0 \\ 0 &< a < 10 \end{aligned}$$

**Step 3: Write the final answer**

For the series to converge,  $0 < a < 10$  and  $-1 < r < 1$ .

### Exercise 1 – 11:

1. What value does  $\left(\frac{2}{5}\right)^n$  approach as  $n$  tends towards  $\infty$ ?
2. Find the sum to infinity of the geometric series  $3 + 1 + \frac{1}{3} + \frac{1}{9} + \dots$
3. Determine for which values of  $x$ , the geometric series  $2 + \frac{2}{3}(x+1) + \frac{2}{9}(x+1)^2 + \dots$  will converge.
4. The sum to infinity of a geometric series with positive terms is  $4\frac{1}{6}$  and the sum of the first two terms is  $2\frac{2}{3}$ . Find  $a$ , the first term, and  $r$ , the constant ratio between consecutive terms.
5. Use the sum to infinity to show that  $0,9 = 1$ .
6. A shrub 110 cm high is planted in a garden. At the end of the first year, the shrub is 120 cm tall. Thereafter the growth of the shrub each year is half of its growth in the previous year. Show that the height of the shrub will never exceed 130 cm. Draw a graph of the relationship between time and growth.  
[IEB, Nov 2003]
7. Find  $p$ :

$$\sum_{k=1}^{\infty} 27p^k = \sum_{t=1}^{12} (24 - 3t)$$

8. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [287Q](#)   2. [287R](#)   3. [287S](#)   4. [287T](#)   5. [287V](#)   6. [287W](#)  
7. [287X](#)



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**Arithmetic sequence**

- common difference ( $d$ ) between any two consecutive terms:  $d = T_n - T_{n-1}$
- general form:  $a + (a + d) + (a + 2d) + \dots$
- general formula:  $T_n = a + (n - 1)d$
- graph of the sequence lies on a straight line

**Quadratic sequence**

- common second difference between any two consecutive terms
- general formula:  $T_n = an^2 + bn + c$
- graph of the sequence lies on a parabola

**Geometric sequence**

- constant ratio ( $r$ ) between any two consecutive terms:  $r = \frac{T_n}{T_{n-1}}$
- general form:  $a + ar + ar^2 + \dots$
- general formula:  $T_n = ar^{n-1}$
- graph of the sequence lies on an exponential curve

**Sigma notation**

$$\sum_{k=1}^n T_k$$

Sigma notation is used to indicate the sum of the terms given by  $T_k$ , starting from  $k = 1$  and ending at  $k = n$ .

**Series**

- the sum of certain numbers of terms in a sequence
- arithmetic series:
  - $S_n = \frac{n}{2}[a + l]$
  - $S_n = \frac{n}{2}[2a + (n - 1)d]$
- geometric series:
  - $S_n = \frac{a(1-r^n)}{1-r}$  if  $r < 1$
  - $S_n = \frac{a(r^n-1)}{r-1}$  if  $r > 1$

**Sum to infinity**

A convergent geometric series, with  $-1 < r < 1$ , tends to a certain fixed number as the number of terms in the sum tends to infinity.

$$S_{\infty} = \sum_{n=1}^{\infty} T_n = \frac{a}{1-r}$$

## Exercise 1 – 12: End of chapter exercises

1. Is  $1 + 2 + 3 + 4 + \dots$  an example of a *finite series* or an *infinite series*?
2. A new soccer competition requires each of 8 teams to play every other team once.
  - a) Calculate the total number of matches to be played in the competition.
  - b) If each of  $n$  teams played each other once, determine a formula for the total number of matches in terms of  $n$ .
3. Calculate:

$$\sum_{k=2}^6 3 \left( \frac{1}{3} \right)^{k+2}$$

4. The first three terms of a convergent geometric series are:  $x + 1$ ;  $x - 1$ ;  $2x - 5$ .
  - a) Calculate the value of  $x$ , ( $x \neq 1$  or  $1$ ).
  - b) Sum to infinity of the series.
5. Write the sum of the first twenty terms of the following series in  $\sum$  notation.

$$6 + 3 + \frac{3}{2} + \frac{3}{4} + \dots$$

6. Determine:

$$\sum_{k=1}^{\infty} 12 \left( \frac{1}{5} \right)^{k-1}$$

7. A man was injured in an accident at work. He receives a disability grant of R 4800 in the first year. This grant increases with a fixed amount each year.
  - a) What is the annual increase if he received a total of R 143 500 over 20 years?
  - b) His initial annual expenditure is R 2600, which increases at a rate of R 400 per year. After how many years will his expenses exceed his income?
8. The length of the side of a square is 4 units. This square is divided into 4 equal, smaller squares. One of the smaller squares is then divided into four equal, even smaller squares. One of the even smaller squares is divided into four, equal squares. This process is repeated indefinitely. Calculate the sum of the areas of all the squares.
9. Thembi worked part-time to buy a Mathematics book which costs R 29,50. On 1 February she saved R 1,60, and every day saves 30 cents more than she saved the previous day. So, on the second day, she saved R 1,90, and so on. After how many days did she have enough money to buy the book?
10. A plant reaches a height of 118 mm after one year under ideal conditions in a greenhouse. During the next year, the height increases by 12 mm. In each successive year, the height increases by  $\frac{5}{8}$  of the previous year's growth. Show that the plant will never reach a height of more than 150 mm.
11. Calculate the value of  $n$  if:

$$\sum_{a=1}^n (20 - 4a) = -20$$

12. Michael saved R 400 during the first month of his working life. In each subsequent month, he saved 10% more than what he had saved in the previous month.

- a) How much did he save in the seventh working month?
- b) How much did he save all together in his first 12 working months?

13. The Cape Town High School wants to build a school hall and is busy with fundraising. Mr. Manuel, an ex-learner of the school and a successful politician, offers to donate money to the school. Having enjoyed mathematics at school, he decides to donate an amount of money on the following basis. He sets a mathematical quiz with 20 questions. For the correct answer to the first question (any learner may answer), the school will receive R 1, for a correct answer to the second question, the school will receive R 2, and so on. The donations 1; 2; 4; ... form a geometric sequence. Calculate, to the nearest Rand:

- a) The amount of money that the school will receive for the correct answer to the 20<sup>th</sup> question.
- b) The total amount of money that the school will receive if all 20 questions are answered correctly.

14. The first term of a geometric sequence is 9, and the ratio of the sum of the first eight terms to the sum of the first four terms is 97 : 81. Find the first three terms of the sequence, if it is given that all the terms are positive.

15. Given the geometric sequence:  $6 + p$ ;  $10 + p$ ;  $15 + p$

- a) Determine  $p$ , ( $p \neq -6$  or  $-10$ ).
- b) Show that the constant ratio is  $\frac{5}{4}$ .
- c) Determine the tenth term of this sequence correct to one decimal place.

16. The second and fourth terms of a convergent geometric series are 36 and 16, respectively. Find the sum to infinity of this series, if all its terms are positive.

17. Evaluate:

$$\sum_{k=2}^5 \frac{k(k+1)}{2}$$

18.  $S_n = 4n^2 + 1$  represents the sum of the first  $n$  terms of a particular series. Find the second term.

19. Determine whether the following series converges for the given values of  $x$ . If it does converge, calculate the sum to infinity.

$$\sum_{p=1}^{\infty} (x+2)^p$$

- a)  $x = -\frac{5}{2}$
- b)  $x = -5$

20. Calculate:

$$\sum_{i=1}^{\infty} 5(4^{-i})$$

21. The sum of the first  $p$  terms of a sequence is  $p(p+1)$ . Find the tenth term.

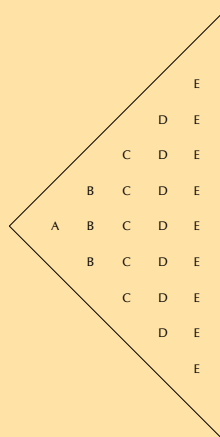
22. The powers of 2 are removed from the following set of positive integers

1; 2; 3; 4; 5; 6; ...; 1998; 1999; 2000

Find the sum of remaining integers.



23. Observe the pattern below:



- If the pattern continues, find the number of letters in the column containing M's.
- If the total number of letters in the pattern is 361, which letter will be in the last column.

24. Write  $0,5\dot{7}$  as a proper fraction.

25. Given:

$$f(x) = \sum_{p=1}^{\infty} \frac{(1+x)^p}{1-x}$$

- For which values of  $x$  will  $f(x)$  converge?
  - Determine the value of  $f\left(-\frac{1}{2}\right)$ .
26. From the definition of a geometric sequence, deduce a formula for calculating the sum of  $n$  terms of the series

$$a^2 + a^4 + a^6 + \dots$$

27. Calculate the tenth term of the series if  $S_n = 2n + 3n^2$ .

28. A theatre is filling up at a rate of 4 people in the first minute, 6 people in the second minute, and 8 people in the third minute and so on. After 6 minutes the theatre is half full. After how many minutes will the theatre be full?

[IEB, Nov 2001]

29. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- |           |           |           |           |           |           |
|-----------|-----------|-----------|-----------|-----------|-----------|
| 1. 287Y   | 2a. 287Z  | 2b. 288Z  | 3. 2883   | 4a. 2884  | 4b. 2885  |
| 5. 2886   | 6. 2887   | 7a. 2888  | 7b. 2889  | 8. 288B   | 9. 288C   |
| 10. 288D  | 11. 288F  | 12a. 288G | 12b. 288H | 13a. 288J | 13b. 288K |
| 14. 288M  | 15a. 288N | 15b. 288P | 15c. 288Q | 16. 288R  | 17. 288S  |
| 18. 288T  | 19a. 288V | 19b. 288W | 20. 288X  | 21. 288Y  | 22. 288Z  |
| 23a. 2892 | 23b. 2893 | 24. 2894  | 25. 2895  | 26. 2896  | 27. 2897  |
| 28. 2898  |           |           |           |           |           |



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## *Functions*

|     |                                       |    |
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## 2.1 Revision

EMCF6

In previous grades we learned about the characteristics of linear, quadratic, hyperbolic and exponential functions. In this chapter we will demonstrate the ability to work with various types of functions and relations including inverses. In particular, we will look at the graphs of the inverses of:

Linear functions:  $y = mx + c$  or  $y = ax + q$

Quadratic functions:  $y = ax^2$

Exponential functions:  $y = b^x$  ( $b > 0, b \neq 1$ )

**Worked example 1: Linear function****QUESTION**

Draw a graph of  $2y + x - 8 = 0$  and determine the significant characteristics of this linear function.

**SOLUTION**

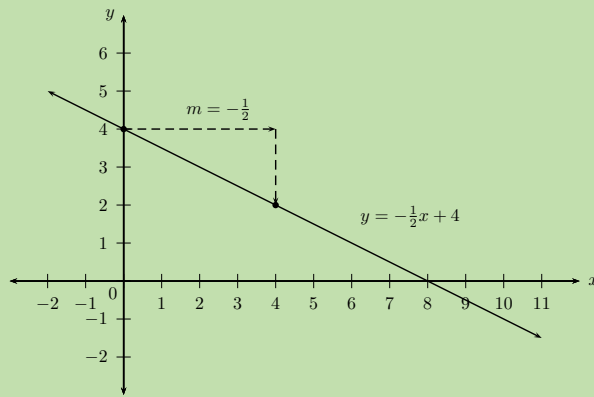
**Step 1: Write the equation in standard form**  $y = mx + c$

$$\begin{aligned} 2y + x - 8 &= 0 \\ 2y &= -x + 8 \\ \therefore y &= -\frac{1}{2}x + 4 \\ \therefore m &= -\frac{1}{2} \\ \text{And } c &= 4 \end{aligned}$$

**Step 2: Draw the straight line graph**

To draw the straight line graph we can use the gradient-intercept method:

$$\begin{aligned} y - \text{intercept} &: (0; 4) \\ m &= -\frac{1}{2} \end{aligned}$$



Alternative method: we can also determine and plot the  $x$ - and  $y$ -intercepts as follows:

For the  $y$ -intercept, let  $x = 0$ :

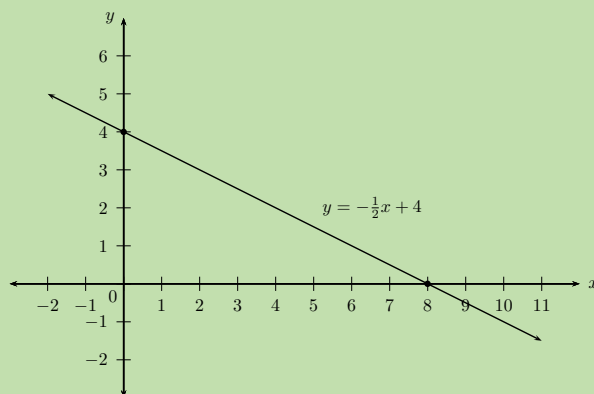
$$\begin{aligned} y &= -\frac{1}{2}(0) + 4 \\ \therefore y &= 0 + 4 \\ &= 4 \end{aligned}$$

This gives the point  $(0; 4)$ .

For the  $x$ -intercept, let  $y = 0$ :

$$\begin{aligned} 0 &= -\frac{1}{2}x + 4 \\ \frac{1}{2}x &= 4 \\ \therefore x &= 8 \end{aligned}$$

This gives the point  $(8; 0)$ .



### Step 3: Determine the characteristics

Gradient:  $-\frac{1}{2}$

Intercepts:  $(0; 4)$  and  $(8; 0)$

Domain:  $\{x : x \in \mathbb{R}\}$

Range:  $\{y : y \in \mathbb{R}\}$

Decreasing function: as  $x$  increases,  $y$  decreases.

## Worked example 2: Quadratic function

### QUESTION

Write the quadratic function  $2y - x^2 + 4 = 0$  in standard form. Draw a graph of the function and state the significant characteristics.

### SOLUTION

**Step 1: Write the equation in standard form**  $y = ax^2 + bx + c$

$$\begin{aligned}2y - x^2 + 4 &= 0 \\2y &= x^2 - 4 \\y &= \frac{1}{2}x^2 - 2\end{aligned}$$

Therefore, we see that:

$$a = \frac{1}{2}; \quad b = 0; \quad c = -2$$

**Step 2: Draw a graph of the parabola**

For the  $y$ -intercept, let  $x = 0$ :

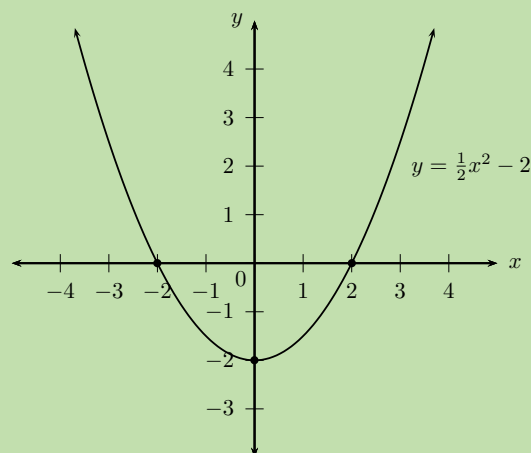
$$\begin{aligned}y &= \frac{1}{2}(0)^2 - 2 \\ \therefore y &= 0 - 2 \\ &= -2\end{aligned}$$

This gives the point  $(0; -2)$ .

For the  $x$ -intercept, let  $y = 0$ :

$$\begin{aligned}0 &= \frac{1}{2}x^2 - 2 \\ 0 &= x^2 - 4 \\ 0 &= (x - 2)(x + 2) \\ \therefore x &= -2 \text{ or } x = 2\end{aligned}$$

This gives the points  $(-2; 0)$  and  $(2; 0)$ .



**Step 3: State the significant characteristics**

Shape:  $a > 0$ , therefore the graph is a “smile”.

Intercepts:  $(-2; 0)$ ,  $(2; 0)$  and  $(0; -2)$

Turning point:  $(0; -2)$

Axes of symmetry:  $x = -\frac{b}{2a} = -\frac{0}{2(\frac{1}{2})} = 0$

Domain:  $\{x : x \in \mathbb{R}\}$

Range:  $\{y : y \geq -2, y \in \mathbb{R}\}$

The function is decreasing for  $x < 0$  and increasing for  $x > 0$ .

**Worked example 3: Exponential function****QUESTION**

Draw the graphs of  $f(x) = 2^x$  and  $g(x) = \left(\frac{1}{2}\right)^x$  on the same set of axes and compare the two functions.

**SOLUTION****Step 1: Examine the functions and determine the information needed to draw the graphs**

Consider the function:  $f(x) = 2^x$

$$\text{If } y = 0 : 2^x = 0$$

$$\text{But } 2^x \neq 0$$

$\therefore$  no solution

$$\text{If } x = 0 : 2^0 = 1$$

This gives the point  $(0; 1)$ .

Asymptotes:  $f(x) = 2^x$  has a horizontal asymptote, the line  $y = 0$ , which is the  $x$ -axis.

|        |               |               |   |   |   |
|--------|---------------|---------------|---|---|---|
| $x$    | -2            | -1            | 0 | 1 | 2 |
| $f(x)$ | $\frac{1}{4}$ | $\frac{1}{2}$ | 1 | 2 | 4 |

Consider the function:  $g(x) = \left(\frac{1}{2}\right)^x$

$$\text{If } y = 0 : \left(\frac{1}{2}\right)^x = 0$$

$$\text{But } \left(\frac{1}{2}\right)^x \neq 0$$

$\therefore$  no solution

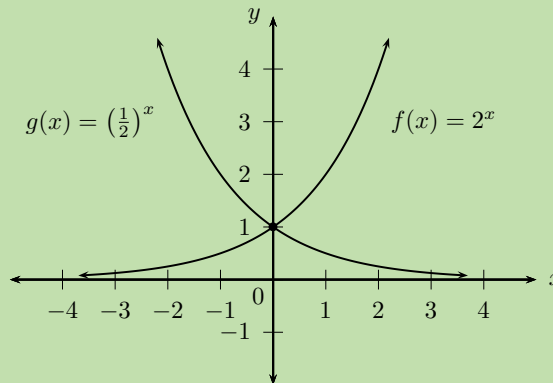
$$\text{If } x = 0 : \left(\frac{1}{2}\right)^0 = 1$$

This gives the point  $(0; 1)$ .

Asymptotes:  $g(x) = \left(\frac{1}{2}\right)^x$  also has a horizontal asymptote at  $y = 0$ .

|        |    |    |   |               |               |
|--------|----|----|---|---------------|---------------|
| $x$    | -2 | -1 | 0 | 1             | 2             |
| $g(x)$ | 4  | 2  | 1 | $\frac{1}{2}$ | $\frac{1}{4}$ |

### Step 2: Draw the graphs of the exponential functions



### Step 3: State the significant characteristics

Symmetry:  $f$  and  $g$  are symmetrical about the  $y$ -axis.

Domain of  $f$  and  $g$ :  $\{x : x \in \mathbb{R}\}$

Range of  $f$  and  $g$ :  $\{y : y > 0, y \in \mathbb{R}\}$

The function  $g$  decreases as  $x$  increases and function  $f$  increases as  $x$  increases. The two graphs intersect at the point  $(0; 1)$ .

### Exercise 2 – 1: Revision

- Sketch the graphs on the same set of axes and determine the following for each function:
  - Intercepts
  - Turning point
  - Axes of symmetry
  - Domain and range
  - Maximum and minimum values
  - $f(x) = 3x^2$  and  $g(x) = -x^2$
  - $j(x) = -\frac{1}{5}x^2$  and  $k(x) = -5x^2$
  - $h(x) = 2x^2 + 4$  and  $l(x) = -2x^2 - 4$
- Given  $f(x) = -3x - 6$  and  $g(x) = mx + c$ . Determine the values of  $m$  and  $c$  if  $g \parallel f$  and  $g$  passes through the point  $(1; 2)$ . Sketch both functions on the same system of axes.
- Given  $m : \frac{x}{2} - \frac{y}{3} = 1$  and  $n : -\frac{y}{3} = 1$ . Determine the  $x$ - and  $y$ -intercepts and sketch both graphs on the same system of axes.
- Given  $p(x) = 3^x$ ,  $q(x) = 3^{-x}$  and  $r(x) = -3^x$ .
  - Sketch  $p$ ,  $q$  and  $r$  on the same system of axes.
  - For each of the functions, determine the intercepts, asymptotes, domain and range.

5. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 2899 1b. 289B 1c. 289C 2. 289D 3. 289F 4a. 289G  
4b. 289H



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## 2.2 Functions and relations

EMCF7

### DEFINITION: Relation

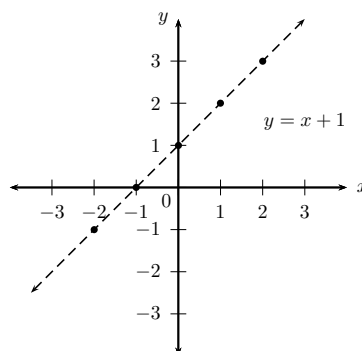
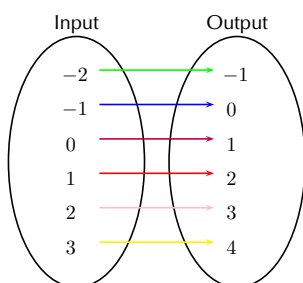
A rule which associates each element of set ( $A$ ) with at least one element in set ( $B$ ).

### DEFINITION: Function

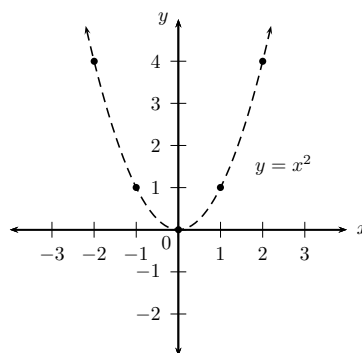
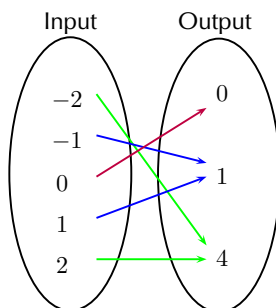
A rule which uniquely associates elements of one set ( $A$ ) with the elements of another set ( $B$ ); each element in set ( $A$ ) maps to only one element in set ( $B$ ).

Functions can be one-to-one relations or many-to-one relations. A many-to-one relation associates two or more values of the independent (input) variable with a single value of the dependent (output) variable. The domain is the set of values to which the rule is applied ( $A$ ) and the range is the set of values (also called the images or function values) determined by the rule.

**Example of a one-to-one function:**  $y = x + 1$

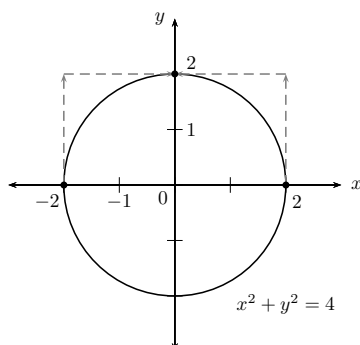


**Example of a many-to-one function:**  $y = x^2$





However, some very common mathematical constructions are not functions. For example, consider the relation  $x^2 + y^2 = 4$ . This relation describes a circle of radius 2 centred at the origin. If we let  $x = 0$ , we see that  $y^2 = 4$  and thus either  $y = 2$  or  $y = -2$ . This is a one-to-many relation because a single  $x$ -value relates to two different  $y$ -values. Therefore  $x^2 + y^2 = 4$  is not a function.



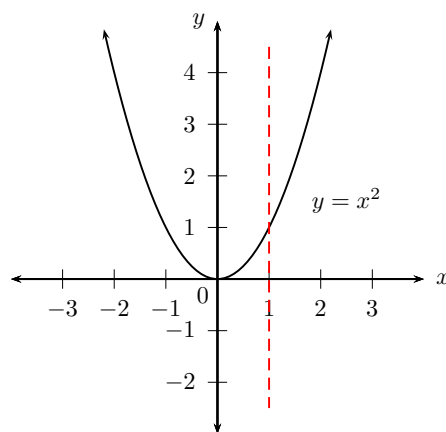
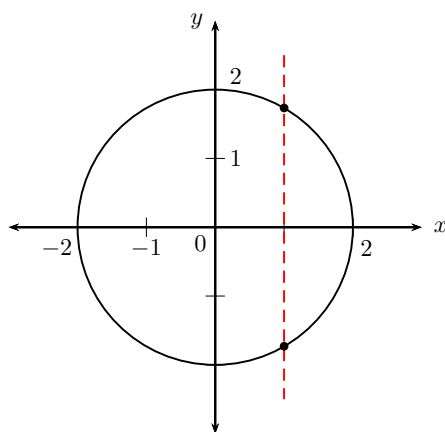
### Vertical line test

Given the graph of a relation, there is a simple test for whether or not the relation is a function. This test is called the vertical line test. If it is possible to draw any vertical line (a line of constant  $x$ ) which crosses the graph of the relation more than once, then the relation is not a function. If more than one intersection point exists, then the intersections correspond to multiple values of  $y$  for a single value of  $x$  (one-to-many).

If any vertical line cuts the graph only once, then the relation is a function (one-to-one or many-to-one).

The red vertical line cuts the circle twice and therefore the circle is not a function.

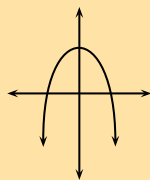
The red vertical line only cuts the parabola once and therefore the parabola is a function.



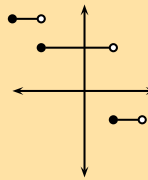
## Exercise 2 – 2: Identifying functions

1. Consider the graphs given below and determine whether or not they are functions:

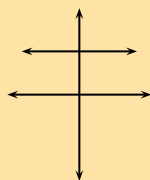
a)



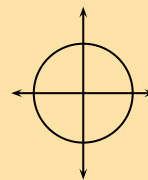
e)



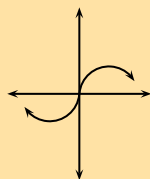
b)



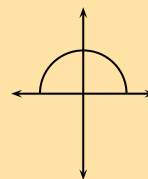
f)



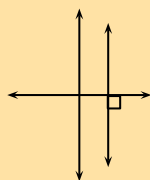
c)



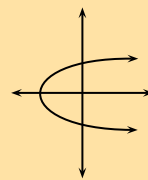
g)



d)



h)



2. Sketch the following and determine whether or not they are functions:

a)  $x + y = 4$

d)  $x^2 + y^2 = 9$

b)  $y = \frac{x}{4}$

e)  $y = \tan x$

c)  $y = 2^x$

3. The table below gives the average per capita income,  $d$ , in a region of the country as a function of  $u$ , the percentage of unemployed people. Write down an equation to show that the average income is a function of the percentage of unemployed people.

| $u$ | 1      | 2      | 3      | 4      |
|-----|--------|--------|--------|--------|
| $d$ | 22 500 | 22 000 | 21 500 | 21 000 |

Check answers online with the exercise code below or click on 'show me the answer'.

- 1a. 289J   1b. 289K   1c. 289M   1d. 289N   1e. 289P   1f. 289Q  
 1g. 289R   1h. 289S   2a. 289T   2b. 289V   2c. 289W   2d. 289X  
 2e. 289Y   3. 289Z



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## Function notation

For the function  $y = f(x)$ ,  $y$  is the dependent variable, because the value of  $y$  (output) depends on the value of  $x$  (input). We say  $x$  is the independent variable, since we can choose  $x$  to be any number. Similarly, if  $g(t) = 2t + 1$ , then  $t$  is the independent variable and  $g$  is the function name.

- If  $h(x) = 3x - 5$  and we need to determine when  $h(x) = 3$ , then we solve for the value of  $x$  such that:

$$\begin{aligned}h(x) &= 3x - 5 \\3 &= 3x - 5 \\8 &= 3x \\\therefore x &= \frac{8}{3}\end{aligned}$$

- If  $h(x) = 3x - 5$  and we need to determine  $h(3)$ , then we calculate the value for  $h(x)$  when  $x = 3$ :

$$\begin{aligned}h(x) &= 3x - 5 \\h(3) &= 3(3) - 5 \\&= 4\end{aligned}$$

## 2.3 Inverse functions

EMCF8

An inverse function is a function which does the “reverse” of a given function. More formally, if  $f$  is a function with domain  $X$ , then  $f^{-1}$  is its inverse function if and only if  $f^{-1}(f(x)) = x$  for every  $x \in X$ .

$y = f(x)$  : indicates a function

$y_1 = f(x_1)$  : indicates we must substitute a specific  $x_1$  value  
into the function to get the corresponding  $y_1$  value

$f^{-1}(y) = x$  : indicates the inverse function

$f^{-1}(y_1) = x_1$  : indicates we must substitute a specific  $y_1$  value  
into the inverse to return the specific  $x_1$  value

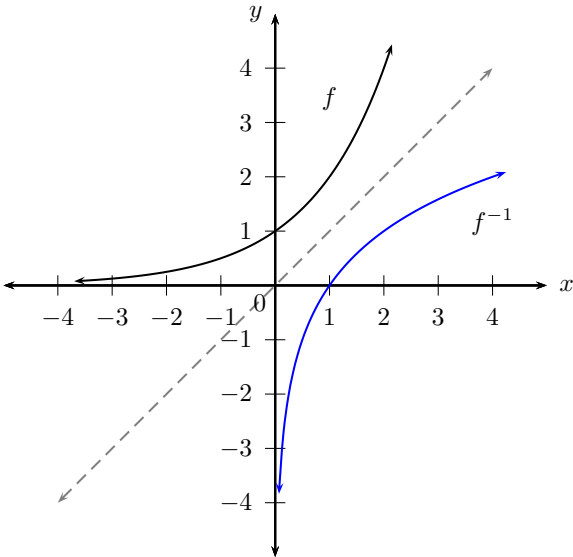
A function must be a one-to-one relation if its inverse is to be a function. If a function  $f$  has an inverse function  $f^{-1}$ , then  $f$  is said to be invertible.

Given the function  $f(x)$ , we determine the inverse  $f^{-1}(x)$  by:

- interchanging  $x$  and  $y$  in the equation;
- making  $y$  the subject of the equation;
- expressing the new equation in function notation.

**Note:** if the inverse is not a function then it cannot be written in function notation. For example, the inverse of  $f(x) = 3x^2$  cannot be written as  $f^{-1}(x) = \pm\sqrt{\frac{1}{3}x}$  as it is not a function. We write the inverse as  $y = \pm\sqrt{\frac{1}{3}x}$  and conclude that  $f$  is not invertible.

If we represent the function  $f$  and the inverse function  $f^{-1}$  graphically, the two graphs are reflected about the line  $y = x$ . Any point on the line  $y = x$  has  $x$ - and  $y$ -coordinates with the same numerical value, for example  $(-3; -3)$  and  $(\frac{4}{5}; \frac{4}{5})$ . Therefore interchanging the  $x$ - and  $y$ -values makes no difference.



This diagram shows an exponential function (black graph) and its inverse (blue graph) reflected about the line  $y = x$  (grey line).

**Important:** for  $f^{-1}$ , the superscript  $-1$  is not an exponent. It is the notation for indicating the inverse of a function. Do not confuse this with exponents, such as  $(\frac{1}{2})^{-1}$  or  $3 + x^{-1}$ .

**Be careful not to confuse the inverse of a function and the reciprocal of a function:**

| Inverse                                          | Reciprocal                                           |
|--------------------------------------------------|------------------------------------------------------|
| $f^{-1}(x)$                                      | $[f(x)]^{-1} = \frac{1}{f(x)}$                       |
| $f(x)$ and $f^{-1}(x)$ symmetrical about $y = x$ | $f(x) \times \frac{1}{f(x)} = 1$                     |
| Example:                                         | Example:                                             |
| $g(x) = 5x \therefore g^{-1}(x) = \frac{x}{5}$   | $g(x) = 5x \therefore \frac{1}{g(x)} = \frac{1}{5x}$ |

► See video: 28B2 at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

**Worked example 4: Inverse of the function  $y = ax + q$** **QUESTION**

Determine the inverse function of  $p(x) = -3x + 1$  and sketch the graphs of  $p(x)$  and  $p^{-1}(x)$  on the same system of axes.

**SOLUTION****Step 1: Determine the inverse of the given function**

- Interchange  $x$  and  $y$  in the equation.
- Make  $y$  the subject of the new equation.
- Express the new equation in function notation.

$$\text{Let } y = -3x + 1$$

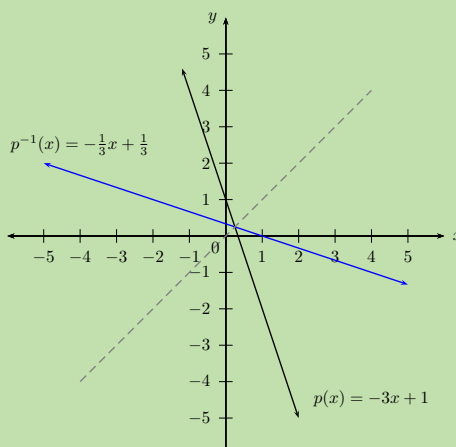
$$\text{Interchange } x \text{ and } y : x = -3y + 1$$

$$x - 1 = -3y$$

$$-\frac{1}{3}(x - 1) = y$$

$$\therefore y = -\frac{x}{3} + \frac{1}{3}$$

Therefore,  $p^{-1}(x) = -\frac{x}{3} + \frac{1}{3}$ .

**Step 2: Sketch the graphs of the same system of axes**

The graph of  $p^{-1}(x)$  is the reflection of  $p(x)$  about the line  $y = x$ . This means that every point on the graph of  $p(x)$  has a mirror image on the graph of  $p^{-1}(x)$ .

**To determine the inverse function of  $y = ax + q$ :**

- (1) Interchange  $x$  and  $y$  :  $x = ay + q$
- (2) Make  $y$  the subject of the equation :  $x - q = ay$   
 $\frac{x}{a} - \frac{q}{a} = \frac{ay}{a}$   
 $\therefore y = \frac{1}{a}x - \frac{q}{a}$

Therefore the inverse of  $y = ax + q$  is  $y = \frac{1}{a}x - \frac{q}{a}$ . If a linear function is invertible, then its inverse will also be linear.

### Worked example 5: Inverses - domain, range and intercepts

#### QUESTION

Determine and sketch the inverse of the function  $f(x) = 2x - 3$ . State the domain, range and intercepts.

#### SOLUTION

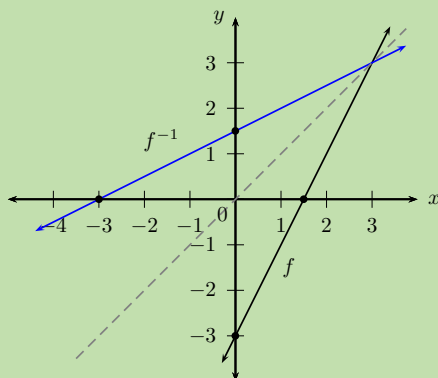
##### Step 1: Determine the inverse of the given function

- Interchange  $x$  and  $y$  in the equation.
- Make  $y$  the subject of the new equation.
- Express the new equation in function notation.

$$\begin{aligned}\text{Let } y &= 2x - 3 \\ \text{Interchange } x \text{ and } y : \quad x &= 2y - 3 \\ x + 3 &= 2y \\ \frac{1}{2}(x + 3) &= y \\ \therefore y &= \frac{x}{2} + \frac{3}{2}\end{aligned}$$

Therefore,  $f^{-1}(x) = \frac{x}{2} + \frac{3}{2}$ .

##### Step 2: Sketch the graphs on the same system of axes



The graph of  $f^{-1}(x)$  is the reflection of  $f(x)$  about the line  $y = x$ .

### Step 3: Determine domain, range and intercepts

$$\text{Domain of } f : \{x : x \in \mathbb{R}\}$$

$$\text{Range of } f : \{y : y \in \mathbb{R}\}$$

$$\text{Intercepts of } f : (0; -3) \text{ and } \left(\frac{3}{2}; 0\right)$$

$$\text{Domain of } f^{-1} : \{x : x \in \mathbb{R}\}$$

$$\text{Range of } f^{-1} : \{y : y \in \mathbb{R}\}$$

$$\text{Intercepts of } f^{-1} : \left(0; \frac{3}{2}\right) \text{ and } (-3; 0)$$

Notice that the intercepts of  $f$  and  $f^{-1}$  are mirror images of each other. In other words, the  $x$ - and  $y$ -values have “swapped” positions. This is true of every point on the two graphs.

### Domain and range

For a function of the form  $y = ax + q$ , the domain is  $\{x : x \in \mathbb{R}\}$  and the range is  $\{y : y \in \mathbb{R}\}$ . When a function is inverted the domain and range are interchanged. Therefore, the domain and range of the inverse of an invertible, linear function will be  $\{x : x \in \mathbb{R}\}$  and  $\{y : y \in \mathbb{R}\}$  respectively.

### Intercepts

The general form of an invertible, linear function is  $y = ax + q$  ( $a \neq 0$ ) and its inverse is  $y = \frac{1}{a}x - \frac{q}{a}$ .

The  $y$ -intercept is obtained by letting  $x = 0$ :

$$\begin{aligned} y &= \frac{1}{a}(0) - \frac{q}{a} \\ y &= -\frac{q}{a} \end{aligned}$$

This gives the point  $(0; -\frac{q}{a})$ .

The  $x$ -intercept is obtained by letting  $y = 0$ :

$$\begin{aligned} 0 &= \frac{1}{a}x - \frac{q}{a} \\ \frac{q}{a} &= \frac{1}{a}x \\ q &= x \end{aligned}$$

This gives the point  $(q; 0)$ .

It is interesting to note that if  $f(x) = ax + q$  ( $a \neq 0$ ), then  $f^{-1}(x) = \frac{1}{a}x - \frac{q}{a}$  and the  $y$ -intercept of  $f(x)$  is the  $x$ -intercept of  $f^{-1}(x)$  and the  $x$ -intercept of  $f(x)$  is the  $y$ -intercept of  $f^{-1}(x)$ .

► See video: 28B3 at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

### Exercise 2 – 3: Inverse of the function $y = ax + q$

- Given  $f(x) = 5x + 4$ , find  $f^{-1}(x)$ .
- Consider the relation  $f(x) = -3x - 7$ .
  - Is the relation a function? Explain your answer.
  - Identify the domain and range.
  - Determine  $f^{-1}(x)$ .
- Sketch the graph of the function  $f(x) = 3x - 1$  and its inverse on the same system of axes. Indicate the intercepts and the axis of symmetry of the two graphs.
  - $T(\frac{4}{3}; 3)$  is a point on  $f$  and  $R$  is a point on  $f^{-1}$ . Determine the coordinates of  $R$  if  $R$  and  $T$  are symmetrical.
- Explain why the line  $y = x$  is an axis of symmetry for a function and its inverse.
  - Will the line  $y = -x$  be an axis of symmetry for a function and its inverse?
- Given  $f^{-1}(x) = -2x + 4$ , determine  $f(x)$ .
  - Calculate the intercepts of  $f(x)$  and  $f^{-1}(x)$ .
  - Determine the coordinates of  $T$ , the point of intersection of  $f(x)$  and  $f^{-1}(x)$ .
  - Sketch the graphs of  $f$  and  $f^{-1}$  on the same system of axes. Indicate the intercepts and point  $T$  on the graph.
  - Is  $f^{-1}$  an increasing or decreasing function?
- More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [28B4](#)   2a. [28B5](#)   2b. [28B6](#)   2c. [28B7](#)   3. [28B8](#)   4. [28B9](#)  
5. [28BB](#)



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## 2.5 Quadratic functions

EMCFC

### Inverse of the function $y = ax^2$

EMCFD

#### Worked example 6: Inverse of the function $y = ax^2$

##### QUESTION

Determine the inverse of the quadratic function  $h(x) = 3x^2$  and sketch both graphs on the same system of axes.

##### SOLUTION

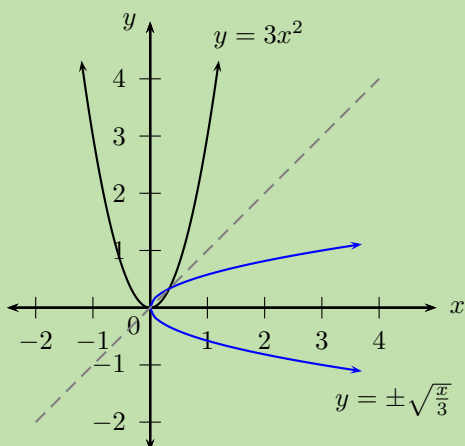
**Step 1: Determine the inverse of the given function  $h(x)$**

- Interchange  $x$  and  $y$  in the equation.
- Make  $y$  the subject of the new equation.

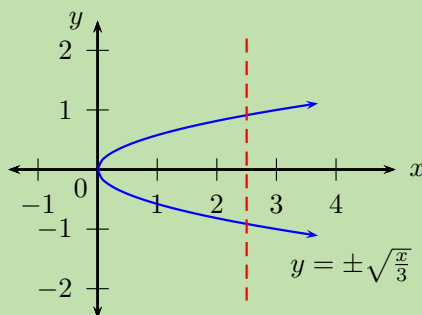


$$\begin{aligned}
 &\text{Let } y = 3x^2 \\
 &\text{Interchange } x \text{ and } y : \quad x = 3y^2 \\
 &\quad \frac{x}{3} = y^2 \\
 &\therefore y = \pm \sqrt{\frac{x}{3}} \quad (x \geq 0)
 \end{aligned}$$

**Step 2: Sketch the graphs on the same system of axes**



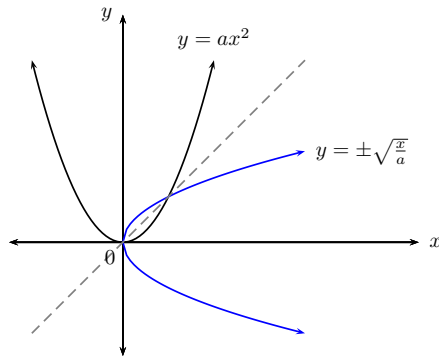
Notice that the inverse does not pass the vertical line test and therefore is not a function.



**To determine the inverse function of  $y = ax^2$ :**

- (1) Interchange  $x$  and  $y$  :  $x = ay^2$
- (2) Make  $y$  the subject of the equation :  $\frac{x}{a} = y^2$   
 $\therefore y = \pm \sqrt{\frac{x}{a}} \quad (x \geq 0)$

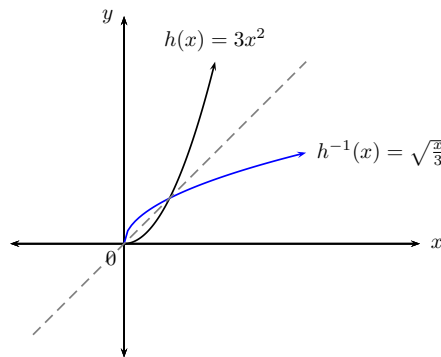
The vertical line test shows that the inverse of a parabola is not a function. However, we can limit the domain of the parabola so that the inverse of the parabola is a function.



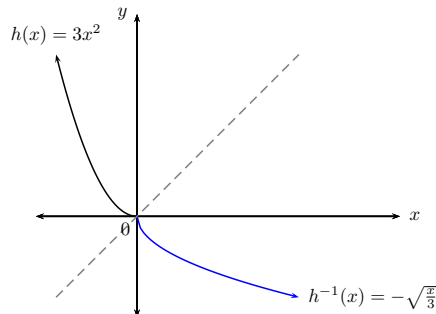
## Domain and range

Consider the previous worked example  $h(x) = 3x^2$  and its inverse  $y = \pm\sqrt{\frac{x}{3}}$ :

- If we restrict the domain of  $h$  so that  $x \geq 0$ , then  $h^{-1}(x) = \sqrt{\frac{x}{3}}$  passes the vertical line test and is a function.

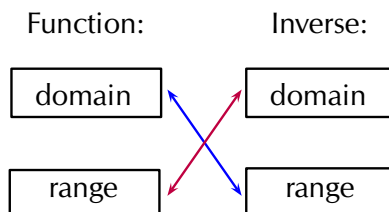


- If the restriction on the domain of  $h$  is  $x \leq 0$ , then  $h^{-1}(x) = -\sqrt{\frac{x}{3}}$  would also be a function.



The domain of the function is equal to the range of the inverse. The range of the function is equal to the domain of the inverse.

Similarly, a restriction on the domain of the function results in a restriction on the range of the inverse and vice versa.



## Worked example 7: Inverses - domain, range and restrictions

### QUESTION

Determine the inverse of  $q(x) = 7x^2$  and sketch both graphs on the same system of axes. Restrict the domain of  $q$  so that the inverse is a function.

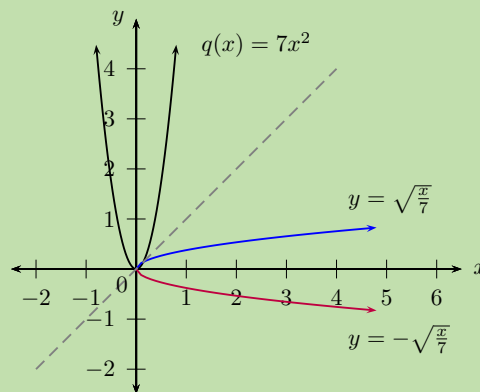
### SOLUTION

#### Step 1: Examine the function and determine the inverse

Determine the inverse of the function:

$$\begin{aligned}\text{Let } y &= 7x^2 \\ \text{Interchange } x \text{ and } y : \quad x &= 7y^2 \\ \frac{x}{7} &= y^2 \\ \therefore y &= \pm \sqrt{\frac{x}{7}} \quad (x \geq 0)\end{aligned}$$

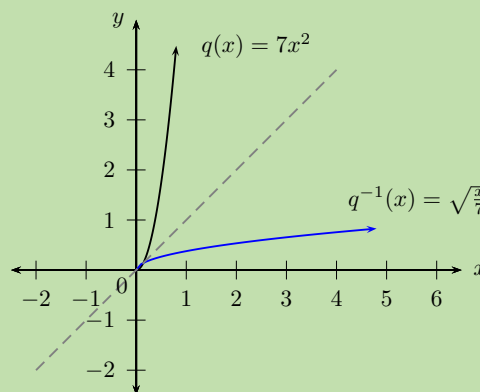
#### Step 2: Sketch both graphs on the same system of axes



#### Step 3: Determine the restriction on the domain

Option 1: Restrict the domain of  $q$  to  $x \geq 0$  so that the inverse will also be a function ( $q^{-1}$ ). The restriction  $x \geq 0$  on the domain of  $q$  will restrict the range of  $q^{-1}$  such that  $y \geq 0$ .

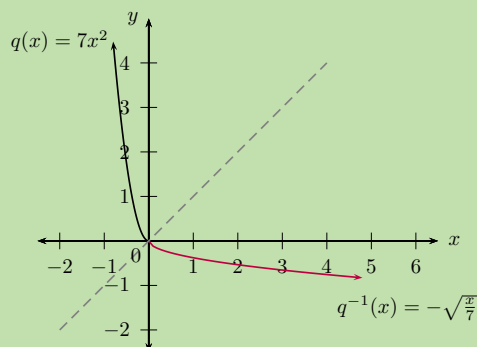
$$\begin{aligned}q : \quad & \text{domain } x \geq 0 \quad \text{range } y \geq 0 \\ q^{-1} : \quad & \text{domain } x \geq 0 \quad \text{range } y \geq 0\end{aligned}$$



OR

Option 2: Restrict the domain of  $q$  to  $x \leq 0$  so that the inverse will also be a function ( $q^{-1}$ ). The restriction  $x \leq 0$  on the domain of  $q$  will restrict the range of  $q^{-1}$  such that  $y \leq 0$ .

$$\begin{aligned} q : & \quad \text{domain } x \leq 0 \quad \text{range } y \geq 0 \\ q^{-1} : & \quad \text{domain } x \geq 0 \quad \text{range } y \leq 0 \end{aligned}$$



### Worked example 8: Inverses - domain, range and restrictions

#### QUESTION

1. Determine the inverse of  $f(x) = -x^2$ .
2. Sketch both graphs on the same system of axes.
3. Restrict the domain of  $f$  so that its inverse is a function.

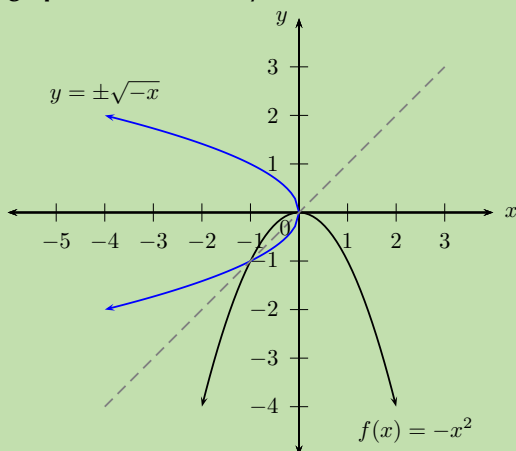
#### SOLUTION

##### Step 1: Determine the inverse of the function

$$\begin{aligned} \text{Let } y &= -x^2 \\ \text{Interchange } x \text{ and } y : & \quad x = -y^2 \\ & \quad -x = y^2 \\ & \quad y = \pm\sqrt{-x} \quad (x \leq 0) \end{aligned}$$

Note:  $\sqrt{-x}$  is only defined if  $x \leq 0$ .

##### Step 2: Sketch both graphs on the same system of axes



The inverse does not pass the vertical line test and is not a function.

**Step 3: Determine the restriction on the domain**

- If  $f(x) = -x^2$ , for  $x \leq 0$ :

$$\begin{array}{ll} f : & \text{domain } x \leq 0 \quad \text{range } y \leq 0 \\ f^{-1} : & \text{domain } x \leq 0 \quad \text{range } y \leq 0 \end{array}$$

- If  $f(x) = -x^2$ , for  $x \geq 0$ :

$$\begin{array}{ll} f : & \text{domain } x \geq 0 \quad \text{range } y \leq 0 \\ f^{-1} : & \text{domain } x \leq 0 \quad \text{range } y \geq 0 \end{array}$$

**Exercise 2 – 4: Inverses - domain, range, intercepts, restrictions**

- Determine the inverse for each of the following functions:

a)  $y = \frac{3}{4}x^2$

c)  $x^2 + 5y = 0$

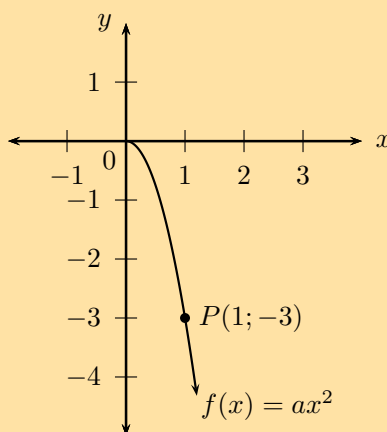
b)  $4y - 8x^2 = 0$

d)  $4y - 9 = (x + 3)(x - 3)$

- Given the function  $g(x) = \frac{1}{2}x^2$  for  $x \geq 0$ .

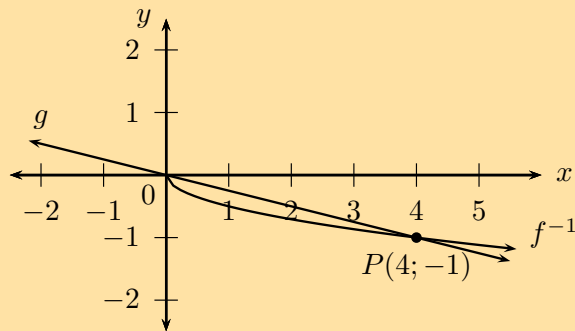
- Find the inverse of  $g$ .
- Draw  $g$  and  $g^{-1}$  on the same set of axes.
- Is  $g^{-1}$  a function? Explain your answer.
- State the domain and range for  $g$  and  $g^{-1}$ .
- Determine the coordinates of the point(s) of intersection of the function and its inverse.

- Given the graph of the parabola  $f(x) = ax^2$  with  $x \geq 0$  and passing through the point  $P(1; -3)$ .



- Determine the equation of the parabola.
- State the domain and range of  $f$ .
- Give the coordinates of the point on  $f^{-1}$  that is symmetrical to the point  $P$  about the line  $y = x$ .

- d) Determine the equation of  $f^{-1}$ .
  - e) State the domain and range of  $f^{-1}$ .
  - f) Draw a graph of  $f^{-1}$ .
4.
    - a) Determine the inverse of  $h(x) = \frac{11}{5}x^2$ .
    - b) Sketch both graphs on the same system of axes.
    - c) Restrict the domain of  $h$  so that the inverse is a function.
  5. The diagram shows the graph of  $g(x) = mx + c$  and  $f^{-1}(x) = a\sqrt{x}$ , ( $x \geq 0$ ). Both graphs pass through the point  $P(4; -1)$ .



- a) Determine the values of  $a$ ,  $c$  and  $m$ .
  - b) Give the domain and range of  $f^{-1}$  and  $g$ .
  - c) For which values of  $x$  is  $g(x) < f(x)$ ?
  - d) Determine  $f$ .
  - e) Determine the coordinates of the point(s) of intersection of  $g$  and  $f$  intersect.
6. More questions. Sign in at Everything Maths online and click 'Practise Maths'.
- Check answers online with the exercise code below or click on 'show me the answer'.
- 1a. 28BC   1b. 28BD   1c. 28BF   1d. 28BG   2. 28BH   3. 28BJ
4. 28BK   5. 28BM



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### Worked example 9: Inverses - average gradient

#### QUESTION

Given:  $h(x) = 2x^2$ ,  $x \geq 0$

1. Determine the inverse,  $h^{-1}$ .
2. Find the point where  $h$  and  $h^{-1}$  intersect.
3. Sketch  $h$  and  $h^{-1}$  on the same set of axes.
4. Use the sketch to determine if  $h$  and  $h^{-1}$  are increasing or decreasing functions.
5. Calculate the average gradient of  $h$  between the two points of intersection.

## SOLUTION

### Step 1: Determine the inverse of the function

$$\begin{aligned}\text{Let } y &= 2x^2 & (x \geq 0) \\ \text{Interchange } x \text{ and } y : & x = 2y^2 & (y \geq 0) \\ & \frac{x}{2} = y^2 \\ & y = \sqrt{\frac{x}{2}} & (x \geq 0, y \geq 0)\end{aligned}$$

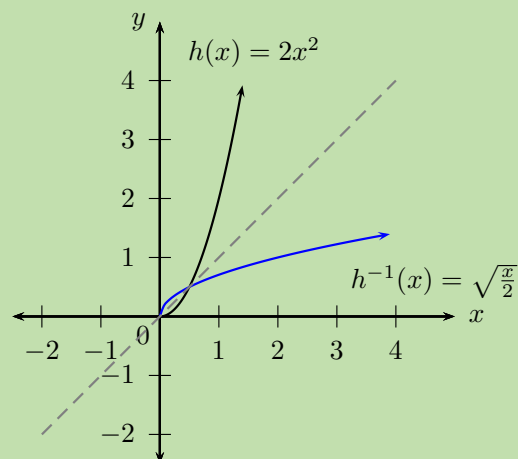
$$\therefore h^{-1}(x) = \sqrt{\frac{x}{2}} \quad (x \geq 0)$$

### Step 2: Determine the point of intersection

$$\begin{aligned}2x^2 &= \sqrt{\frac{x}{2}} \\ (2x^2)^2 &= \left(\sqrt{\frac{x}{2}}\right)^2 \\ 4x^4 &= \frac{x}{2} \\ 8x^4 &= x \\ 8x^4 - x &= 0 \\ x(8x^3 - 1) &= 0 \\ \therefore x &= 0 \text{ or } 8x^3 - 1 = 0\end{aligned}$$
$$\begin{aligned}\text{If } x &= 0, \quad y = 0 \\ \text{If } 8x^3 - 1 &= 0 \\ 8x^3 &= 1 \\ x^3 &= \frac{1}{8} \\ \therefore x &= \frac{1}{2} \\ \text{If } x &= \frac{1}{2}, \quad y = \frac{1}{2}\end{aligned}$$

Therefore, this gives the points A(0; 0) and B  $\left(\frac{1}{2}; \frac{1}{2}\right)$ .

### Step 3: Sketch both graphs on the same system of axes



#### Step 4: Examine the graphs

From the graphs, we see that both  $h$  and  $h^{-1}$  pass the vertical line test and therefore are functions.

$h$  : as  $x$  increases,  $y$  also increases, therefore  $h$  is an increasing function.

$h^{-1}$  : as  $x$  increases,  $y$  also increases, therefore  $h^{-1}$  is an increasing function.

#### Step 5: Calculate the average gradient

Calculate the average gradient of  $h$  between the points  $A(0;0)$  and  $B(\frac{1}{2}; \frac{1}{2})$ .

$$\begin{aligned}\text{Average gradient:} &= \frac{y_B - y_A}{x_B - x_A} \\ &= \frac{\frac{1}{2} - 0}{\frac{1}{2} - 0} \\ &= 1\end{aligned}$$

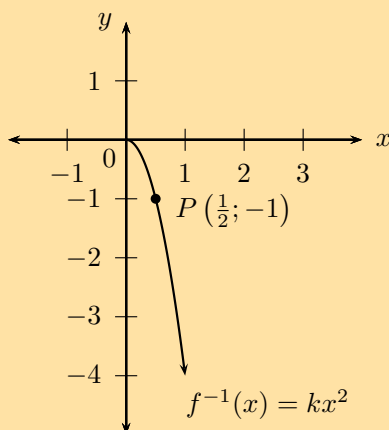
Note: this is also the average gradient of  $h^{-1}$  between the points  $A$  and  $B$ .

#### Exercise 2 – 5: Inverses - average gradient, increasing and decreasing functions

1.
  - a) Sketch the graph of  $y = x^2$  and label a point other than the origin on the graph.
  - b) Find the equation of the inverse of  $y = x^2$ .
  - c) Sketch the graph of the inverse on the same system of axes.
  - d) Is the inverse a function? Explain your answer.
  - e)  $P(2;4)$  is a point on  $y = x^2$ . Determine the coordinates of  $Q$ , the point on the graph of the inverse which is symmetrical to  $P$  about the line  $y = x$ .
  - f) Determine the average gradient between:
    - i. the origin and  $P$ ;
    - ii. the origin and  $Q$ .

Interpret the answers.

2. Given the function  $f^{-1}(x) = kx^2$ ,  $x \geq 0$ , which passes through the point  $P(\frac{1}{2}; -1)$ .





- Find the value of  $k$ .
- State the domain and range of  $f^{-1}$ .
- Find the equation of  $f$ .
- State the domain and range of  $f$ .
- Sketch the graphs of  $f$  and  $f^{-1}$  on the same system of axes.
- Is  $f$  an increasing or decreasing function?

3. Given:  $g(x) = \frac{5}{2}x^2, x \geq 0$ .

- Find  $g^{-1}(x)$ .
- Calculate the point(s) where  $g$  and  $g^{-1}$  intersect.
- Sketch  $g$  and  $g^{-1}$  on the same set of axes.
- Use the sketch to determine if  $g$  and  $g^{-1}$  are increasing or decreasing functions.
- Calculate the average gradient of  $g^{-1}$  between the two points of intersection.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28BN   2. 28BP   3. 28BQ



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## 2.6 Exponential functions

EMCFF

### Revision of exponents

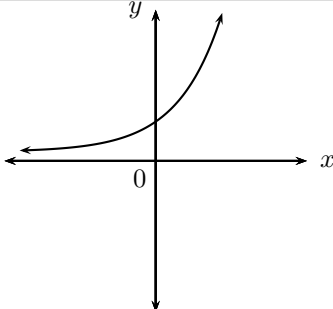
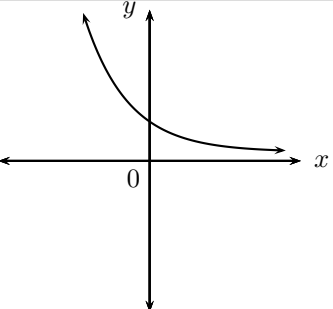
base  $\longleftarrow b^n \longrightarrow$  exponent/index

An exponent indicates the number of times a certain number (the base) is multiplied by itself. The exponent, also called the index or power, indicates the number of times the multiplication is repeated. For example,  $10^3 = 10 \times 10 \times 10 = 1000$ .

### Graphs of the exponential function $f(x) = b^x$

The value of  $b$  affects the direction of the graph:

- If  $b > 1$ ,  $f(x)$  is an increasing function.
- If  $0 < b < 1$ ,  $f(x)$  is a decreasing function.
- If  $b \leq 0$ ,  $f(x)$  is not defined.

| $y = b^x$ | $b > 1$                                                                             | $0 < b < 1$                                                                          | $b \leq 0$  |
|-----------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------|
|           |  |  | Not defined |

Investigation: Determining the inverse

| Function                         | Type of function | Inverse:                | Inverse:             |
|----------------------------------|------------------|-------------------------|----------------------|
|                                  |                  | interchange $x$ and $y$ | make $y$ the subject |
| $y = \frac{x}{3} + 10$           |                  |                         |                      |
| $y = \frac{x^2}{3}$              |                  |                         |                      |
| $y = (10)^x$                     |                  |                         |                      |
| $y = \left(\frac{1}{3}\right)^x$ |                  |                         |                      |

Consider the exponential function

$$y = b^x$$

To determine the inverse of the exponential function, we interchange the  $x$ - and  $y$ -variables:

$$x = b^y$$

For straight line functions and parabolic functions, we could easily manipulate the inverse to make  $y$  the subject of the formula. For the inverse of an exponential function, however,  $y$  is the index and we do not know a method of solving for the index.

To resolve this problem, mathematicians defined the logarithmic function. The logarithmic function allows us to rewrite the expression  $x = b^y$  with  $y$  as the subject of the formula:

$$y = \log_b x$$

This means that  $x = b^y$  is the same as  $y = \log_b x$  and both are the inverse of the exponential function  $y = b^x$ .

Logarithms

**DEFINITION:** *Logarithm*

If  $x = b^y$ , then  $y = \log_b(x)$ , where  $b > 0$ ,  $b \neq 1$  and  $x > 0$ .

Note that the brackets around the number ( $x$ ) are not compulsory, we use them to avoid confusion.

The logarithm of a number ( $x$ ) with a certain base ( $b$ ) is equal to the exponent ( $y$ ), the value to which that certain base must be raised to equal the number ( $x$ ).

For example,  $\log_2(8)$  means the power of 2 that will give 8. Since  $2^3 = 8$ , we see that  $\log_2(8) = 3$ . Therefore the exponential form is  $2^3 = 8$  and the logarithmic form is  $\log_2 8 = 3$ .

### Restrictions on the definition of logarithms

| Restriction: | Reason:                                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------|
| $b > 0$      | If $b$ is a negative number, then $b^y$ will oscillate between:<br>positive values if $y$ is even<br>negative values if $y$ is odd |
| $b \neq 1$   | Since $1^{(\text{any value})} = 1$                                                                                                 |
| $x > 0$      | Since $(\text{positive number})^{(\text{any value})} > 0$                                                                          |

### Investigation: Exponential and logarithmic form

Discuss the following statements and determine whether they are true or false:

1.  $p = a^n$  is the inverse of  $p = \log_a n$ .
2.  $y = 2^x$  is a one-to-one function, therefore  $y = \log_2 x$  is also a one-to-one function.
3.  $x = \log_5 y$  is the inverse of  $5^x = y$ .
4.  $k = b^t$  is the same as  $t = \log_b k$ .

**To determine the inverse function of  $y = b^x$ :**

- (1) Interchange  $x$  and  $y$  :  $x = b^y$
- (2) Make  $y$  the subject of the equation :  $y = \log_b x$

Therefore, if we have the exponential function  $f(x) = b^x$ , then the inverse is the logarithmic function  $f^{-1}(x) = \log_b x$ .

The “common logarithm” has a base 10 and can be written as  $\log_{10} x = \log x$ . In other words, the log symbol written without a base is interpreted as the logarithm to base 10. For example,  $\log 25 = \log_{10} 25$ .

► See video: 28BR at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

### Worked example 10: Exponential form to logarithmic form

#### QUESTION

Write the following exponential expressions in logarithmic form and express each in words:

1.  $5^2 = 25$
2.  $10^{-3} = 0,001$
3.  $p^x = q$

#### SOLUTION

##### Step 1: Determine the inverse of the given exponential expressions

Remember:  $m = a^n$  is the same as  $n = \log_a m$ .

1.  $2 = \log_5 25$
2.  $-3 = \log_{10} (0,001)$
3.  $x = \log_p q$

##### Step 2: Express in words

1. 2 is the power to which 5 must be raised to give the number 25.
2.  $-3$  is the power to which 10 must be raised to give the decimal number 0,001.
3.  $x$  is the power to which  $p$  must be raised to give  $q$ .

### Worked example 11: Logarithmic form to exponential form

#### QUESTION

Write the following logarithmic expressions in exponential form:

1.  $\log_2 128 = 7$
2.  $-2 = \log_3 \left(\frac{1}{9}\right)$
3.  $z = \log_w k$

#### SOLUTION

##### Step 1: Determine the inverse of the given logarithmic expressions

For  $n = \log_a m$ , we can write  $m = a^n$ .

1.  $2^7 = 128$
2.  $3^{-2} = \frac{1}{9}$
3.  $w^z = k$

## Exercise 2 – 6: Finding the inverse of $y = b^x$

1. Write the following in logarithmic form:

a)  $16 = 2^4$

b)  $3^{-5} = \frac{1}{243}$

c)  $(1,7)^3 = 4,913$

d)  $y = 2^x$

e)  $q = 4^5$

f)  $4 = y^9$

g)  $9 = (x - 4)^p$

h)  $3 = m^{(a+4)}$

2. Express each of the following logarithms in words and then write in exponential form:

a)  $\log_2 32 = 5$

b)  $\log \frac{1}{1000} = -3$

c)  $\log 0,1 = -1$

d)  $\log_d c = b$

e)  $\log_5 1 = 0$

f)  $\log_3 \frac{1}{81} = -4$

g)  $\log 100$

h)  $\log_{\frac{1}{2}} 16$

3. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. [28BS](#) 1b. [28BT](#) 1c. [28BV](#) 1d. [28BW](#) 1e. [28BX](#) 1f. [28BY](#)

1g. [28BZ](#) 1h. [28C2](#) 2a. [28C3](#) 2b. [28C4](#) 2c. [28C5](#) 2d. [28C6](#)

2e. [28C7](#) 2f. [28C8](#) 2g. [28C9](#) 2h. [28CB](#)



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## Logarithm bases

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From the definition of a logarithm we know that the base of a logarithm must be a positive number and it cannot be equal to 1. The value of the base influences the value of the logarithm. For example,  $\log_2 2$  is not the same as  $\log 2$  and  $\log_f 11$  is not the same as  $\log_g 11$ , ( $f \neq g$ ).

We often calculate the "common logarithm", which has a base 10 and can be written as  $\log_{10} x = \log x$ . For example,  $\log 8 = \log_{10} 8$ .

The "natural logarithm", which has a base  $e$  (an irrational number between 2,71 and 2,72), can be written as  $\log_e x = \ln x$ . For example,  $\log_e 5 = \ln 5$ .

### Special logarithmic values

- $\log_a 1 = 0$

Given the exponential form  $a^n = x$

we define the logarithmic function  $\log_a x = n$

So then for  $a^0 = 1$

we can write  $\log_a 1 = 0$

- $\log_a a = 1$

From the general exponential form  $a^n = x$

we define the logarithmic function  $\log_a x = n$

Since  $a^1 = a$

we can write  $\log_a a = 1$

In earlier grades, we used the following exponential laws for working with exponents:

- $a^m \times a^n = a^{m+n}$
- $\frac{a^m}{a^n} = a^{m-n}$
- $(ab)^n = a^n b^n$
- $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
- $(a^m)^n = a^{mn}$

where  $a > 0$ ,  $b > 0$  and  $m, n \in \mathbb{Z}$ .

The logarithmic laws are based on the exponential laws and make working with logarithms much easier.

#### Logarithmic laws:

- $\log_a x^b = b \log_a x \quad (x > 0)$
- $\log_a x = \frac{\log_b x}{\log_b a} \quad (b > 0 \text{ and } b \neq 1)$
- $\log_a xy = \log_a x + \log_a y \quad (x > 0 \text{ and } y > 0)$
- $\log_a \frac{x}{y} = \log_a x - \log_a y \quad (x > 0 \text{ and } y > 0)$

The last two logarithmic laws in the list above are not covered in this section. They are discussed at the end of the chapter and are included for enrichment only.

#### IMPORTANT: PROOFS ARE NOT REQUIRED FOR EXAMS

##### Logarithmic law:

$$\log_a x^b = b \log_a x \quad (x > 0)$$

$$\text{Let } \log_a x = m \dots (1) \quad (x > 0)$$

$$\therefore x = a^m$$

$$\therefore (x)^b = (a^m)^b$$

$$\therefore x^b = a^{bm}$$

$$\text{Change to logarithmic form: } \log_a (x^b) = bm$$

$$\text{And subst: } m = \log_a x$$

$$\therefore \log_a x^b = b \log_a x$$

In words: the logarithm of a number which is raised to a power is equal to the value of the power multiplied by the logarithm of the number.

🔗 See video: [28CC](https://www.everythingmaths.co.za) at [www.everythingmaths.co.za](https://www.everythingmaths.co.za)

**Worked example 12: Applying the logarithmic law**  $\log_a x^b = b \log_a x$ **QUESTION**

Determine the value of  $\log_3 27^4$ .

**SOLUTION**

**Step 1: Use the logarithmic law to simplify the expression**

$$\begin{aligned}\log_3 27^4 &= 4 \log_3 27 \\ &= 4 \log_3 3^3 \\ &= (4 \times 3) \log_3 3 \\ &= 12(1) \\ &= 12\end{aligned}$$

**Step 2: Write the final answer**

$$\log_3 27^4 = 12$$

**Special case:**

$$\log_a \sqrt[b]{x} = \frac{\log_a x}{b} \quad (x > 0 \text{ and } b > 0)$$

The following is a special case of the logarithmic law  $\log_a x^b = b \log_a x$ :

$$\begin{aligned}\log_a \sqrt[b]{x} &= \log_a x^{\frac{1}{b}} \\ &= \frac{1}{b} \log_a x \\ &= \frac{\log_a x}{b}\end{aligned}$$

**Exercise 2 – 7: Applying the logarithmic law:**  $\log_a x^b = b \log_a x$ 

Simplify the following:

- |                         |                                  |
|-------------------------|----------------------------------|
| 1. $\log_8 10^{10}$     | 7. $\log_2 \sqrt[4]{8}$          |
| 2. $\log_{16} x^y$      | 8. $\log_5 \frac{1}{5}$          |
| 3. $\log_3 \sqrt{5}$    | 9. $\log_2 8^5$                  |
| 4. $\log_z y^z$         | 10. $\log_4 16 \times \log_3 81$ |
| 5. $\log_y \sqrt[x]{y}$ | 11. $(\log_5 25)^2$              |
| 6. $\log_p p^q$         | 12. $\log_2 0,125$               |

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28CD   2. 28CF   3. 28CG   4. 28CH   5. 28CJ   6. 28CK  
7. 28CM   8. 28CN   9. 28CP   10. 28CQ   11. 28CR   12. 28CS



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**Logarithmic law:**  $\log_a x = \frac{\log_b x}{\log_b a}$  ( $b > 0$  and  $b \neq 1$ )

It is often necessary or convenient to convert a logarithm from one base to another base. This is referred to as a **change of base**.

$$\text{Let } \log_a x = m$$

$$\therefore x = a^m$$

$$\text{Consider the fraction: } \frac{\log_b x}{\log_b a}$$

$$\begin{aligned} \text{Substitute } x = a^m : \quad \frac{\log_b x}{\log_b a} &= \frac{\log_b a^m}{\log_b a} \\ &= m \left( \frac{\log_b a}{\log_b a} \right) \\ &= m(1) \end{aligned}$$

$$\therefore \frac{\log_b x}{\log_b a} = \log_a x$$

**Special applications:**

$$\begin{aligned} (1) \quad \log_a x &= \frac{\log_x x}{\log_x a} \\ \therefore \log_a x &= \frac{1}{\log_x a} \end{aligned}$$

$$\begin{aligned} (2) \quad \log_a \frac{1}{x} &= \log_a x^{-1} \\ \therefore \log_a \frac{1}{x} &= -\log_a x \end{aligned}$$

► See video: 28CT at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

**Worked example 13: Applying the logarithmic law**  $\log_a x = \frac{\log_b x}{\log_b a}$

**QUESTION**

Show:  $\log_2 8 = \frac{\log 8}{\log 2}$

**SOLUTION**

**Step 1: Simplify the right-hand side of the equation**

$$\begin{aligned} \text{RHS} &= \frac{\log 8}{\log 2} \\ &= \frac{\log 2^3}{\log 2} \\ &= 3 \left( \frac{\log 2}{\log 2} \right) \\ &= 3(1) \\ &= 3 \end{aligned}$$



**Step 2: Simplify the left-hand side of the equation**

$$\begin{aligned}\text{LHS} &= \log_2 8 \\ &= \log_2 2^3 \\ &= 3 \log_2 2 \\ &= 3(1) \\ &= 3\end{aligned}$$

**Step 3: Write the final answer**

We have shown that  $\log_2 8 = \frac{\log 8}{\log 2} = 3$ .

**Worked example 14: Applying the logarithmic law**  $\log_a x = \frac{\log_b x}{\log_b a}$

**QUESTION**

If  $a = \log 2$  and  $b = \log 3$ , express the following in terms of  $a$  and  $b$ :

1.  $\log_3 2$
2.  $\log_2 \frac{10}{3}$

**SOLUTION**

**Step 1: Use a change of base to simplify the expressions**

$$\begin{aligned}\log_3 2 &= \frac{\log 2}{\log 3} \\ &= \frac{a}{b}\end{aligned}$$

$$\begin{aligned}\log_2 \frac{10}{3} &= \frac{\log \frac{10}{3}}{\log 2} \\ &= \frac{\log 10 - \log 3}{\log 2} \\ &= \frac{1 - b}{a}\end{aligned}$$

**Exercise 2 – 8: Applying the logarithmic law:**  $\log_a x = \frac{\log_b x}{\log_b a}$

1. Convert the following:

- |                                  |                             |
|----------------------------------|-----------------------------|
| a) $\log_2 4$ to base 8          | d) $\log_2 8$ to base 8     |
| b) $\log_{10} 14$ to base 2      | e) $\log_y x$ to base $x$   |
| c) $\log 4\frac{1}{2}$ to base 2 | f) $\log_{10} 2x$ to base 2 |

2. Simplify the following using a change of base:

- |                                   |                 |
|-----------------------------------|-----------------|
| a) $\log_2 10 \times \log_{10} 2$ | b) $\log_5 100$ |
|-----------------------------------|-----------------|

3. If  $\log 3 = 0,477$  and  $\log 2 = 0,301$ , determine (correct to 2 decimal places):

a)  $\log_2 3$

b)  $\log_3 2000$

4. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. [28CV](#) 1b. [28CW](#) 1c. [28CX](#) 1d. [28CY](#) 1e. [28CZ](#) 1f. [28D2](#)

2a. [28D3](#) 2b. [28D4](#) 3a. [28D5](#) 3b. [28D6](#)



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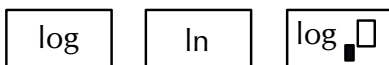
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## Logarithms using a calculator

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### Calculating a logarithmic value

There are many different types and models of scientific calculators. It is very important to be familiar with your own calculator and the different function buttons. Some calculators only have two buttons for logarithms: one for calculating the common logarithm (base is equal to 10) and another for calculating the natural log (base is equal to  $e$ ). Newer models will have a third button which allows the user to calculate the logarithm of a number to a certain base.



### Worked example 15: Using a calculator: logarithm function

#### QUESTION

Use a calculator to determine the following values (correct to 3 decimal places):

1.  $\log 9$
2.  $\log 0,3$
3.  $\log \frac{3}{4}$
4.  $\log (-2)$

#### SOLUTION

##### Step 1: Use the common logarithm function on your calculator

Make sure that you are familiar with the "LOG" function on your calculator. Notice that the base for each of the logarithms given above is 10.

1.    0,954...
2.      -0,522...
3.        -0,124...
4.       ERROR

**Step 2: Write the final answer**

1.  $\log 9 = 0,954$
2.  $\log 0,3 = -0,523$
3.  $\log \frac{3}{4} = -0,125$
4.  $\log(-2) = \text{undefined}$

**Worked example 16: Using a calculator: inverse logarithm function**

**QUESTION**

Use a calculator to determine the following values (correct to 3 decimal places):

1.  $\log x = 1,7$
2.  $\log t = \frac{2}{7}$
3.  $\log y = -3$

**SOLUTION**

**Step 1: Use the second function and common logarithm function on your calculator**

For each of the logarithms given above we need to calculate the inverse of the logarithm (sometimes called the antilog). Make sure that you are familiar with the “2nd F” button on your calculator.

Notice that by pressing the “2nd F” button and then the “LOG” button, we are using the “ $10^x$ ” function on the calculator, which is correct since exponentials are the inverse of logarithms.

1.       50,118...
2.         1,930...
3.        0,001

**Step 2: Write the final answer**

1.  $x = 50,119$
2.  $t = 1,930$
3.  $y = 0,001$

### Worked example 17: Using a calculator: change of base

#### QUESTION

Use a calculator to find  $\log_2 5$  correct to two decimal places.

#### SOLUTION

**Step 1:**

$$\log_2 5 = \frac{\log 5}{\log 2}$$

**Step 2: Use a change of base to convert given logarithm to base 10**

$$\log_2 5 = \frac{\log 5}{\log 2}$$

**Step 3: Use the common logarithm function on your calculator**

|     |   |   |     |   |   |          |
|-----|---|---|-----|---|---|----------|
| log | 5 | ÷ | log | 2 | = | 2,321... |
|-----|---|---|-----|---|---|----------|

**Step 4: Write the final answer**

$$\log_2 5 = 2,32$$

#### Important:

- Do not write down an intermediate step when doing this type of calculation:

$$\begin{aligned}\log_2 5 &= \frac{\log 5}{\log 2} \\ &= \frac{0,7}{0,3} \quad (\text{this step can cause rounding off errors}) \\ &= 2,33\end{aligned}$$

Perform the calculation in one step on your calculator:

$$\begin{aligned}\log_2 5 &= \frac{\log 5}{\log 2} \\ &= 2,32\end{aligned}$$

- Do not round off before the final answer as this can affect the accuracy of the answer.
- Be sure that you determine the correct sequence and order of operations when using a calculator.

## Exercise 2 – 9: Logarithms using a calculator

1. Calculate the following (correct to three decimal places):

- |                       |                           |
|-----------------------|---------------------------|
| a) $\log 3$           | g) $\log(-6)$             |
| b) $\log 30$          | h) $\log_3 4$             |
| c) $\log 300$         | i) $\log 0,01$            |
| d) $\log 0.66$        | j) $\log_2 15$            |
| e) $\log \frac{1}{4}$ | k) $\log_4 10$            |
| f) $\log 852$         | l) $\log_{\frac{1}{2}} 6$ |

2. Use a calculator to determine the value of  $x$  (correct to two decimal places). Check your answer by changing to exponential form.

- |                     |                               |
|---------------------|-------------------------------|
| a) $\log x = 0,6$   | g) $\log x = \frac{2}{5}$     |
| b) $\log x = -2$    | h) $\log x = -\frac{6}{5}$    |
| c) $\log x = 1,8$   | i) $\log_2 x = 0,25$          |
| d) $\log x = 5$     | j) $\log_5 x = -0,1$          |
| e) $\log x = -0,5$  | k) $\log_{\frac{1}{4}} x = 2$ |
| f) $\log x = 0,076$ | l) $\log_7 x = 0,3$           |

3. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|
| 1a. 28D7 | 1b. 28D8 | 1c. 28D9 | 1d. 28DB | 1e. 28DC | 1f. 28DD |
| 1g. 28DF | 1h. 28DG | 1i. 28DH | 1j. 28DJ | 1k. 28DK | 1l. 28DM |
| 2a. 28DN | 2b. 28DP | 2c. 28DQ | 2d. 28DR | 2e. 28DS | 2f. 28DT |
| 2g. 28DV | 2h. 28DW | 2i. 28DX | 2j. 28DY | 2k. 28DZ | 2l. 28F2 |



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## Exponential and logarithmic graphs

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### Worked example 18: Graphs of the inverse of $y = b^x$

#### QUESTION

On the same system of axes, draw the graphs of  $f(x) = 10^x$  and its inverse  $f^{-1}(x) = \log x$ . Investigate the properties of  $f$  and  $f^{-1}$ .

#### SOLUTION

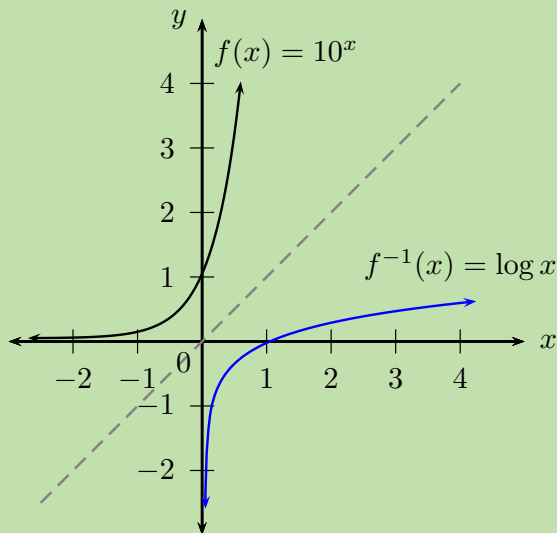
**Step 1: Determine the properties of  $f(x)$**

- Function:  $y = 10^x$
- Shape: increasing graph

- Intercept(s):  $(0; 1)$
- Asymptote(s): horizontal asymptote at  $x$ -axis, line  $y = 0$
- Domain:  $\{x : x \in \mathbb{R}\}$
- Range:  $\{y : y > 0, y \in \mathbb{R}\}$

### Step 2: Draw the graphs

The graph of the inverse  $f^{-1}$  is the reflection of  $f$  about the line  $y = x$ .



### Step 3: Determine the properties of $f^{-1}(x)$

- Function:  $y = \log x$
- Shape: increasing graph
- Intercept(s):  $(1; 0)$
- Asymptote(s): vertical asymptote at  $y$ -axis, line  $x = 0$
- Domain:  $\{x : x > 0, x \in \mathbb{R}\}$
- Range:  $\{y : y \in \mathbb{R}\}$

Notice that the inverse is a function:  $f^{-1}(x) = \log x$  is a one-to-one function since every input value is associated with only one output value.

The exponential function and the logarithmic function are inverses of each other:

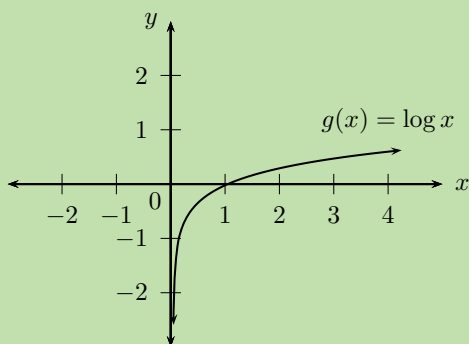
- the domain of the function is equal to the range of the inverse
- the range of the function is equal to the domain of the inverse
- the  $y$ -intercept of the function is equal to the  $x$ -intercept of the inverse
- the  $x$ -intercept of the function is equal to the  $y$ -intercept of the inverse
- the asymptote for the function is  $y = 0$  and the asymptote for the inverse is  $x = 0$
- the graphs are reflected about the line  $y = x$

**QUESTION**

1. Draw a sketch of  $g(x) = \log_{10} x$ .
2. Reflect the graph of  $g$  about the  $x$ -axis to give the graph  $h$ .
3. Investigate the properties of  $h$ .
4. Use  $g$  and  $h$  to suggest a general conclusion.

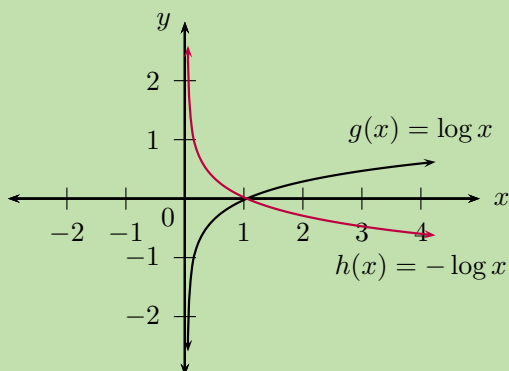
**SOLUTION**

**Step 1: Sketch the graph of  $g(x) = \log_{10} x$**



**Step 2: Reflect  $g$  about the  $x$ -axis**

An easy method for reflecting a graph about a certain line is to imagine folding the Cartesian plane along that line and the reflected graph is pressed onto the plane.



**Step 3: Investigate the properties of  $h$**

- Function: passes the vertical line test
- Shape: decreasing graph
- Intercept(s): (1; 0)
- Asymptote(s): vertical asymptote at  $y$ -axis, line  $x = 0$
- Domain:  $\{x : x > 0, x \in \mathbb{R}\}$
- Range:  $\{y : y \in \mathbb{R}\}$

Since  $h(x)$  is symmetrical to  $g(x)$  about the  $x$ -axis, this means that every  $y$ -value of  $g$  corresponds to a  $y$ -value of the opposite sign for  $h$ .

$$\text{Given } g(x) = \log_{10} x$$

$$\therefore h(x) = -\log_{10} x$$

$$\text{Let } y = -\log_{10} x$$

$$-y = \log_{10} x$$

$$\therefore 10^{-y} = x$$

$$\left(\frac{1}{10}\right)^y = x$$

$$\therefore y = \log_{\frac{1}{10}} x$$

$$\therefore h(x) = -\log_{10} x = \log_{\frac{1}{10}} x$$

#### Step 4: General conclusion

From this example of  $g$  and  $h$  we see that:

$$-\log_m p = \log_{\frac{1}{m}} p$$

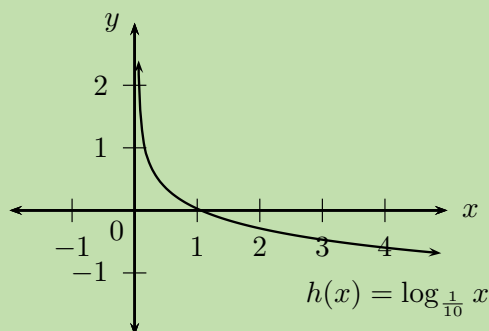
#### Worked example 20: Graph of $y = \log_b x$

##### QUESTION

1. Draw a sketch of  $h(x) = \log_{\frac{1}{10}} x$ .
2. Draw the graph of  $r(x)$ , the reflection of  $h$  about the line  $y = x$ .
3. Investigate the properties of  $r$ .
4. Write down the new equation if  $h$  is shifted 1 unit upwards and 2 units to the right.

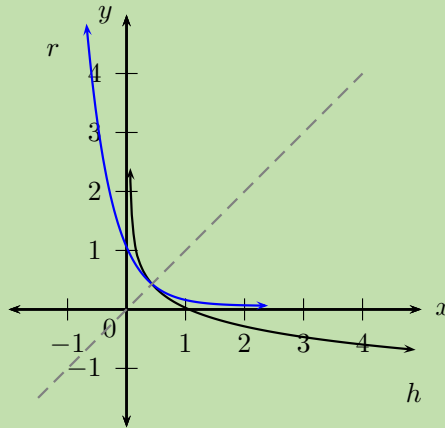
##### SOLUTION

**Step 1: Sketch the graph of  $h(x) = \log_{\frac{1}{10}} x$**





**Step 2: Reflect  $h$  about the line  $y = x$**



**Step 3: Investigate the properties of  $r$**

- Function: passes the vertical line test
- Shape: decreasing graph
- Intercept(s): (0; 1)
- Asymptote(s): horizontal asymptote at  $x$ -axis, line  $y = 0$
- Domain:  $\{x : x \in \mathbb{R}\}$
- Range:  $\{y : y > 0, y \in \mathbb{R}\}$

Since  $h(x)$  is symmetrical to  $r(x)$  about the line  $y = x$ , this means that  $r$  is the inverse of  $h$ .

$$h(x) = \log_{\frac{1}{10}} x$$

$$\text{Let } y = \log_{\frac{1}{10}} x$$

$$\text{Inverse: } x = \log_{\frac{1}{10}} y$$

$$\therefore \left(\frac{1}{10}\right)^x = y$$

$$10^{-x} = y$$

$$\therefore r(x) = h^{-1}(x) = 10^{-x}$$

Therefore,  $r(x)$  is an exponential function of the form  $y = b^x$  with  $0 < b < 1$ . In words, the base  $b$  is a positive fraction with a value between 0 and 1.

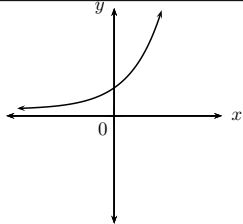
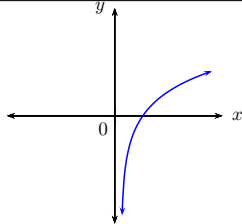
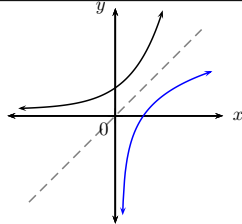
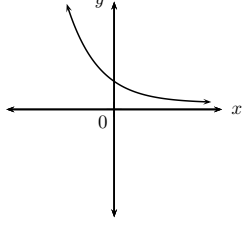
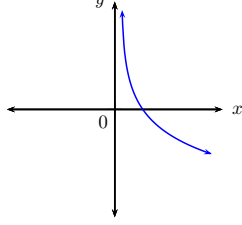
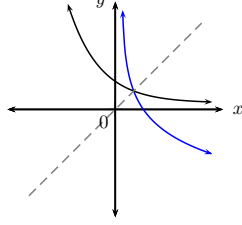
**Step 4: Vertical and horizontal shifts**

If  $h$  is shifted 1 unit upwards and 2 units to the right, then the new equation will be:

$$y = \log_{\frac{1}{10}} (x - 2) + 1$$

The vertical asymptote is  $x = 2$  and the horizontal asymptote is  $y = 1$ .

**Summary of graphs:  $y = b^x$  and  $y = \log_b x$**

|                  | Exponential function                                                              | Logarithmic function                                                              | Axis of symmetry                                                                   |
|------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|                  | $y = b^x$                                                                         | $y = \log_b x$                                                                    | $y = x$                                                                            |
| $b > 1$          |  |  |  |
| $0 < b < 1$      |  |  |  |
| Axis of symmetry | $y$ -axis, $x = 0$                                                                | $x$ -axis, $y = 0$                                                                |                                                                                    |

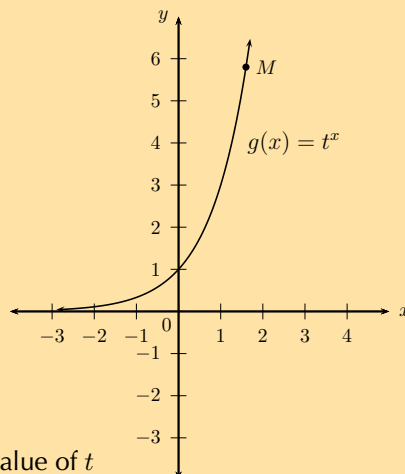
► See video: 28F3 at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

**Exercise 2 – 10: Graphs and inverses of  $y = \log_b x$**

1. Given  $f(x) = \left(\frac{1}{5}\right)^x$ .

- Sketch the graphs of  $f$  and  $f^{-1}$  on the same system of axes. Label both graphs clearly.
- State the intercept(s) for each graph.
- Label  $P$ , the point of intersection of  $f$  and  $f^{-1}$ .
- State the domain, range and asymptote(s) of each function.

2. Given  $g(x) = t^x$  with  $M\left(1\frac{3}{5}; 5\frac{4}{5}\right)$  a point on the graph of  $g$ .



- Determine the value of  $t$
- Find the inverse of  $g$ .

- c) Use symmetry about the line  $y = x$  to sketch the graphs of  $g$  and  $g^{-1}$  on the same system of axes.
- d) Point  $N$  lies on the graph of  $g^{-1}$  and is symmetrical to point  $M$  about the line  $y = x$ . Determine the coordinates of  $N$ .

3. More questions. Sign in at Everything Maths online and click 'Practise Maths'. Check answers online with the exercise code below or click on 'show me the answer'.

1. 28F4    2. 28F5



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## Applications of logarithms

EMCFP

Logarithms have many different applications:

- seismologists use logarithms to calculate the magnitude of earthquakes
- financial institutions make use of logarithms to calculate the length of loan repayments
- scientists use logarithms to determine the rate of radioactive decay
- biologists use logarithms to calculate population growth rates
- scientists use logarithms to determine pH levels

$$\text{pH} = -\log_{10} [H^+]$$

▶ See video: 28F6 at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

### Worked example 21: Population growth

#### QUESTION

The population of a city grows by 5% every two years. How long will it take for the city's population to triple in size?

#### SOLUTION

**Step 1: Write down a suitable formula and the known values**

$$A = P(1 + i)^n$$

- Let  $P = x$
- The population triples in size, so  $A = 3x$
- Growth rate  $i = \frac{5}{100}$
- Growth rate is given for a 2 year period, so we use  $\frac{n}{2}$

**Step 2: Substitute known values and solve for  $n$**

$$\begin{aligned} 3x &= x \left( 1 + \frac{5}{100} \right)^{\frac{n}{2}} \\ 3 &= (1,05)^{\frac{n}{2}} \end{aligned}$$

**Step 3: Method 1: take the logarithm of both sides of the equation**

$$\begin{aligned}\log 3 &= \log (1,05)^{\frac{n}{2}} \\ \log 3 &= \frac{n}{2} \times \log 1,05 \\ 2 \times \frac{\log 3}{\log 1,05} &= n \\ 45,034\dots &= n\end{aligned}$$

**Step 4: Method 2: change from exponential form to logarithmic form**

$$\begin{aligned}\frac{n}{2} &= \log_{1,05} 3 \\ &= \frac{\log 3}{\log 1,05} \\ n &= 2 \times \frac{\log 3}{\log 1,05} \\ n &= 45,034\dots\end{aligned}$$

**Step 5: Write the final answer**

It will take approximately 45 years for the city's population to triple in size.

**Exercise 2 – 11: Applications of logarithms**

1. The population of Uppington grows 6% every 3 years. How long will it take to triple in size?

Give your answer in years and round to the nearest integer.

2. An ant population of 36 ants doubles every month.

- Determine a formula that describes the growth of the population.
- Calculate how long it will take for the ant population to reach a quarter of a million ants.

3. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28F7    2. 28F8



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► See presentation: 28F9 at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

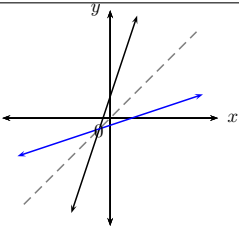
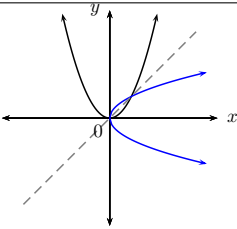
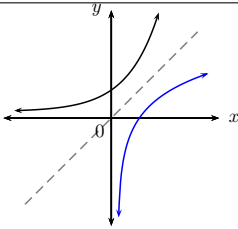
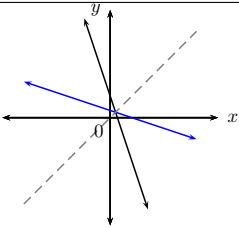
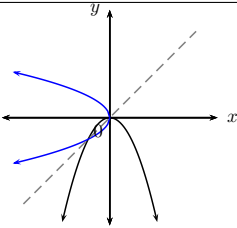
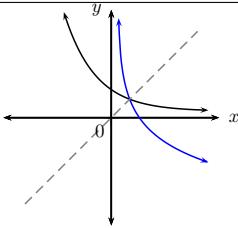
- Function: a rule which uniquely associates elements of one set  $A$  with the elements of another set  $B$ ; each element in set  $A$  maps to only one element in set  $B$ .
- Functions can be one-to-one relations or many-to-one relations. A many-to-one relation associates two or more values of the independent (input) variable with a single value of the dependent (output) variable.
- Vertical line test: if it is possible to draw any vertical line which crosses the graph of the relation more than once, then the relation is not a function.
- Given the invertible function  $f(x)$ , we determine the inverse  $f^{-1}(x)$  by:
  - replacing every  $x$  with  $y$  and  $y$  with  $x$ ;
  - making  $y$  the subject of the equation;
  - expressing the new equation in function notation.

If we represent the function  $f$  and the inverse function  $f^{-1}$  graphically, the two graphs are reflected about the line  $y = x$ .

- The domain of the function is equal to the range of the inverse. The range of the function is equal to the domain of the inverse.
- The inverse function of a straight line is also a straight line. Vertical and horizontal lines are exceptions.
- The inverse of a parabola is not a function. However, we can limit the domain of the parabola so that the inverse of the parabola is a function.
- The inverse of the exponential function  $f(x) = b^x$ , ( $b > 0, b \neq 1$ ) is the logarithmic function  $f^{-1}(x) = \log_b x$ .
- The “common logarithm” has a base 10 and can be written as  $\log_{10} x = \log x$ . The log symbol written without a base means log base 10.

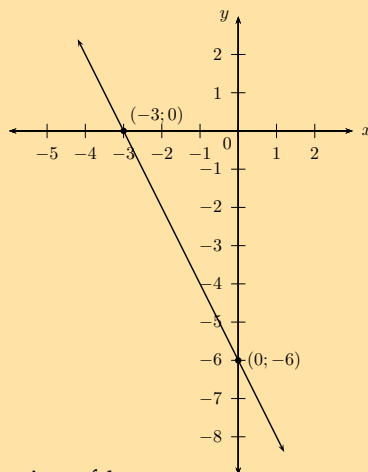
#### Logarithmic laws:

- $\log_a x^b = b \log_a x$  ( $x > 0$ )
- $\log_a x = \frac{\log_b x}{\log_b a}$  ( $b > 0$  and  $b \neq 1$ )
- $\log_a xy = \log_a x + \log_a y$  ( $x > 0$  and  $y > 0$ )
- $\log_a \frac{x}{y} = \log_a x - \log_a y$  ( $x > 0$  and  $y > 0$ )

|                     | Straight line function                                                            | Quadratic function                                                                | Exponential function                                                               |
|---------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Formula             | $y = ax + q$                                                                      | $y = ax^2$                                                                        | $y = b^x$                                                                          |
| Inverse             | $y = \frac{x}{a} - \frac{q}{a}$                                                   | $y = \pm \sqrt{\frac{x}{a}}$                                                      | $y = \log_b x$                                                                     |
| Inverse a function? | yes                                                                               | no                                                                                | yes                                                                                |
| Graphs              |  |  |  |
|                     |  |  |  |

## Exercise 2 – 12: End of chapter exercises

1. Given the straight line  $h$  with intercepts  $(-3; 0)$  and  $(0; -6)$ .



- Determine the equation of  $h$ .
  - Find  $h^{-1}$ .
  - Draw both graphs on the same system of axes.
  - Calculate the coordinates of  $S$ , the point of intersection of  $h$  and  $h^{-1}$ .
  - State the property regarding the point of intersection that will always be true for a function and its inverse.
2. The inverse of a function is  $f^{-1}(x) = 2x + 4$ .
- Determine  $f$ .
  - Draw  $f$  and  $f^{-1}$  on the same set of axes. Label each graph clearly.
  - Is  $f^{-1}$  an increasing or decreasing function? Explain your answer.

3.  $f(x) = 2x^2$ .

- Draw the graph of  $f$  and state its domain and range.
- Determine the inverse and state its domain and range.

4. Given the function  $f(x) = \left(\frac{1}{4}\right)^x$ .

- Sketch the graphs of  $f$  and  $f^{-1}$  on the same system of axes.
- Determine if the point  $\left(-\frac{1}{2}; 2\right)$  lies on the graph of  $f$ .
- Write  $f^{-1}$  in the form  $y = \dots$
- If the graphs of  $f$  and  $f^{-1}$  intersect at  $\left(\frac{1}{2}; P\right)$ , determine the value of  $P$ .
- Give the equation of the new graph,  $G$ , if the graph of  $f^{-1}$  is shifted 2 units to the left.
- Give the asymptote(s) of  $G$ .

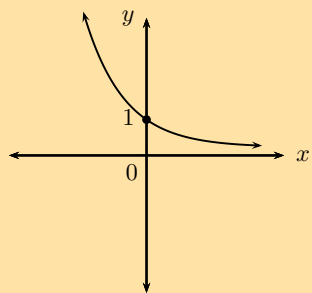
5. Consider the function  $h(x) = 3^x$ .

- Write down the inverse in the form  $h^{-1}(x) = \dots$
- State the domain and range of  $h^{-1}$ .
- Sketch the graphs of  $h$  and  $h^{-1}$  on the same system of axes, label all intercepts.
- For which values of  $x$  will  $h^{-1}(x) < 0$ ?

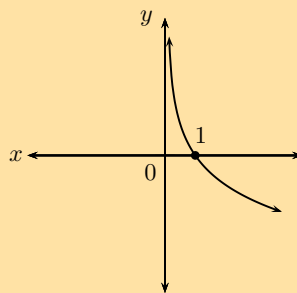
6. Consider the functions  $f(x) = 2^x$  and  $g(x) = x^2$ .

- Sketch the graphs of  $f$  and  $g$  on the same system of axes.
- Determine whether or not  $f$  and  $g$  intersect at a point where  $x = -1$ .
- How many solutions does the equation  $2^x = x^2$  have?

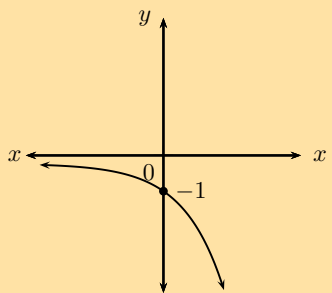
7. Below are three graphs and six equations. Write down the equation that best matches each of the graphs.



Graph 1



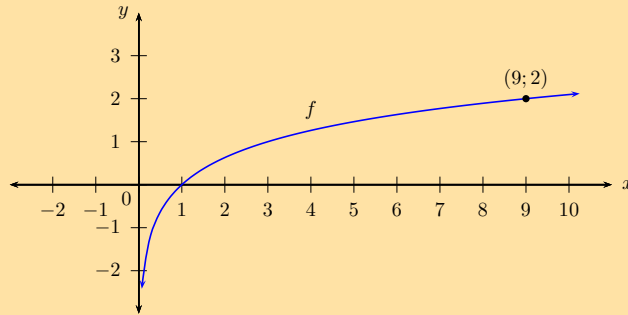
Graph 2



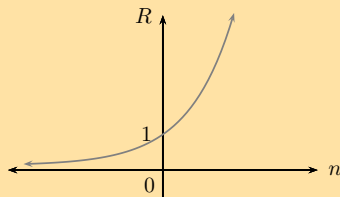
Graph 3

- $y = \log_3 x$
- $y = -\log_3 x$
- $y = \log_{\frac{1}{3}} x$
- $y = 3^x$
- $y = 3^{-x}$
- $y = -3^x$

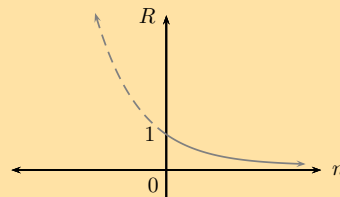
8. Given the graph of the function  $f : y = \log_b x$  passing through the point  $(9; 2)$ .



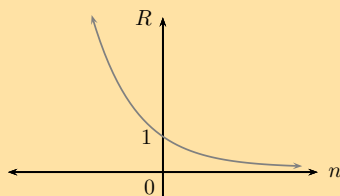
- Show that  $b = 3$ .
  - Determine the value of  $a$  if  $(a; -1)$  lies on  $f$ .
  - Write down the new equation if  $f$  is shifted 2 units upwards.
  - Write down the new equation if  $f$  is shifted 1 units to the right.
9.
  - If the rhino population in South Africa starts to decrease at a rate of 7% per annum, determine how long it will take for the current rhino population to halve in size? Give your answer to the nearest integer.
  - Which of the following graphs best illustrates the rhino population's decline? Motivate your answer.



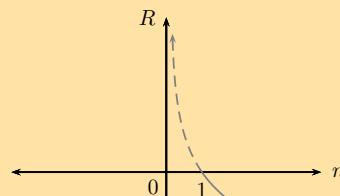
Graph A



Graph C



Graph B



Graph D

Important note: the graphs above have been drawn as a continuous curve to show a trend. Rhino population numbers are discrete values and should be plotted points.

10. At 8 a.m. a local celebrity tweets about his new music album to 100 of his followers. Five minutes later, each of his followers retweet his message to two of their friends. Five minutes after that, each friend retweets the message to another two friends. Assume this process continues.
- Determine a formula that describes this retweeting process.



- b) Calculate how many retweets of the celebrity's message are sent an hour after his original tweet.

$$1 \text{ hour} = 60 \text{ minutes} = 12 \times 5, \text{ therefore } n = 12.$$

- c) How long will it take for the total number of retweets to exceed 200 million?

11. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- 1a. 28FB   1b. 28FC   1c. 28FD   1d. 28FF   1e. 28FG   2. 28FH  
 3a. 28FJ   3b. 28FK   4. 28FM   5. 28FN   6. 28FP   7. 28FQ  
 8. 28FR   9. 28FS   10. 28FT



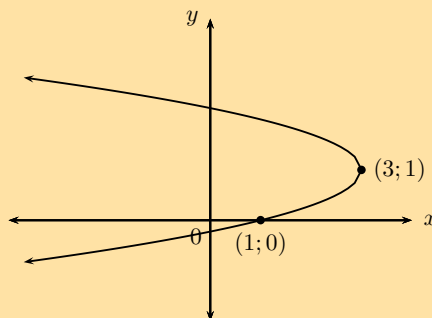
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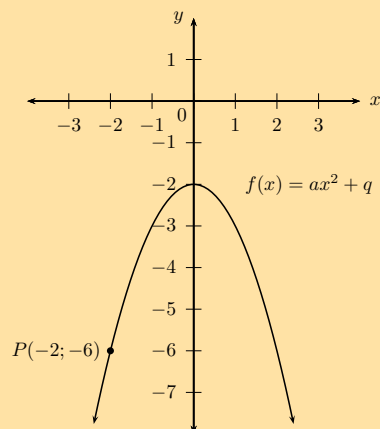
### Exercise 2 – 13: Inverses (ENRICHMENT ONLY)

1.
  - a) Given:  $g(x) = -1 + \sqrt{x}$ , find the inverse of  $g(x)$  in the form  $g^{-1}(x) = \dots$
  - b) Draw the graph of  $g^{-1}$ .
  - c) Use symmetry to draw the graph of  $g$  on the same set of axes.
  - d) Is  $g^{-1}$  a function?
  - e) Give the domain and range of  $g^{-1}$ .
2. The graph of the inverse of  $f$  is shown below:



- a) Find the equation of  $f$ , given that  $f$  is a parabola of the form  $y = (x + p)^2 + q$ .
  - b) Will  $f$  have a maximum or a minimum value?
  - c) State the domain, range and axis of symmetry of  $f$ .
3. Given:  $k(x) = 2x^2 + 1$ 
  - a) If  $(q; 3)$  lies on  $k$ , determine the value(s) of  $q$ .
  - b) Sketch the graph of  $k$ , label the point(s)  $(q; 3)$  on the graph.
  - c) Find the equation of the inverse of  $k$  in the form  $y = \dots$
  - d) Sketch  $k$  and  $y = \sqrt{\frac{x-1}{2}}$  on the same system of axes.
  - e) Determine the coordinates of the point on the graph of the inverse that is symmetrical to  $(q; 3)$  about the line  $y = x$ .

4. The sketch shows the graph of a parabola  $f(x) = ax^2 + q$  passing through the point  $P(-2; -6)$ .



- Determine the equation of  $f$ .
  - Determine and investigate the inverse.
  - Sketch the inverse and discuss the characteristics of the graph.
5. Given the function  $H : y = x^2 - 9$ .
- Determine the algebraic formula for the inverse of  $H$ .
  - Draw graphs of  $H$  and its inverse on the same system of axes. Indicate intercepts and turning points.
  - Is the inverse a function? Give reasons.
  - Show algebraically and graphically the effect of restricting the domain of  $H$  to  $\{x : x \leq 0\}$ .
6. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28FV   2. 28FW   3. 28FX   4a. 28FY   4b. 28FZ   4c. 28G2  
5. 28G3



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**NOTE: THIS SECTION IS NOT PART OF THE CURRICULUM**

## Laws of logarithms

EMCFS

**Logarithmic law:**

$$\log_a xy = \log_a x + \log_a y \quad (x > 0 \text{ and } y > 0)$$

$$\begin{aligned} \text{Let } \log_a(x) = m &\implies x = a^m \dots (1) & (x > 0) \\ \text{and } \log_a(y) = n &\implies y = a^n \dots (2) & (y > 0) \end{aligned}$$

$$\begin{aligned} \text{Then } (1) \times (2) : \quad x \times y &= a^m \times a^n \\ \therefore xy &= a^{m+n} \end{aligned}$$

Now we change from the exponential form back to logarithmic form:

$$\begin{aligned} \log_a xy &= m + n \\ \text{But } m &= \log_a(x) \text{ and } n = \log_a(y) \\ \therefore \log_a xy &= \log_a(x) + \log_a(y) \end{aligned}$$

In words: the logarithm of a product is equal to the sum of the logarithms of the factors.

**Worked example 22: Applying the logarithmic law**  $\log_a xy = \log_a x + \log_a y$ **QUESTION**

Simplify:  $\log 5 + \log 2 - \log 30$

**SOLUTION****Step 1: Use the logarithmic law to simplify the expression**

We combine the first two terms since the product of 5 and 2 is equal to 10, which is always useful when simplifying logarithms.

$$\begin{aligned} \log 5 + \log 2 - \log 30 &= (\log 5 + \log 2) - \log 30 \\ &= \log(5 \times 2) - \log 30 \\ &= \log 10 - \log 30 \\ &= 1 - \log 30 \end{aligned}$$

We expand the last term to simplify the expression further:

$$\begin{aligned} &= 1 - \log(3 \times 10) \\ &= 1 - (\log 3 + \log 10) \\ &= 1 - (\log 3 + 1) \\ &= 1 - \log 3 - 1 \\ &= -\log 3 \end{aligned}$$

**Step 2: Write the final answer**

$$\log 5 + \log 2 - \log 30 = -\log 3$$

**Exercise 2 – 14: Applying logarithmic law:**  $\log_a xy = \log_a (x) + \log_a (y)$

1. Simplify the following, if possible:

- |                            |                        |
|----------------------------|------------------------|
| a) $\log_8 (10 \times 10)$ | d) $\log_{16} (x + y)$ |
| b) $\log_2 14$             | e) $\log_2 2xy$        |
| c) $\log_2 (8 \times 5)$   | f) $\log (5 + 2)$      |

2. Write the following as a single term, if possible:

- |                                         |                               |
|-----------------------------------------|-------------------------------|
| a) $\log 15 + \log 2$                   | e) $\log 7 \times \log 2$     |
| b) $\log 1 + \log 5 + \log \frac{1}{5}$ | f) $\log_2 7 + \log_3 2$      |
| c) $1 + \log_3 4$                       | g) $\log_a p + \log_a q$      |
| d) $(\log x)(\log y) + \log x$          | h) $\log_a p \times \log_a q$ |

3. Simplify the following:

- |                                  |                                                                     |
|----------------------------------|---------------------------------------------------------------------|
| a) $\log x + \log y + \log z$    | c) $\log 125 + \log 2 + \log 8$                                     |
| b) $\log ab + \log bc + \log cd$ | d) $\log_4 \frac{3}{8} + \log_4 \frac{10}{3} + \log_4 \frac{16}{5}$ |

4. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|
| 1a. 28G4 | 1b. 28G5 | 1c. 28G6 | 1d. 28G7 | 1e. 28G8 | 1f. 28G9 |
| 2a. 28GB | 2b. 28GC | 2c. 28GD | 2d. 28GF | 2e. 28GG | 2f. 28GH |
| 2g. 28GJ | 2h. 28GK | 3a. 28GM | 3b. 28GN | 3c. 28GP | 3d. 28GQ |



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**Logarithmic law:**

$$\log_a \frac{x}{y} = \log_a x - \log_a y \quad (x > 0 \text{ and } y > 0)$$

$$\text{Let } \log_a (x) = m \implies x = a^m \dots (1) \quad (x > 0)$$

$$\text{and } \log_a (y) = n \implies y = a^n \dots (2) \quad (y > 0)$$

$$\begin{aligned} \text{Then } (1) \div (2) : \quad \frac{x}{y} &= \frac{a^m}{a^n} \\ \therefore \frac{x}{y} &= a^{m-n} \end{aligned}$$

Now we change from the exponential form back to logarithmic form:

$$\log_a \frac{x}{y} = m - n$$

$$\text{But } m = \log_a (x) \text{ and } n = \log_a (y)$$

$$\therefore \log_a \frac{x}{y} = \log_a (x) - \log_a (y)$$

In words: the logarithm of a quotient is equal to the difference of the logarithms of the numerator and the denominator.

**Worked example 23: Applying the logarithmic law**  $\log_a \frac{x}{y} = \log_a x - \log_a y$ **QUESTION**

Simplify:  $\log 40 - \log 4 + \log_5 \frac{8}{5}$

**SOLUTION****Step 1: Use the logarithmic law to simplify the expression**

We combine the first two terms since both terms have the same base and the quotient of 40 and 4 is equal to 10:

$$\begin{aligned} \log 40 - \log 4 + \log_5 \frac{8}{5} &= (\log 40 - \log 4) + \log_5 \frac{8}{5} \\ &= \left( \log \frac{40}{4} \right) + \log_5 \frac{8}{5} \\ &= \log 10 + \log_5 \frac{8}{5} \\ &= 1 + \log_5 \frac{8}{5} \end{aligned}$$

We expand the last term to simplify the expression further:

$$\begin{aligned} &= 1 + (\log_5 8 - \log_5 5) \\ &= 1 + \log_5 8 - 1 \\ &= \log_5 8 \end{aligned}$$

**Step 2: Write the final answer**

$$\log 40 - \log 4 + \log_5 \frac{8}{5} = \log_5 8$$

**Exercise 2 – 15: Applying logarithmic law:**  $\log_a \frac{x}{y} = \log_a x - \log_a y$ 

1. Expand and simplify the following:

a)  $\log \frac{100}{3}$

b)  $\log_2 7\frac{1}{2}$

c)  $\log_{16} \frac{x}{y}$

d)  $\log_{16} (x - y)$

e)  $\log_5 \frac{5}{8}$

f)  $\log_x \frac{y}{r}$

2. Write the following as a single term:

a)  $\log 10 - \log 50$

b)  $\log_3 36 - \log_3 4$

c)  $\log_a p - \log_a q$

d)  $\log_a (p - q)$

e)  $\log 15 - \log_2 5$

f)  $\log 15 - \log 5$

3. Simplify the following:

a)  $\log 450 - \log 9 - \log 5$

b)  $\log \frac{4}{5} - \log \frac{3}{25} - \log \frac{1}{15}$

4. Vini and Dirk complete their mathematics homework and check each other's answers. Compare the two methods shown below and decide if they are correct or incorrect:

**Question:**

Simplify the following:

$$\log m - \log n - \log p - \log q$$

**Vini's answer:**

$$\begin{aligned}\log m - \log n - \log p - \log q &= (\log m - \log n) - \log p - \log q \\ &= \left( \log \frac{m}{n} - \log p \right) - \log q \\ &= \log \left( \frac{m}{n} \times \frac{1}{p} \right) - \log q \\ &= \log \frac{m}{np} - \log q \\ &= \log \frac{m}{np} \times \frac{1}{q} \\ &= \log \frac{m}{npq}\end{aligned}$$

**Dirk's answer:**

$$\begin{aligned}\log m - \log n - \log p - \log q &= \log m - (\log n + \log p + \log q) \\ &= \log m - \log (n \times p \times q) \\ &= \log m - \log (npq) \\ &= \log \frac{m}{npq}\end{aligned}$$

5. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 28GR 1b. 28GS 1c. 28GT 1d. 28GV 1e. 28GW 1f. 28GX

2a. 28GY 2b. 28GZ 2c. 28H2 2d. 28H3 2e. 28H4 2f. 28H5

3a. 28H6 3b. 28H7 4. 28H8



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### Useful summary:

- |                    |                             |
|--------------------|-----------------------------|
| 1. $\log 1 = 0$    | 5. $\log \frac{1}{10} = -1$ |
| 2. $\log 10 = 1$   | 6. $\log 0,1 = -1$          |
| 3. $\log 100 = 2$  | 7. $\log 0,01 = -2$         |
| 4. $\log 1000 = 3$ | 8. $\log 0,001 = -3$        |

## Simplification of logarithms

EMCFT

### Worked example 24: Simplification of logarithms

#### QUESTION

Simplify (without a calculator):  $3 \log 3 + \log 125$

#### SOLUTION

**Step 1: Apply the appropriate logarithmic laws to simplify the expression**

$$\begin{aligned} 3 \log 3 + \log 125 &= 3 \log 3 + \log 5^3 \\ &= 3 \log 3 + 3 \log 5 \\ &= 3 (\log 3 + \log 5) \\ &= 3 \log (3 \times 5) \\ &= 3 \log 15 \end{aligned}$$

**Step 2: Write the final answer**

We cannot simplify any further, therefore  $3 \log 3 + \log 125 = 3 \log 15$ .

**Important:** all the algebraic manipulation techniques ( $\times$ ,  $\div$ ,  $+$ ,  $-$ , factorisation etc.) also apply for logarithmic expressions. Always be aware of the number of terms in an expression as this will help to determine how to simplify.

### Exercise 2 – 16: Simplification of logarithms

Simplify the following without using a calculator:

- |                                  |                                                |
|----------------------------------|------------------------------------------------|
| 1. $8^{\frac{2}{3}} + \log_2 32$ | 3. $\log_2 8 - \log 1 + \log_4 \frac{1}{4}$    |
| 2. $2 \log 3 + \log 2 - \log 5$  | 4. $\log_8 1 - \log_5 \frac{1}{25} + \log_3 9$ |

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28H9   2. 28HB   3. 28HC   4. 28HD



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**Worked example 25: Solving logarithmic equations****QUESTION**

Solve for  $p$ :

$$18 \log p - 36 = 0$$

**SOLUTION**

**Step 1: Make  $\log p$  the subject of the equation**

$$\begin{aligned} 18 \log p - 36 &= 0 \\ 18 \log p &= 36 \\ \frac{18 \log p}{18} &= \frac{36}{18} \\ \therefore \log p &= 2 \end{aligned}$$

**Step 2: Change from logarithmic form to exponential form**

$$\begin{aligned} \log p &= 2 \\ \therefore p &= 10^2 \\ &= 100 \end{aligned}$$

**Step 3: Write the final answer**

$$p = 100$$

**Worked example 26: Solving logarithmic equations****QUESTION**

Solve for  $n$  (correct to the nearest integer):

$$(1,02)^n = 2$$

**SOLUTION**



### Step 1: Change from exponential form to logarithmic form

$$(1,02)^n = 2$$
$$\therefore n = \log_{1,02} 2$$

### Step 2: Use a change of base to solve for $n$

$$n = \frac{\log 2}{\log 1,02}$$
$$\therefore n = 35,00 \dots$$

### Step 3: Write the final answer

$$n = 35$$

## Exercise 2 – 17: Solving logarithmic equations

1. Determine the value of  $a$  (correct to 2 decimal places):

a)  $\log_3 a - \log 1,2 = 0$

e)  $2^{(a+1)} = 0,7$

b)  $\log_2 (a - 1) = 1,5$

f)  $(1,03)^{\frac{a}{2}} = 2,65$

c)  $\log_2 a - 1 = 1,5$

g)  $(9)^{(1-2a)} = 101$

d)  $3^a = 2,2$

2. Given  $y = 3^x$ .

a) Write down the equation of the inverse of  $y = 3^x$  in the form  $y = \dots$

b) If  $6 = 3^p$ , determine the value of  $p$  (correct to one decimal place).

c) Draw the graph of  $y = 3^x$  and its inverse. Plot the points  $A(p; 6)$  and  $B(6; p)$ .

3. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. [28HF](#)   1b. [28HG](#)   1c. [28HH](#)   1d. [28HJ](#)   1e. [28HK](#)   1f. [28HM](#)

1g. [28HN](#)   2. [28HP](#)



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- The logarithm of a number ( $x$ ) with a certain base ( $a$ ) is equal to the exponent ( $y$ ), the value to which that certain base must be raised to equal the number ( $x$ ).  
If  $x = a^y$ , then  $y = \log_a(x)$ , where  $a > 0$ ,  $a \neq 1$  and  $x > 0$ .
- Logarithms and exponentials are inverses of each other.  
 $f(x) = \log_a x$  and  $f^{-1}(x) = a^x$
- Common logarithm:  $\log a$  means  $\log_{10} a$
- The "LOG" function on your calculator uses a base of 10.
- Natural logarithm:  $\ln$  uses a base of  $e$ .
- Special values:
  - $a^0 = 1$        $\log_a 1 = 0$
  - $a^1 = a$        $\log_a a = 1$
- Logarithmic laws:
  - $\log_a xy = \log_a x + \log_a y$       ( $x > 0$  and  $y > 0$ )
  - $\log_a \frac{x}{y} = \log_a x - \log_a y$       ( $x > 0$  and  $y > 0$ )
  - $\log_a x^b = b \log_a x$       ( $x > 0$ )
  - $\log_a x = \frac{\log_b x}{\log_b a}$       ( $b > 0$  and  $b \neq 1$ )
- Special reciprocal applications:
  - $\log_a x = \frac{1}{\log_x a}$
  - $\log_a \frac{1}{x} = -\log_a x$

### Exercise 2 – 18: Logarithms (ENRICHMENT ONLY)

1. State whether the following are true or false. If false, change the statement so that it is true.

- |                                             |                                                     |
|---------------------------------------------|-----------------------------------------------------|
| a) $\log t + \log d = \log(t + d)$          | h) $\log_p q = \frac{1}{\log_q p}$                  |
| b) If $p^q = r$ , then $q = \log_r p$       | i) $2 \log_2 a + 3 \log a = 5 \log a$               |
| c) $\log \frac{A}{B} = \log A - \log B$     | j) $5 \log x + 10 \log x = 5 \log x^3$              |
| d) $\log A - B = \frac{\log A}{\log B}$     | k) $\frac{\log_n a}{\log_n b} = \log_n \frac{a}{b}$ |
| e) $\log_{\frac{1}{2}} x = -\log_2 x$       | l) $\log(A + B) = \log A + \log B$                  |
| f) $\log_k m = \frac{\log_p k}{\log_p m}$   | m) $\log 2a^3 = 3 \log 2a$                          |
| g) $\log_n \sqrt{b} = \frac{1}{2} \log_n b$ | n) $\frac{\log_n a}{\log_n b} = \log_n(a - b)$      |

2. Simplify the following without using a calculator:

- |                           |                                  |
|---------------------------|----------------------------------|
| a) $\log 7 - \log 0,7$    | c) $\log \frac{1}{3} + \log 300$ |
| b) $\log 8 \times \log 1$ | d) $2 \log 3 + \log 2 - \log 6$  |

3. Given  $\log 5 = 0,7$ . Find the value of the following without using a calculator:

- a)  $\log 50$
- b)  $\log 20$
- c)  $\log 25$
- d)  $\log_2 5$
- e)  $10^{0,7}$

4. Given  $A = \log_8 1 - \log_5 \frac{1}{25} + \log_3 9$ .

- a) Without using a calculator, show that  $A = 4$ .
- b) Now solve for  $x$  if  $\log_2 x = A$ .
- c) Let  $f(x) = \log_2 x$ . Draw the graph of  $f$  and  $f^{-1}$ . Indicate the point  $(x; A)$  on the graph.

5. Solve for  $x$  if  $\frac{35^x}{7^x} = 15$ . Give answer correct to two decimal places.

6. Given  $f(x) = 5 \times (1,5)^x$  and  $g(x) = \left(\frac{1}{4}\right)^x$ .

- a) For which integer values of  $x$  will  $f(x) < 295$ .
- b) For which values of  $x$  will  $g(x) \geq 2,7 \times 10^{-7}$ . Give answer to the nearest integer.

7. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- 1a. [28HQ](#)   1b. [28HR](#)   1c. [28HS](#)   1d. [28HT](#)   1e. [28HV](#)   1f. [28HW](#)
- 1g. [28HX](#)   1h. [28HY](#)   1i. [28HZ](#)   1j. [28J2](#)   1k. [28J3](#)   1l. [28J4](#)
- 1m. [28J5](#)   1n. [28J6](#)   2a. [28J7](#)   2b. [28J8](#)   2c. [28J9](#)   2d. [28JB](#)
- 3a. [28JC](#)   3b. [28JD](#)   3c. [28JF](#)   3d. [28JG](#)   3e. [28JH](#)   4. [28JJ](#)
- 5. [28JK](#)   6. [28JM](#)



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## *Finance*

|     |                                                |     |
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In earlier grades we studied simple interest and compound interest, together with the concept of depreciation. Nominal and effective interest rates were also described.

- Simple interest:  $A = P(1 + in)$
- Compound interest:  $A = P(1 + i)^n$
- Simple depreciation:  $A = P(1 - in)$
- Compound depreciation:  $A = P(1 - i)^n$
- Nominal and effective annual interest rates:  $1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m$

In this chapter we look at different types of annuities, sinking funds and pyramid schemes. We also look at how to critically analyse investment and loan options and how to make informed financial decisions.

Financial planning is very important and it allows people to achieve certain goals, such as supporting a family, attending university, buying a house, and saving enough money for retirement. Prudent financial planning includes making a budget, opening a savings account, wisely investing savings and planning for retirement.

### 3.1 Calculating the period of an investment

EMCFX

For calculations using the simple interest formula, we solve for  $n$ , the time period of an investment or loan, by simply rearranging the formula to make  $n$  the subject. For compound interest calculations, where  $n$  is an exponent in the formula, we need to use our knowledge of logarithms to determine the value of  $n$ .

$$A = P(1 + i)^n$$

$A$  = accumulated amount

$P$  = principal amount

$i$  = interest rate written as a decimal

$n$  = time period

Solving for  $n$ :

$$A = P(1 + i)^n$$

$$\frac{A}{P} = (1 + i)^n$$

$$\text{Use definition: } n = \log_{(1+i)} \left( \frac{A}{P} \right)$$

$$\text{Change of base: } n = \frac{\log \left( \frac{A}{P} \right)}{\log(1 + i)}$$

### Worked example 1: Determining the value of $n$

#### QUESTION

Thembile invests R 3500 into a savings account which pays 7,5% per annum compounded yearly. After an unknown period of time his account is worth R 4044,69. For how long did Thembile invest his money?

#### SOLUTION

**Step 1: Write down the compound interest formula and the known values**

$$A = P(1 + i)^n$$

$$A = 4044,69$$

$$P = 3500$$

$$i = 0,075$$

**Step 2: Substitute the values and solve for  $n$**

$$A = P(1 + i)^n$$

$$4044,69 = 3500(1 + 0,075)^n$$

$$\frac{4044,69}{3500} = 1,075^n$$

$$\therefore n = \log_{(1,075)} \left( \frac{4044,69}{3500} \right)$$

$$= \frac{\log \frac{4044,69}{3500}}{\log 1,075}$$

$$= 2,00 \dots$$

**Step 3: Write final answer**

The R 3500 was invested for 2 years.

### Worked example 2: Duration of investments

#### QUESTION

Margo has R 12 000 to invest and needs the money to grow to at least R 30 000 to pay for her daughter's studies. If it is invested at a compound interest rate of 9% per annum, determine how long (in full years) her money must be invested?

#### SOLUTION

**Step 1: Write down the compound interest formula and the known values**

$$A = P(1 + i)^n$$

$$A = 30\,000$$

$$P = 12\,000$$

$$i = 0,09$$

**Step 2: Substitute the values and solve for  $n$** 

$$\begin{aligned}A &= P(1 + i)^n \\30\,000 &= 12\,000(1 + 0,09)^n \\ \frac{5}{2} &= (1,09)^n \\ \therefore n &= \log_{1,09} \left( \frac{5}{2} \right) \quad (\text{use definition}) \\ &= \frac{\log \frac{5}{2}}{\log 1,09} \quad (\text{change of base}) \\ &= 10,632 \dots\end{aligned}$$

**Step 3: Write the final answer**

In this case we round up, because 10 years will not yet deliver the required R 30 000. Therefore the money must be invested for at least 11 years.

**Exercise 3 – 1: Determining the period of an investment**

1. Nzuzo invests R 80 000 at an interest rate of 7,5% per annum compounded yearly. How long will it take for his investment to grow to R 100 000?
2. Sally invests R 120 000 at an interest rate of 12% per annum compounded quarterly. How long will it take for her investment to double?
3. When Banele was still in high school he deposited R 2250 into a savings account with an interest rate of 6,99% per annum compounded yearly. How long ago did Banele open the account if the balance is now R 2882,53? Write the answer as a combination of years and months.
4. The annual rate of depreciation of a vehicle is 15%. A new vehicle costs R 122 000. After how many years will the vehicle be worth less than R 40 000?
5. Some time ago, a man opened a savings account at KMT South Bank and deposited an amount of R 2100. The balance of his account is now R 3160,59. If the account gets 8,52% compound interest p.a., determine how many years ago the man made the deposit.
6. Mr. and Mrs. Dlamini want to save money for their son's university fees. They deposit R 7000 in a savings account with a fixed interest rate of 6,5% per year compounded annually. How long will take for this deposit to double in value?
7. A university lecturer retires at the age of 60. She has saved R 300 000 over the years.
  - a) She decides not to let her savings decrease at a rate faster than 15% per year. How old will she be when the value of her savings is less than R 50 000?
  - b) If she doesn't use her savings and invests all her money in an investment account that earns a fixed interest rate of 5,95% per annum, how long will it take for her investment to grow to R 390 000?

8. Simosethu puts R 450 into a bank account at the Bank of Upington. Simosethu's account pays interest at a rate of 7,11% p.a. compounded monthly. After how many years will the bank account have a balance of R 619,09?
9. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28JP   2. 28JQ   3. 28JR   4. 28JS   5. 28JT   6. 28JV  
7. 28JW   8. 28JX



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## 3.2 Annuities

EMCFY

### DEFINITION: *Annuity*

A number of equal payments made at regular intervals for a certain amount of time. An annuity is subject to a rate of interest.

- **Future value annuity** - regular equal deposits/payments are made into a savings account or investment fund to provide an accumulated amount at the end of the time period. The amount accumulating in the fund earns compound interest at a certain rate.
- **Present value annuity** - regular equal payments/installments are made to pay back a loan or bond over a given time period. The reducing balance of the loan is usually charged compound interest at a certain rate.

For investment funds, pension funds, loan repayments, mortgage bonds (home loan) and other types of annuities, payments are typically made each month. To "default" on a payment means that a payment for a certain month was not paid. The period of an investment is also referred to as the term of an investment.

## 3.3 Future value annuities

EMCFZ

For future value annuities, we regularly save the same amount of money into an account, which earns a certain rate of compound interest, so that we have money for the future.

### Worked example 3: Future value annuities

#### QUESTION

At the end of each year for 4 years, Kobus deposits R 500 into an investment account. If the interest rate on the account is 10% per annum compounded yearly, determine the value of his investment at the end of the 4 years.



## SOLUTION

**Step 1: Write down the given information and the compound interest formula**

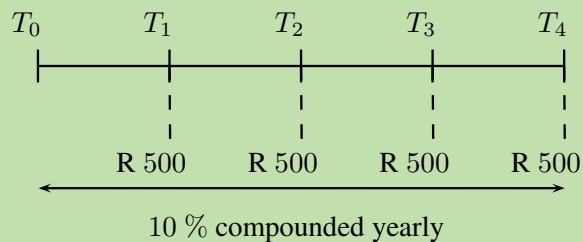
$$A = P(1 + i)^n$$

$$P = 500$$

$$i = 0,1$$

$$n = 4$$

**Step 2: Draw a timeline**



The first deposit in the account earns the highest amount of interest (three interest payments) and the last deposit earns the least interest (no interest payments).

We can summarize this information in the table below:

|        | Deposit | No. of interest payments | Calculation      | Accumulated amount |
|--------|---------|--------------------------|------------------|--------------------|
| Year 1 | R 500   | 3                        | $500(1 + 0,1)^3$ | R 665,50           |
| Year 2 | R 500   | 2                        | $500(1 + 0,1)^2$ | R 605,00           |
| Year 3 | R 500   | 1                        | $500(1 + 0,1)^1$ | R 550,00           |
| Year 4 | R 500   | 0                        | $500(1 + 0,1)^0$ | R 500,00           |
| Total  |         |                          |                  | R 2320,50          |

## Deriving the formula

EMCG2

Note: for this section it is important to be familiar with the formulae for the sum of a geometric series (Chapter 1):

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{for } r > 1$$

$$S_n = \frac{a(1 - r^n)}{1 - r} \quad \text{for } r < 1$$

In the worked example above, the total value of Kobus' investment at the end of the four year period is calculated by summing the accumulated amount for each deposit:

$$\begin{aligned} \text{R } 2320,50 &= \text{R } 500,00 + \text{R } 550,00 + \text{R } 605,00 + \text{R } 665,50 \\ &= 500(1 + 0,1)^0 + 500(1 + 0,1)^1 + 500(1 + 0,1)^2 + 500(1 + 0,1)^3 \end{aligned}$$

We notice that this is a geometric series with a constant ratio  $r = 1 + 0,1$ .

Using the formula for the sum of a geometric series:

$$a = 500$$

$$r = 1,1$$

$$n = 4$$

$$\begin{aligned} S_n &= \frac{a(r^n - 1)}{r - 1} \\ &= \frac{500(1,1^4 - 1)}{1,1 - 1} \\ &= 2320,50 \end{aligned}$$

We can therefore use the formula for the sum of a geometric series to derive a formula for the future value ( $F$ ) of a series of ( $n$ ) regular payments of an amount ( $x$ ) which are subject to an interest rate ( $i$ ):

$$a = x$$

$$r = 1 + i$$

$$\begin{aligned} S_n &= \frac{a(r^n - 1)}{r - 1} \\ \therefore F &= \frac{x[(1 + i)^n - 1]}{(1 + i) - 1} \\ &= \frac{x[(1 + i)^n - 1]}{i} \end{aligned}$$

**Future value of payments:**

$$F = \frac{x[(1 + i)^n - 1]}{i}$$

If we are given the future value of a series of payments, then we can calculate the value of the payments by making  $x$  the subject of the above formula.

**Payment amount:**

$$x = \frac{F \times i}{[(1 + i)^n - 1]}$$

### Worked example 4: Future value annuities

#### QUESTION

Ciza decides to start saving money for the future. At the end of each month she deposits R 900 into an account at Harringstone Mutual Bank, which earns 8,25% interest p.a. compounded monthly.

1. Determine the balance of Ciza's account after 29 years.
2. How much money did Ciza deposit into her account over the 29 year period?
3. Calculate how much interest she earned over the 29 year period.

#### SOLUTION

**Step 1: Write down the given information and the future value formula**

$$F = \frac{x [(1 + i)^n - 1]}{i}$$

$$x = 900$$

$$i = \frac{0,0825}{12}$$

$$n = 29 \times 12 = 348$$

**Step 2: Substitute the known values and use a calculator to determine  $F$**

$$\begin{aligned} F &= \frac{900 \left[ \left( 1 + \frac{0,0825}{12} \right)^{348} - 1 \right]}{\frac{0,0825}{12}} \\ &= \text{R } 1\,289\,665,06 \end{aligned}$$

Remember: do not round off at any of the interim steps of a calculation as this will affect the accuracy of the final answer.

**Step 3: Calculate the total value of deposits into the account**

Ciza deposited R 900 each month for 29 years:

$$\begin{aligned} \text{Total deposits:} &= \text{R } 900 \times 12 \times 29 \\ &= \text{R } 313\,200 \end{aligned}$$

**Step 4: Calculate the total interest earned**

$$\begin{aligned} \text{Total interest} &= \text{final account balance} - \text{total value of all deposits} \\ &= \text{R } 1\,289\,665,06 - \text{R } 313\,200 \\ &= \text{R } 976\,465,06 \end{aligned}$$

### Useful tips for solving problems:

1. Timelines are very useful for summarising the given information in a visual way.
2. When payments are made more than once per annum, we determine the total number of payments ( $n$ ) by multiplying the number of years by  $p$ :

| Term                      | $p$ |
|---------------------------|-----|
| yearly / annually         | 1   |
| half-yearly / bi-annually | 2   |
| quarterly                 | 4   |
| monthly                   | 12  |
| weekly                    | 52  |
| daily                     | 365 |

3. If a nominal interest rate ( $i^{(m)}$ ) is given, then use the following formula to convert it to an effective interest rate:

$$1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m$$

### Worked example 5: Calculating the monthly payments

#### QUESTION

Kosma is planning a trip to Canada to visit her friend in two years' time. She makes an itinerary for her holiday and she expects that the trip will cost R 25 000. How much must she save at the end of every month if her savings account earns an interest rate of 10,7% per annum compounded monthly?

#### SOLUTION

##### Step 1: Write down the given information and the future value formula

$$F = \frac{x[(1+i)^n - 1]}{i}$$

To determine the monthly payment amount, we make  $x$  the subject of the formula:

$$x = \frac{F \times i}{[(1+i)^n - 1]}$$

$$F = 25\,000$$

$$i = \frac{0,107}{12}$$

$$n = 2 \times 12 = 24$$

##### Step 2: Substitute the known values and calculate $x$

$$\begin{aligned} x &= \frac{25\,000 \times \frac{0,107}{12}}{\left[\left(1 + \frac{0,107}{12}\right)^{24} - 1\right]} \\ &= \text{R } 938,80 \end{aligned}$$

##### Step 3: Write the final answer

Kosma must save R 938,80 each month so that she can afford her holiday.

### Worked example 6: Determining the value of an investment

#### QUESTION

Simon starts to save for his retirement. He opens an investment account and immediately deposits R 800 into the account, which earns 12,5% per annum compounded monthly. Thereafter, he deposits R 800 at the end of each month for 20 years. What is the value of his retirement savings at the end of the 20 year period?

#### SOLUTION

**Step 1: Write down the given information and the future value formula**

$$F = \frac{x [(1 + i)^n - 1]}{i}$$

$$x = 800$$

$$i = \frac{0,125}{12}$$

$$n = 1 + (20 \times 12) = 241$$

Note that we added one extra month to the 20 years because Simon deposited R 800 immediately.

**Step 2: Substitute the known values and calculate  $F$**

$$\begin{aligned} F &= \frac{800 \left[ \left( 1 + \frac{0,125}{12} \right)^{241} - 1 \right]}{\frac{0,125}{12}} \\ &= \text{R } 856\,415,66 \end{aligned}$$

**Step 3: Write the final answer**

Simon will have saved R 856 415,66 for his retirement.

### Exercise 3 – 2: Future value annuities

- Shelly decides to start saving money for her son's future. At the end of each month she deposits R 500 into an account at Durban Trust Bank, which earns an interest rate of 5,96% per annum compounded quarterly.
  - Determine the balance of Shelly's account after 35 years.
  - How much money did Shelly deposit into her account over the 35 year period?
  - Calculate how much interest she earned over the 35 year period.
- Gerald wants to buy a new guitar worth R 7400 in a year's time. How much must he deposit at the end of each month into his savings account, which earns a interest rate of 9,5% p.a. compounded monthly?

3. A young woman named Grace has just started a new job, and wants to save money for the future. She decides to deposit R 1100 into a savings account every month. Her money goes into an account at First Mutual Bank, and the account earns 8,9% interest p.a. compounded every month.
  - a) How much money will Grace have in her account after 29 years?
  - b) How much money did Grace deposit into her account by the end of the 29 year period?
4. Ruth decides to save for her retirement so she opens a savings account and immediately deposits R 450 into the account. Her savings account earns 12% per annum compounded monthly. She then deposits R 450 at the end of each month for 35 years. What is the value of her retirement savings at the end of the 35 year period?
5. Musina MoneyLenders offer a savings account with an interest rate of 6,13% p.a. compounded monthly. Monique wants to save money so that she can buy a house when she retires. She decides to open an account and make regular monthly deposits. Her goal is to end up with R 750 000 in her account after 35 years.
  - a) How much must Monique deposit into her account each month in order to reach her goal?
  - b) How much money, to the nearest rand, did Monique deposit into her account by the end of the 35 year period?
6. Lerato plans to buy a car in five and a half years' time. She has saved R 30 000 in a separate investment account which earns 13% per annum compound interest. If she doesn't want to spend more than R 160 000 on a vehicle and her savings account earns an interest rate of 11% p.a. compounded monthly, how much must she deposit into her savings account each month?
7.
  - a) Every Monday Harold puts R 30 into a savings account at the King Bank, which accrues interest of 6,92% p.a. compounded weekly. How long will it take Harold's account to reach a balance of R 4397,53. Give the answer as a number of years and days to the nearest integer.
  - b) How much interest will Harold receive from the bank during the period of his investment?
8. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- 1a. 28JY    1b. 28JZ    1c. 28K2    2. 28K3    3. 28K4    4. 28K5  
 5. 28K6    6. 28K7    7. 28K8



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## Sinking funds

EMCG3

Vehicles, equipment, machinery and other similar assets, all depreciate in value as a result of usage and age. Businesses often set aside money for replacing outdated equipment or old vehicles in accounts called sinking funds. Regular deposits, and sometimes lump sum deposits, are made into these accounts so that enough money will have accumulated by the time a new machine or vehicle needs to be purchased.

## Worked example 7: Sinking funds

### QUESTION

Wellington Courier Company buys a delivery truck for R 296 000. The value of the truck depreciates on a reducing-balance basis at 18% per annum. The company plans to replace this truck in seven years' time and they expect the price of a new truck to increase annually by 9%.

1. Calculate the book value of the delivery truck in seven years' time.
2. Determine the minimum balance of the sinking fund in order for the company to afford a new truck in seven years' time.
3. Calculate the required monthly deposits if the sinking fund earns an interest rate of 13% per annum compounded monthly.

### SOLUTION

#### Step 1: Determine the book value of the truck in seven years' time

$$P = 296\,000$$

$$i = 0,18$$

$$n = 7$$

$$\begin{aligned}A &= P(1 - i)^n \\&= 296\,000(1 - 0,18)^7 \\&= R\,73\,788,50\end{aligned}$$

#### Step 2: Determine the minimum balance of the sinking fund

Calculate the price of a new truck in seven years' time:

$$P = 296\,000$$

$$i = 0,09$$

$$n = 7$$

$$\begin{aligned}A &= P(1 + i)^n \\&= 296\,000(1 + 0,09)^7 \\&= R\,541\,099,58\end{aligned}$$

Therefore, the balance of the sinking fund ( $F$ ) must be greater than the cost of a new truck in seven years' time minus the money from the sale of the old truck:

$$\begin{aligned}F &= R\,541\,099,58 - R\,73\,788,50 \\&= R\,467\,311,08\end{aligned}$$

#### Step 3: Calculate the required monthly payment into the sinking fund

$$x = \frac{F \times i}{[(1 + i)^n - 1]}$$

$$F = 467\,311,08$$

$$i = \frac{0,13}{12}$$

$$n = 7 \times 12 = 84$$

Substitute the values and calculate  $x$ :

$$\begin{aligned}x &= \frac{467\,311,08 \times \frac{0,13}{12}}{\left[\left(1 + \frac{0,13}{12}\right)^{84} - 1\right]} \\&= \text{R } 3438,77\end{aligned}$$

Therefore, the company must deposit R 3438,77 each month.

### Exercise 3 – 3: Sinking funds

1. Mfethu owns his own delivery business and he will need to replace his truck in 6 years' time. Mfethu deposits R 3100 into a sinking fund each month, which earns 5,3% interest p.a. compounded monthly.
  - a) How much money will be in the fund in 6 years' time, when Mfethu wants to buy the new truck?
  - b) If a new truck costs R 285 000 in 6 years' time, will Mfethu have enough money to buy it?
2. Atlantic Transport Company buys a van for R 265 000. The value of the van depreciates on a reducing-balance basis at 17% per annum. The company plans to replace this van in five years' time and they expect the price of a new van to increase annually by 12%.
  - a) Calculate the book value of the van in five years' time.
  - b) Determine the amount of money needed in the sinking fund for the company to be able to afford a new van in five years' time.
  - c) Calculate the required monthly deposits if the sinking fund earns an interest rate of 11% per annum compounded monthly.
3. Tonya owns Freeman Travel Company and she will need to replace her computer in 7 years' time. Tonya creates a sinking fund so that she will be able to afford a new computer, which will cost R 8450. The sinking fund earns interest at a rate of 7,67% p.a. compounded each quarter.
  - a) How much money must Tonya save quarterly so that there will be enough money in the account to buy the new computer?
  - b) How much interest (to the nearest rand) does the bank pay into the account by the end of the 7 year period?
4. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28K9    2. 28KB    3. 28KC



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For present value annuities, regular equal payments/installments are made to pay back a loan or bond over a given time period. The reducing balance of the loan is usually charged compound interest at a certain rate. In this section we learn how to determine the present value of a series of payments.

Consider the following example:

Kate needs to withdraw R 1000 from her bank account every year for the next three years. How much must she deposit into her account, which earns 10% per annum, to be able to make these withdrawals in the future? We will assume that these are the only withdrawals and that there are no bank charges on her account.

To calculate Kate's deposit, we make  $P$  the subject of the compound interest formula:

$$\begin{aligned} A &= P(1+i)^n \\ \frac{A}{(1+i)^n} &= P \\ \therefore P &= A(1+i)^{-n} \end{aligned}$$

We determine how much Kate must deposit for the first withdrawal:

$$\begin{aligned} P &= 1000(1+0,1)^{-1} \\ &= 909,09 \end{aligned}$$

We repeat this calculation to determine how much must be deposited for the second and third withdrawals:

$$\begin{aligned} \text{Second withdrawal: } P &= 1000(1+0,1)^{-2} \\ &= 826,45 \end{aligned}$$

$$\begin{aligned} \text{Third withdrawal: } P &= 1000(1+0,1)^{-3} \\ &= 751,31 \end{aligned}$$

Notice that for each year's withdrawal, the deposit required gets smaller and smaller because it will be in the bank account for longer and therefore earn more interest. Therefore, the total amount is:

$$\text{R } 909,09 + \text{R } 826,45 + \text{R } 751,31 = \text{R } 2486,85$$

We can check these calculations by determining the accumulated amount in Kate's bank account after each withdrawal:

|                                | Calculation                            | Accumulated amount    |
|--------------------------------|----------------------------------------|-----------------------|
| Initial deposit                |                                        | R 2486,85             |
| Amount after one year          | $= 2486,85(1+0,1)$                     | $= \text{R } 2735,54$ |
| Amount after first withdrawal  | $= \text{R } 2735,54 - \text{R } 1000$ | $= \text{R } 1735,54$ |
| Amount after two years         | $= 1735,54(1+0,1)$                     | $= \text{R } 1909,09$ |
| Amount after second withdrawal | $= \text{R } 1909,09 - \text{R } 1000$ | $= \text{R } 909,09$  |
| Amount after three years       | $= 909,09(1+0,1)$                      | $= \text{R } 1000$    |
| Amount after third withdrawal  | $= \text{R } 1000 - \text{R } 1000$    | $= \text{R } 0$       |

Completing this table for a three year period does not take too long. However, if Kate needed to make annual payments for 20 years, then the calculation becomes very repetitive and time-consuming. Therefore, we need a more efficient method for performing these calculations.

## Deriving the formula

EMCG5

In the example above, Kate needed to deposit:

$$\begin{aligned} \text{R } 2486,55 &= \text{R } 909,09 + \text{R } 826,45 + \text{R } 751,31 \\ &= 1000(1 + 0,1)^{-1} + 1000(1 + 0,1)^{-2} + 1000(1 + 0,1)^{-3} \end{aligned}$$

We notice that this is a geometric series with a constant ratio  $r = (1 + 0,1)^{-1}$ .

Using the formula for the sum of a geometric series:

$$\begin{aligned} a &= 1000(1 + 0,1)^{-1} \\ r &= (1 + 0,1)^{-1} \\ n &= 3 \\ S_n &= \frac{a(1 - r^n)}{1 - r} \quad (\text{for } r < 1) \\ &= \frac{1000(1 + 0,1)^{-1} [1 - ((1 + 0,1)^{-1})^3]}{1 - (1 + 0,1)^{-1}} \\ &= \frac{1000 [1 - (1 + 0,1)^{-3}]}{(1 + 0,1)[1 - (1 + 0,1)^{-1}]} \\ &= \frac{1000 [1 - (1 + 0,1)^{-3}]}{(1 + 0,1) - 1} \\ &= \frac{1000 [1 - (1 + 0,1)^{-3}]}{0,1} \\ &= 2486,85 \end{aligned}$$

We can therefore use the formula for the sum of a geometric series to derive a formula for the present value ( $P$ ) of a series of ( $n$ ) regular payments of an amount ( $x$ ) which are subject to an interest rate ( $i$ ):

$$a = x(1+i)^{-1}$$

$$r = (1+i)^{-1}$$

$$S_n = \frac{a(1-r^n)}{1-r} \quad (\text{for } r < 1)$$

$$\therefore P = \frac{x(1+i)^{-1} [1 - ((1+i)^{-1})^n]}{1 - (1+i)^{-1}}$$

$$= \frac{x[1 - (1+i)^{-n}]}{(1+i)[1 - (1+i)^{-1}]}$$

$$= \frac{x[1 - (1+i)^{-n}]}{1+i-1}$$

$$= \frac{x[1 - (1+i)^{-n}]}{i}$$

**Present value of a series of payments:**

$$P = \frac{x[1 - (1+i)^{-n}]}{i}$$

If we are given the present value of a series of payments, we can calculate the value of the payments by making  $x$  the subject of the above formula.

**Payment amount:**

$$x = \frac{P \times i}{[1 - (1+i)^{-n}]}$$

### Worked example 8: Present value annuities

#### QUESTION

Andre takes out a student loan for his first year of civil engineering. The loan agreement states that the repayment period is equal to 1,5 years for every year of financial assistance granted and that the loan is subject to an interest rate of 10,5% p.a. compounded monthly.

1. If Andre pays a monthly installment of R 1446,91, calculate the loan amount.
2. Determine how much interest Andre will have paid on his student loan at the end of the 18 months.

### **SOLUTION**

---

**Step 1: Write down the given information and the present value formula**

$$P = \frac{x[1 - (1 + i)^{-n}]}{i}$$

$$x = 1446,91$$

$$i = \frac{0,105}{12}$$

$$n = 1,5 \times 12 = 18$$

Substitute the known values and determine  $P$ :

$$\begin{aligned} P &= \frac{1446,91 \left[ 1 - \left( 1 + \frac{0,105}{12} \right)^{-18} \right]}{\frac{0,105}{12}} \\ &= \text{R } 24\,000,14 \end{aligned}$$

Therefore, Andre took out a student loan for R 24 000.

**Step 2: Calculate the total amount of interest**

At the end of the 18 month period:

$$\begin{aligned} \text{Total amount repaid for loan:} &= \text{R } 1446,91 \times 18 \\ &= \text{R } 26\,044,38 \end{aligned}$$

$$\begin{aligned} \text{Total amount of interest:} &= \text{R } 26\,044,38 - \text{R } 24\,000 \\ &= \text{R } 2044,38 \end{aligned}$$

### **Worked example 9: Calculating the monthly payments**

#### **QUESTION**

---

Hristo wants to buy a small wine farm worth R 8 500 000. He plans to sell his current home for R 3 400 000 which he will use as a deposit for the purchase of the farm. He secures a loan with HBP Bank with a repayment period of 10 years and an interest rate of 9,5% compounded monthly.

1. Calculate his monthly repayments.
2. Determine how much interest Hristo will have paid on his loan by the end of the 10 years.

#### **SOLUTION**

---

**Step 1: Write down the given information and the present value formula**

$$P = \frac{x[1 - (1 + i)^{-n}]}{i}$$

To determine the monthly repayment amount, we make  $x$  the subject of the formula:

$$x = \frac{P \times i}{[1 - (1 + i)^{-n}]}$$

$$P = R\ 8\ 500\ 000 - R\ 3\ 400\ 000 = R\ 5\ 100\ 000$$

$$i = \frac{0,095}{12}$$

$$n = 10 \times 12 = 120$$

**Step 2: Substitute the known values and calculate  $x$**

$$\begin{aligned} x &= \frac{5\ 100\ 000 \times \frac{0,095}{12}}{\left[1 - \left(1 + \frac{0,095}{12}\right)^{-120}\right]} \\ &= R\ 65\ 992,75 \end{aligned}$$

Therefore, Hristo must pay R 65 992,75 per month to repay his loan over the 10 year period.

**Step 3: Calculate the total amount of interest**

At the end of the 10 year period:

$$\begin{aligned} \text{Total amount repaid for loan:} &= R\ 65\ 992,75 \times 10 \times 12 \\ &= R\ 7\ 919\ 130 \end{aligned}$$

$$\begin{aligned} \text{Total amount of interest:} &= R\ 7\ 919\ 130 - R\ 5\ 100\ 000 \\ &= R\ 2\ 819\ 130 \end{aligned}$$

### Prime lending rate

The prime lending rate is a benchmark rate at which private banks lend out money to the public. It is used as a reference rate for determining interest rates on many types of loans, including small business loans, home loans and personal loans. Some interest rates may be expressed as a percentage above or below prime rate. For calculations in this chapter, we will assume the prime lending rate is 8,5% per annum.

### Worked example 10: Calculating the outstanding balance of a loan

#### QUESTION

A school sells its old bus and uses the proceeds as a 15% deposit for the purchase of the new bus, which costs R 330 000. To finance the balance of the purchase, the school takes out a loan that is subject to an interest rate of prime + 1% compounded monthly. The repayment period of the loan is 3 years.

1. Calculate the monthly repayments.
2. Determine the balance of the loan at the end of the first year, immediately after the 12<sup>th</sup> payment.

## **SOLUTION**

### **Step 1: Write down the given information and the present value formula**

$$P = \frac{x[1 - (1 + i)^{-n}]}{i}$$

To determine the monthly repayment amount, we make  $x$  the subject of the formula:

$$x = \frac{P \times i}{[1 - (1 + i)^{-n}]}$$

$$P = \text{R } 330\,000 - \left( \frac{15}{100} \times \text{R } 330\,000 \right)$$

$$= \text{R } 330\,000 - \text{R } 49\,500$$

$$= \text{R } 280\,500$$

$$i = \frac{0,095}{12}$$

$$n = 3 \times 12 = 36$$

### **Step 2: Substitute the known values and calculate $x$**

$$x = \frac{280\,500 \times \frac{0,095}{12}}{\left[ 1 - \left( 1 + \frac{0,095}{12} \right)^{-36} \right]}$$
$$= \text{R } 8985,24$$

Therefore, the school must pay R 8985,24 per month to repay the loan over the 3 year period.

### **Step 3: Calculate the balance of the loan at the end of the first year**

We can calculate the balance of the loan at the end of the first year by determining the present value of the remaining 24 payments:

$$P = \frac{8985,24 \left[ 1 - \left( 1 + \frac{0,095}{12} \right)^{-24} \right]}{\frac{0,095}{12}}$$
$$= \text{R } 195\,695,07$$

Therefore, the school must still pay R 195 695,07 of the loan.

Alternative method: we can also calculate the balance of the loan at the end of the first year by determining the accumulated amount of the loan for the first year less the future value of the first 12 payments:

$$\text{Balance} = A(\text{loan and accrued interest}) - F(\text{first year's payments and interest})$$

$$= 280\,500 \left( 1 + \frac{0,095}{12} \right)^{12} - \frac{8985,24 \left[ \left( 1 + \frac{0,095}{12} \right)^{12} - 1 \right]}{\frac{0,095}{12}}$$

$$= \text{R } 308\,338,94 \dots - \text{R } 112\,643,79 \dots$$

$$= \text{R } 195\,695,15$$

Therefore, the school must still pay R 195 695,15 of the loan.

(Note the difference of 8 cents due to rounding).

### Exercise 3 – 4: Present value annuities

1. A property costs R 1 800 000. Calculate the monthly repayments if the interest rate is 14% p.a. compounded monthly and the loan must be paid off in 20 years' time.
2. A loan of R 4200 is to be returned in two equal annual installments. If the rate of interest is 10% compounded annually, calculate the amount of each installment.
3. Stefan and Marna want to buy a flat that costs R 1,2 million. Their parents offer to put down a 20% payment towards the cost of the house. They need to get a mortgage for the balance. What is the monthly repayment amount if the term of the home loan is 30 years and the interest is 7,5% p.a. compounded monthly?
4.
  - a) Ziyanda arranges a bond for R 17 000 from Langa Bank. If the bank charges 16,0% p.a. compounded monthly, determine Ziyanda's monthly repayment if she is to pay back the bond over 9 years.
  - b) What is the total cost of the bond?
5. Dullstroom Bank offers personal loans at an interest rate of 15,63% p.a. compounded twice a year. Lubabale borrows R 3000 and must pay R 334,93 every six months until the loan is fully repaid.
  - a) How long will it take Lubabale to repay the loan?
  - b) How much interest will Lubabale pay for this loan?
6. Likengkeng has just started a new job and wants to buy a car that costs R 232 000. She visits the Soweto Savings Bank, where she can arrange a loan with an interest rate of 15,7% p.a. compounded monthly. Likengkeng has enough money saved to pay a deposit of R 50 000. She arranges a loan for the balance of the payment, which is to be paid over a period of 6 years.
  - a) What is Likengkeng's monthly repayment on her loan?
  - b) How much will the car cost Likengkeng?
7. Anathi is a wheat farmer and she needs to buy a new holding tank which costs R 219 450. She bought her old tank 14 years ago for R 196 000. The value of the old grain tank has depreciated at a rate of 12,1% per year on a reducing balance, and she plans to trade it in for its current value. Anathi will then need to arrange a loan for the balance of the cost of the new grain tank.

Orsmond bank offers loans with an interest rate of 9,71% p.a. compounded monthly for any loan up to R 170 000 and 9,31% p.a. compounded monthly for a loan above that amount. The loan agreement allows Anathi a grace period for the first six months (no payments are made) and it states that the loan must be repaid over 30 years.

  - a) Determine the monthly repayment amount.
  - b) What is the total amount of interest Anathi will pay for the loan?
  - c) How much money would Anathi have saved if she did not take the six month grace period?
8. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28KD   2. 28KF   3. 28KG   4. 28KH   5. 28KJ   6. 28KK  
7. 28KM



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In the next worked example we consider the effects of the duration of the repayment period on the total amount repaid (the amount borrowed plus the accrued interest) for a loan.

### Worked example 11: Repayment periods

#### QUESTION

David and Julie take out a home loan of R 2,6 million with an interest rate of 10% per annum compounded monthly.

1. Calculate the monthly repayments for a repayment period of 30 years.
2. Calculate the interest paid on the loan at the end of the 30 year period.
3. Determine the monthly repayments for a repayment period of 20 years.
4. Determine the interest paid on the loan at the end of the 20 year period.
5. What is the difference in the monthly repayment amounts?
6. Comment on the difference in the interest paid for the two different time periods.

#### SOLUTION

**Step 1: Consider a 30 year repayment period on the loan**

$$x = \frac{P \times i}{[1 - (1 + i)^{-n}]}$$

$$P = \text{R } 2\,600\,000$$

$$i = \frac{0,1}{12}$$

$$n = 30 \times 12 = 360$$

$$\begin{aligned} x &= \frac{2\,600\,000 \times \frac{0,1}{12}}{\left[1 - \left(1 + \frac{0,1}{12}\right)^{-360}\right]} \\ &= \text{R } 22\,816,86 \end{aligned}$$

Therefore, the monthly repayment is R 22 816,86 for a 30 year period.

At the end of the 30 years, David and Julie will have paid a total amount of:

$$\begin{aligned} &= 30 \times 12 \times \text{R } 22\,816,86 \\ &= \text{R } 8\,214\,069,60 \end{aligned}$$

The total amount of interest on the loan:

$$\begin{aligned} \text{Interest} &= \text{total amount paid} - \text{loan amount} \\ &= \text{R } 8\,214\,069,60 - \text{R } 2\,600\,000 \\ &= \text{R } 5\,614\,069,60 \end{aligned}$$

We notice that the interest on the loan is more than double the amount borrowed.



**Step 2: Consider a 20 year repayment period on the loan**

$$x = \frac{P \times i}{[1 - (1 + i)^{-n}]}$$

$$P = \text{R } 2\,600\,000$$

$$i = \frac{0,1}{12}$$

$$n = 20 \times 12 = 240$$

$$\begin{aligned} x &= \frac{2\,600\,000 \times \frac{0,1}{12}}{\left[1 - \left(1 + \frac{0,1}{12}\right)^{-240}\right]} \\ &= \text{R } 25\,090,56 \end{aligned}$$

Therefore, the monthly repayment is R 25 090,56 for a 20 year period.

At the end of the 20 years, David and Julie will have paid a total amount of:

$$\begin{aligned} &= 20 \times 12 \times \text{R } 25\,090,56 \\ &= \text{R } 6\,021\,734,40 \end{aligned}$$

The total amount of interest on the loan:

$$\begin{aligned} \text{Interest} &= \text{total amount paid} - \text{loan amount} \\ &= \text{R } 6\,021\,734,40 - \text{R } 2\,600\,000 \\ &= \text{R } 3\,421\,734,40 \end{aligned}$$

We notice that the interest on the loan is about 1,3 times the borrowed amount.

**Step 3: Consider the difference in the repayment and interest amounts**

$$\begin{aligned} \text{Difference in repayments} &= \text{R } 25\,090,56 - \text{R } 22\,816,86 \\ &= \text{R } 2\,273,70 \end{aligned}$$

It is also very interesting to look at the difference in the total interest paid:

$$\begin{aligned} \text{Difference in interest} &= \text{R } 5\,614\,069,60 - \text{R } 3\,421\,734,40 \\ &= \text{R } 2\,192\,335,20 \end{aligned}$$

Therefore, by paying an extra R 2273,70 each month over a shorter repayment period, David and Julie could save more than R 2 million on the repayment of their home loan.

When considering taking out a loan, it is advisable to investigate and compare a few options offered by financial institutions. It is very important to make informed decisions regarding personal finances and to make sure that the monthly repayment amount

is serviceable (payable). A credit rating is an estimate of a person's ability to fulfill financial commitments based on their previous payment history. Defaulting on a loan can affect a person's credit rating and their chances of taking out another loan in the future.

### Worked example 12: Analysing investment opportunities

#### QUESTION

Marlene wants to start saving for a deposit on a house. She can afford to invest between R 400 and R 600 each month and gets information from four different investment firms. Each firm quotes a different interest rate and a prescribed monthly installment amount. She plans to buy a house in 7 years' time. Calculate which company offers the best investment opportunity for Marlene.

|                     | Interest rate (compounded monthly) | Monthly payment |
|---------------------|------------------------------------|-----------------|
| TBS Investments     | 13,5% p.a.                         | R 450           |
| Taylor Anderson     | 13% p.a.                           | R 555           |
| PHK                 | 12,5% p.a.                         | R 575           |
| Simfords Consulting | 11% p.a.                           | R 600           |

#### SOLUTION

##### Step 1: Consider the different investment options

To compare the different investment options, we need to calculate the following for each option at the end of the seven year period:

- The future value of the monthly payments.
- The total amount paid into the investment fund.
- The total interest earned.

$$F = \frac{x [(1 + i)^n - 1]}{i}$$

##### TBS Investments:

$$F = \frac{450 \left[ \left( 1 + \frac{0,135}{12} \right)^{84} - 1 \right]}{\frac{0,135}{12}}$$

$$= \text{R } 62\,370,99$$

$$\begin{aligned} \text{Total amount (T)} : &= 7 \times 12 \times \text{R } 450 \\ &= \text{R } 37\,800 \end{aligned}$$

$$\begin{aligned} \text{Total interest (I)} : &= \text{R } 62\,370,99 - \text{R } 37\,800 \\ &= \text{R } 24\,570,99 \end{aligned}$$

**Taylor Anderson:**

$$F = \frac{555 \left[ \left( 1 + \frac{0,13}{12} \right)^{84} - 1 \right]}{\frac{0,13}{12}}$$

$$= \text{R } 75\,421,65$$

$$\text{Total amount } (T) : = 7 \times 12 \times \text{R } 555$$

$$= \text{R } 46\,620$$

$$\text{Total interest } (I) : = \text{R } 75\,421,65 - \text{R } 46\,620$$

$$= \text{R } 28\,801,65$$

**PHK:**

$$F = \frac{575 \left[ \left( 1 + \frac{0,125}{12} \right)^{84} - 1 \right]}{\frac{0,125}{12}}$$

$$= \text{R } 76\,619,96$$

$$\text{Total amount } (T) : = 7 \times 12 \times \text{R } 575$$

$$= \text{R } 48\,300$$

$$\text{Total interest } (I) : = \text{R } 76\,619,96 - \text{R } 48\,300$$

$$= \text{R } 28\,319,96$$

**Simfords Consulting:**

$$F = \frac{600 \left[ \left( 1 + \frac{0,11}{12} \right)^{84} - 1 \right]}{\frac{0,11}{12}}$$

$$= \text{R } 75\,416,96$$

$$\text{Total amount } (T) : = 7 \times 12 \times \text{R } 600$$

$$= \text{R } 50\,400$$

$$\text{Total interest } (I) : = \text{R } 75\,416,96 - \text{R } 50\,400$$

$$= \text{R } 25\,016,96$$

**Step 2: Draw a table of the results to compare the answers**

|                     | <i>F</i>    | <i>T</i> | <i>I</i>    |
|---------------------|-------------|----------|-------------|
| TBS Investments     | R 62 370,99 | R 37 800 | R 24 570,99 |
| Taylor Anderson     | R 75 421,65 | R 46 620 | R 28 801,65 |
| PHK                 | R 76 619,96 | R 48 300 | R 28 319,96 |
| Simfords Consulting | R 75 416,96 | R 50 400 | R 25 016,96 |

**Step 3: Make a conclusion**

An investment with PHK would provide Marlene with the highest deposit (R 76 619,96) for her house at the end of the 7 year period. However, we notice that an investment with Taylor Anderson would earn the highest amount of interest (R 28 801,65) and is therefore the better investment option.

**QUESTION**

William wants to take out a loan of R 750 000, so he approaches three different banks. He plans to start repaying the loan immediately and he calculates that he can afford a monthly repayment amount between R 5500 and R 7000.

Calculate which of the three options would be best for William.

- West Bank offers a repayment period of 30 years and an interest rate of prime compounded monthly.
- AcuBank offers a repayment period of 20 years and an interest rate of prime +0,5% compounded monthly.
- FinTrust Bank offers a repayment period of 15 years and an interest rate of prime +2% compounded monthly.

**SOLUTION****Step 1: Consider the different loan options**

To compare the different loan options, we need to calculate the following for each option:

- The monthly payment amount.
- The total amount paid to repay the loan.
- The amount of interest on the loan.

$$x = \frac{P \times i}{[1 - (1 + i)^{-n}]}$$

**West Bank:**

$$x = \frac{750\,000 \times \frac{0,085}{12}}{[1 - (1 + \frac{0,085}{12})^{-360}]}$$

$$= \text{R } 5766,85$$

$$\text{Total amount } (T) : = 30 \times 12 \times \text{R } 5766,85$$

$$= \text{R } 2\,076\,066$$

$$\text{Total interest } (I) : = \text{R } 2\,076\,066 - \text{R } 750\,000$$

$$= \text{R } 1\,326\,066$$

**AcuBank:**

$$x = \frac{750\,000 \times \frac{0,09}{12}}{[1 - (1 + \frac{0,09}{12})^{-240}]}$$

$$= \text{R } 6747,94$$

$$\text{Total amount } (T) : = 20 \times 12 \times \text{R } 6747,94$$

$$= \text{R } 1\,619\,505,60$$

$$\text{Total interest } (I) : = \text{R } 1\,619\,505,60 - \text{R } 750\,000$$

$$= \text{R } 869\,505,60$$

**FinTrust Bank:**

$$x = \frac{750\,000 \times \frac{0,105}{12}}{\left[1 - \left(1 + \frac{0,105}{12}\right)^{-180}\right]}$$
$$= R\,8290,49$$

$$\text{Total amount } (T) : = 15 \times 12 \times R\,8290,49$$
$$= R\,1\,492\,288,20$$

$$\text{Total interest } (I) : = R\,1\,492\,288,20 - R\,750\,000$$
$$= R\,742\,288,20$$

**Step 2: Draw a table of the results to compare the answers**

|               | $x$       | $T$            | $I$            |
|---------------|-----------|----------------|----------------|
| West Bank     | R 5766,85 | R 2 076 066,00 | R 1 326 066,00 |
| AcuBank       | R 6747,94 | R 1 619 505,60 | R 869 505,60   |
| FinTrust Bank | R 8290,49 | R 1 492 288,20 | R 742 288,20   |

**Step 3: Make a conclusion**

A loan from FinTrust Bank would accumulate the lowest amount of interest but the monthly repayment amounts are not within William's budget. Although West Bank offers the lowest interest rate and monthly repayment amount, the interest earned on the loan is very high as a result of the longer repayment period. If we assume that William must repay the loan over the given time periods, then AcuBank offers the best option.

However, we know that William can afford to pay more than R 5766,85 per month, and if the bank allows him to pay back the loan earlier, he should consider taking out a loan with West Bank and take advantage of the lower interest rate.

**Exercise 3 – 5: Analysing investment and loan options**

1. Cokisa is 31 years old and starting to plan for her future. She has been thinking about her retirement and wants to open an annuity so that she will have money when she retires. Her intention is to retire when she is 65 years old. Cokisa visits the Trader's Bank of Tembisa and learns that there are two investment options from which she can choose:
  - Option A: 7,76% p.a. compounded once every four months
  - Option B: 7,78% p.a. compounded half-yearly
  - a) Which is the better investment option for Cokisa if the amount she will deposit will always be the same?
  - b) Cokisa opens an account and starts saving R 4000 every four months. How much money (to the nearest rand) will she have saved when she reaches her planned retirement?

2. Phoebe wants to take out a home loan of R 1,6 million. She approaches three different banks for their loan options:

- Bank A offers a repayment period of 30 years and an interest rate of 12% per annum compounded monthly.
- Bank B offers a repayment period of 20 years and an interest rate of 14% per annum compounded monthly.
- Bank C offers a repayment period of 30 years and an interest rate of 14% per annum compounded monthly.

If Phoebe intends to start her monthly repayments immediately, calculate which of the three options would be best for her.

3. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 28KN    2. 28KP



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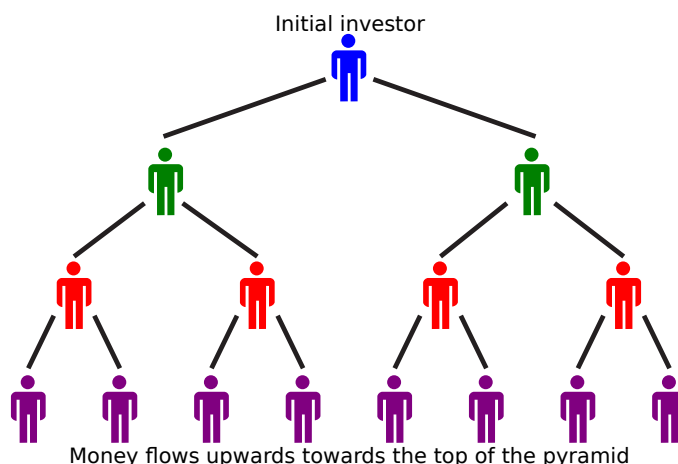
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## Pyramid schemes

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A pyramid scheme is a moneymaking scheme that promises investors unusually high returns on their investment. The concept of a pyramid scheme is quite simple and should be easy to identify, however, it is often cleverly disguised as a legitimate business. In some cases a product is offered and in others the scheme is marketed as a highly profitable investment opportunity. Unfortunately, many of these schemes have cost millions of people their savings. Pyramid schemes are illegal in South Africa.

- "Ponzi schemes" are named after Charles Ponzi, a Italian businessman who lived in the U.S. and Canada. One of his investment schemes involved buying postal reply coupons in other countries and redeeming them for a higher value in the United States. He promised investors 100% profit within 90 days of their investment. As the scheme grew, Ponzi paid off early investors with money from investors who joined the scheme at a later stage. Ponzi's fraudulent scheme was exposed and investors lost millions of dollars. He was sent to prison for a number of years.
- South African Adriaan Nieuwoudt started a pyramid scheme commonly referred to as the "Kubus" scheme. Participants bought a biological substance called an activator that was supposed to be used in beauty products. The activator was used to grow cultures in milk that were then dried and ground up and resold to new participants. The scheme had thousands of investors and took in approximately R 140 million before it was declared an illegal lottery.



A pyramid scheme starts with one person, who finds other people to invest their money in the scheme. These people then enrol more people into the scheme and the base of the pyramid grows. The money invested by new investors goes to participants closer to the top of the pyramid. This is unsustainable because it requires that more and more people join the scheme and the growth must end at some point because there are a finite number of people. Therefore, most of the investors lose their money when the scheme collapses.

The South African Reserve Bank has launched a public awareness campaign, “Beware of oMashayana (crooks)”, to help educate people about pyramid schemes and how to identify them.

### ***Beware of oMashayana***

*So, you think you’ve found the perfect investment? Regular high returns with no risk? Be careful. Don’t lose all your money. If it sounds too good to be true, it’s probably a Pyramid or Ponzi scheme.*

### ***What is a Ponzi scheme?***

*A Ponzi con-artist will only ask you to give them money that they say will be invested in a scheme or project, such as supposed property developments; bridging finance; foreign-exchange transactions; venture capital to other companies; or share units. The scheme operator promises to give you back much more money than you have given them initially, in a very short space of time.*

### ***What is a pyramid scheme?***

*A pyramid con-artist will offer you the chance to make quick money for yourself, often by selling something. You pay a joining fee, buy the product and then sell it. They tell you that the more people you get to sell for you, the more money you will make. It’s easy for the first people to introduce new members, but soon everyone is part of the scheme and it gets harder to find new members to join. For example, the con-artist recruits 6 people who each pay a R 100 fee. Each of those people has to recruit 6 people. There are now up to 36 people. Now, those 36 people each have to sign up 6 people – this equals 216, and by the time you have to get to level 10, you have to get 60 million people to join to keep the scheme going. This can never work.*

### ***What is the difference between a pyramid and a Ponzi scheme?***

*The main difference is that with a pyramid scheme you have to work or sell to recruit investors, while with a Ponzi scheme the con-artist will only ask you to invest in something (for example, property development). Both schemes are illegal.*

**What can I do to protect myself?**

You are your own best protection. It is your responsibility to make sure that you never give your money to any company or person that is not registered as a deposit-taking institution in terms of the Banks Act.

**Why is handing my money over so risky?**

When you hand over your money (notes and coins) to another person who then loses the money, steals it or goes bankrupt, you only have an unsecured claim against that person or their estate and you might not get all your money back.

**Why is it safer to hand my money to a bank?**

Banks and investment companies have to be registered so that they can be regulated and supervised, to make sure that your money is safe. Unregulated and unsupervised persons and groups don't follow these rules and your money is at great risk with them.

## 3.6 Summary

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- Always keep the rate of interest per time unit and the time period in the same units.
- Simple interest:  $A = P(1 + in)$
- Compound interest:  $A = P(1 + i)^n$
- Simple depreciation:  $A = P(1 - in)$
- Compound depreciation:  $A = P(1 - i)^n$
- Nominal and effective annual interest rates:  $1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m$
- **Future value of payments:**

$$F = \frac{x[(1 + i)^n - 1]}{i}$$

Payment amount:

$$x = \frac{F \times i}{[(1 + i)^n - 1]}$$

- **Present value of a series of payments:**

$$P = \frac{x[1 - (1 + i)^{-n}]}{i}$$

Payment amount:

$$x = \frac{P \times i}{[1 - (1 + i)^{-n}]}$$



### Exercise 3 – 6: End of chapter exercises

1. Mpumelelo deposits R 500 into a savings account, which earns interest at 6,81% p.a. compounded quarterly. How long will it take for the savings account to have a balance of R 749,77?
2. How much interest will Gavin pay on a loan of R 360 000 for 5 years at 10,3% per annum compounded monthly?
3. Wingfield school will need to replace a number of old classroom desks in 6 years' time. The principal has calculated that the new desks will cost R 44 500. The school establishes a sinking fund to pay for the new desks and immediately deposits an amount of R 6300 into the fund, which accrues interest at a rate of 6,85% p.a. compounded monthly.
  - a) How much money should the school save every month so that the sinking fund will have enough money to cover the cost of the desks?
  - b) How much interest does the fund earn over the period of 6 years?
4. Determine how many years (to the nearest integer) it will take for the value of a motor vehicle to decrease to 25% of its original value if the rate of depreciation, based on the reducing-balance method, is 21% per annum.
5. Angela has just started a new job, and wants to save money for her retirement. She decides to deposit R 1300 into a savings account once each month. Her money goes into an account at Pinelands Mutual Bank, and the account receives 6,01% interest p.a. compounded once each month.
  - a) How much money will Angela have in her account after 30 years?
  - b) How much money did Angela deposit into her account after 30 years?
6.
  - a) Nicky has been working at Meyer and Associates for 5 years and gets an increase in her salary. She opens a savings account at Langebaan Bank and begins making deposits of R 350 every month. The account earns 5,53% interest p.a., compounded monthly. Her plan is to continue saving on a monthly schedule until she retires. However, after 8 years she stops making the monthly payments and leaves the account to continue growing. How much money will Nicky have in her account 29 years after she first opened it?
  - b) Calculate the difference between the total deposits made into the account and the amount of interest paid by the bank.
7.
  - a) Every three months Louis puts R 500 into an annuity. His account earns an interest rate of 7,51% p.a. compounded quarterly. How long will it take Louis's account to reach a balance of R 13 465,87?
  - b) How much interest will Louis receive from his investment?
8. A dairy farmer named Kayla needs to buy new equipment for her dairy farm which costs R 200 450. She bought her old equipment 12 years ago for R 167 000. The value of the old equipment depreciates at a rate of 12,2% per year on a reducing balance. Kayla will need to arrange a bond for the remaining cost of the new equipment.

An agency which supports farmers offers bonds at a special interest rate of 10,01% p.a. compounded monthly for any loan up to R 175 000 and 9,61% p.a. compounded monthly for a loan above that amount. Kayla arranges a bond such that she will not need to make any payments on the loan in the first six months (called a 'grace period') and she must pay the loan back over 20 years.

  - a) Determine the monthly payment.
  - b) What is the total amount of interest Kayla will pay for the bond?

c) By what factor is the interest she pays greater than the value of the loan?  
Give the answer correct to one decimal place.

9. Thabo invests R 8500 in a special banking product which will pay 1% per annum for 1 month, then 2% per annum for the next 2 months, then 3% per annum for the next 3 months, 4% per annum for the next 4 months, and 0% for the rest of the year. If the bank charges him R 75 to open the account, how much can he expect to get back at the end of the year?
10. Thabani and Lungelo are both using Harper Bank for their savings. Lungelo makes a deposit of  $x$  at an interest rate of  $i$  for six years. Three years after Lungelo made his first deposit, Thabani makes a deposit of  $3x$  at an interest rate of 8% per annum. If after 6 years their investments are equal, calculate the value of  $i$  (correct to three decimal places). If the sum of their investment is R 20 000, determine how much Thabani earned in 6 years.
11. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [28KQ](#) 2. [28KR](#) 3. [28KS](#) 4. [28KT](#) 5. [28KV](#) 6. [28KW](#)  
7. [28KX](#) 8. [28KY](#) 9. [28KZ](#) 10. [28M2](#)



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